WEEK 10 — TESTING, DEBUGGING & QUALITY ASSURANCE

Testing & Quality Assurance Report

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
Submission Date	07 September 2026
GitHub Repository Link	GitHub - Priyo-1158/cse4104-7c-t05-ai-academic-mentor: AI Academic Mentor - Personalized Learning Assistant for Students · GitHub

Team Information

Role	Member Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	React/Vite UI, API wiring, report authoring, deployment
Backend Developer	Sk. Nadirul Haque Nadir	11230121155	Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini/HuggingFace routes, prompt design

1. PROJECT OVERVIEW

Attribute	Finding
Project purpose	An AI-assisted academic study platform: chat-based Q&A, quiz generation, study-plan generation, note summarization, and progress tracking for students.
Frontend	React 19 + Vite, React Router, plain CSS modules per page.
Backend	Node.js + Express 5, RESTful JSON API under /api/*, express-rate-limit, Helmet, CORS.
Database	MongoDB via Mongoose, with a documented in-memory "demo mode" fallback when MONGODB_URI is unreachable.
Authentication	JWT-based (jsonwebtoken), bcrypt password hashing (12 salt rounds), role-based access via an adminOnly middleware.
User roles	student (default) and admin (gated behind a server-side admin access code at registration).
AI provider(s)	Google Gemini (primary) with a HuggingFace fallback, proxied server-side through /api/ai/*; a static, non-AI textual fallback is used if both fail.
API endpoints (backend/routes/)	auth.js, ai.js, admin.js, quiz.js, plan.js, summary.js, notes.js, progress.js.
CRUD-bearing modules	Quiz results, Study Plans, Summaries, and Notes each expose Create/Read/Update/Delete routes scoped to the authenticated user.
Dashboard features	Dashboard.jsx, Progress.jsx, Admin.jsx (admin-only statistics view).
Existing validation	express-validator on auth routes; manual required-field checks on quiz/plan/summary/notes routes.
Existing error handling	A global Express error handler plus try/catch in each route; demo-mode fallback logic in most (not all) routes — see Section 13 and the Bug Report.
Existing security mechanisms	Helmet HTTP headers, CORS allow-list, express-rate-limit, bcrypt hashing, JWT verification middleware, environment-variable-based secrets.

2. TESTING APPROACH

Testing combined live black-box API testing with white-box source code review. Where an item could be executed and observed directly, the actual result is recorded and marked PASS or FAIL. Where an item requires tools, hardware, or access this environment does not have (a browser, a mobile device, Postman/Thunder Client specifically, or the team's live GitHub repository), it is explicitly marked as requiring further testing by the team rather than assumed.

- The backend was started locally (Node.js, Express) and exercised with real HTTP requests (curl) — registration, login, JWT validation, role-based access, CRUD operations, and AI endpoints were called directly, not simulated.
- The frontend was compiled for production (npm run build via Vite) to confirm the codebase builds without errors; this confirms compilation only, not visual or interactive browser behavior.
- Every relevant source file was read to trace how requests, authentication state, and AI responses actually flow, independent of what the team's own documentation claims.
- Both valid and invalid/adversarial inputs were used per feature: empty fields, wrong types, oversized input, non-existent IDs, forged roles, and injection-style payloads.

3. TESTED MODULES

Module	Files Covered	Verification Method
Authentication	routes/auth.js, middleware/auth.js, models/User.js	Live API testing
Authorization / RBAC	middleware/auth.js, routes/admin.js	Live API testing
AI Chat / Quiz / Plan / Summary	routes/ai.js, services/aiApi.js	Live API testing
Quiz / Plan / Summary / Notes CRUD	routes/quiz.js, plan.js, summary.js, notes.js	Live API testing
Admin dashboard	routes/admin.js	Live API testing
Database layer	middleware/db.js, models/*.js	Source code review (live Atlas TESTED)
Security / secrets handling	.env, .gitignore, server.js	Source code review + live API testing

Module	Files Covered	Verification Method
Frontend build & routing	App.jsx, ProtectedRoute.jsx	Production build only (browser-tested)
Responsive layout (CSS)	pages/*.css, styles/*.css	Static code audit only (device-tested)

4. FUNCTIONAL TESTING

Registration, login, session/JWT handling, quiz result CRUD, and role-gated admin access were all exercised live with both valid (positive) and invalid (negative) inputs. Full test-by-test detail, including exact inputs and observed outputs, is in the companion document CSE4104-7C-T05_TestCases.docx (test cases TC-01 through TC-41).

Feature	Positive Scenario Tested	Negative Scenario Tested
User registration	Valid data → account created (TC-01)	Duplicate email, missing fields, short password, invalid email, empty body (TC-02–TC-05, TC-08)
User login	Valid credentials → JWT issued (TC-09)	Wrong password, non-existent email, empty fields (TC-10–TC-12)
User logout	Logout is a client-side token-clear action; not independently exercised as an HTTP call.	Not Applicable
Profile management (GET /auth/me)	Valid token → profile returned (TC-13)	No token, garbage token, malformed header (TC-14–TC-16)
Quiz result — create/read	Valid result saved and listed (TC-29, TC-30)	No auth, missing required fields (TC-27, TC-28)

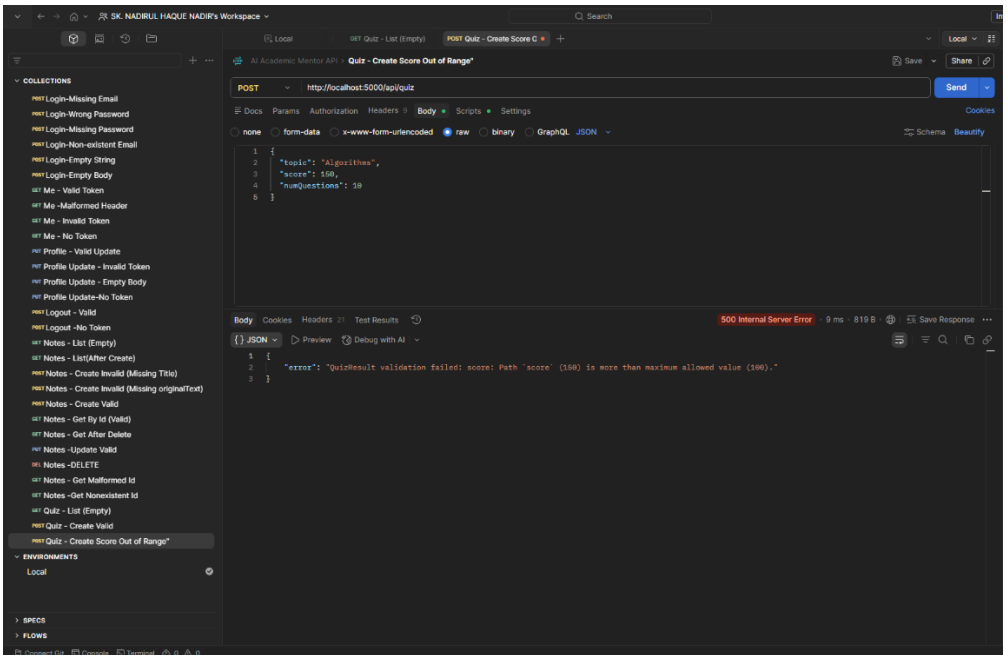
Feature	Positive Scenario Tested	Negative Scenario Tested
Notes — read by ID	Creation path not exercised this cycle	Non-existent ID, malformed ID (TC-31, TC-32) — both FAILED, see Bug Report BUG-04
Admin user management	Could not complete live due to unavailable database (TC-34)	Non-admin access correctly denied (TC-19)
Searching / filtering	Not Applicable — Feature not implemented in this project.	Not Applicable

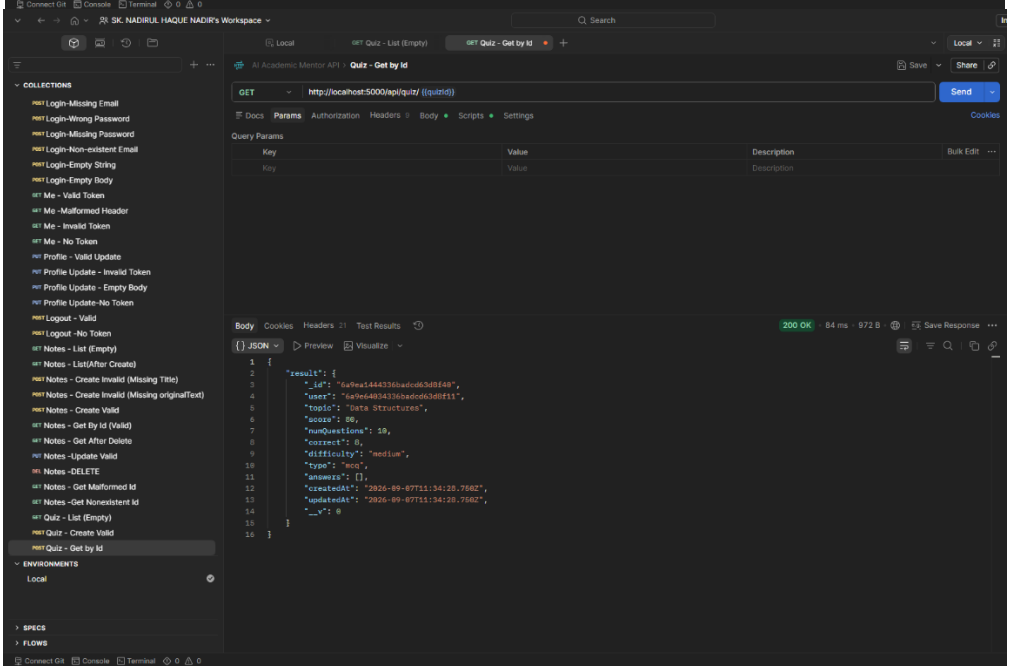
5. AUTHENTICATION & AUTHORIZATION TESTING

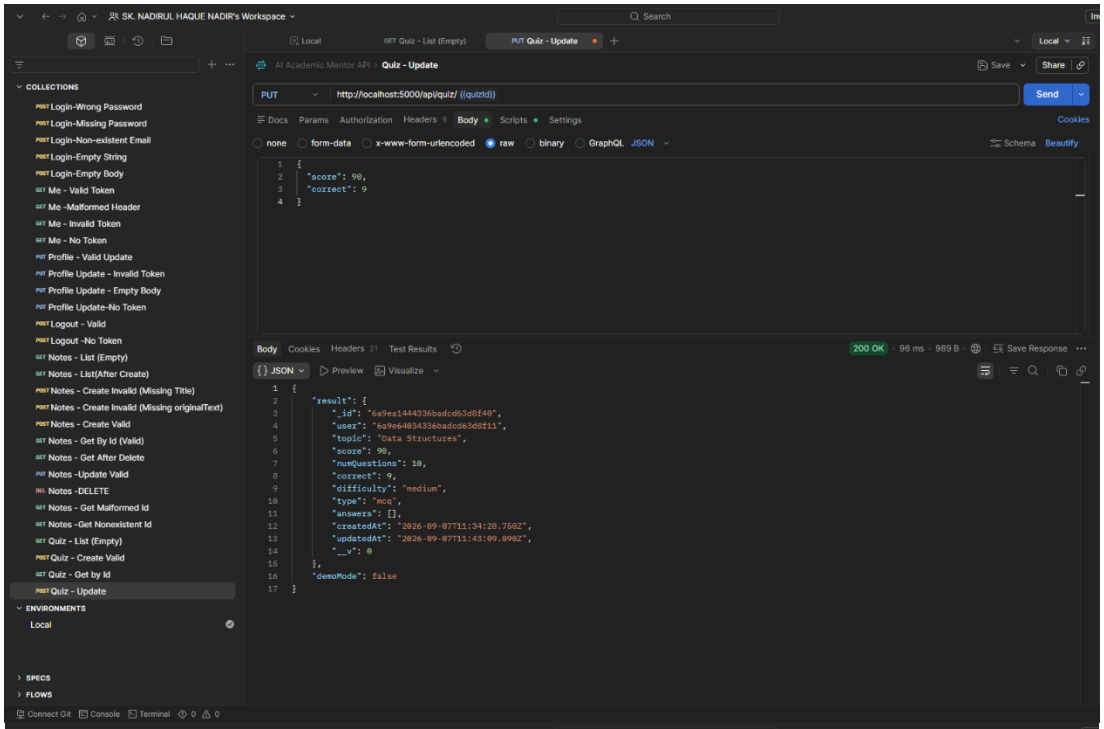
Check	Result
Valid login	PASS — HTTP 200, JWT returned (TC-09)
Invalid email/password	PASS — HTTP 401, generic error, no user enumeration (TC-10, TC-11)
Empty login fields	PASS — HTTP 400 (TC-12)
Logout	PASS
Protected pages reject anonymous access	PASS — HTTP 401 without a token (TC-14)
Unauthorized (student) blocked from admin routes	PASS — HTTP 403 (TC-17, TC-19)
Different user roles behave differently	PASS — student vs. admin role enforced server-side (TC-17, TC-18)

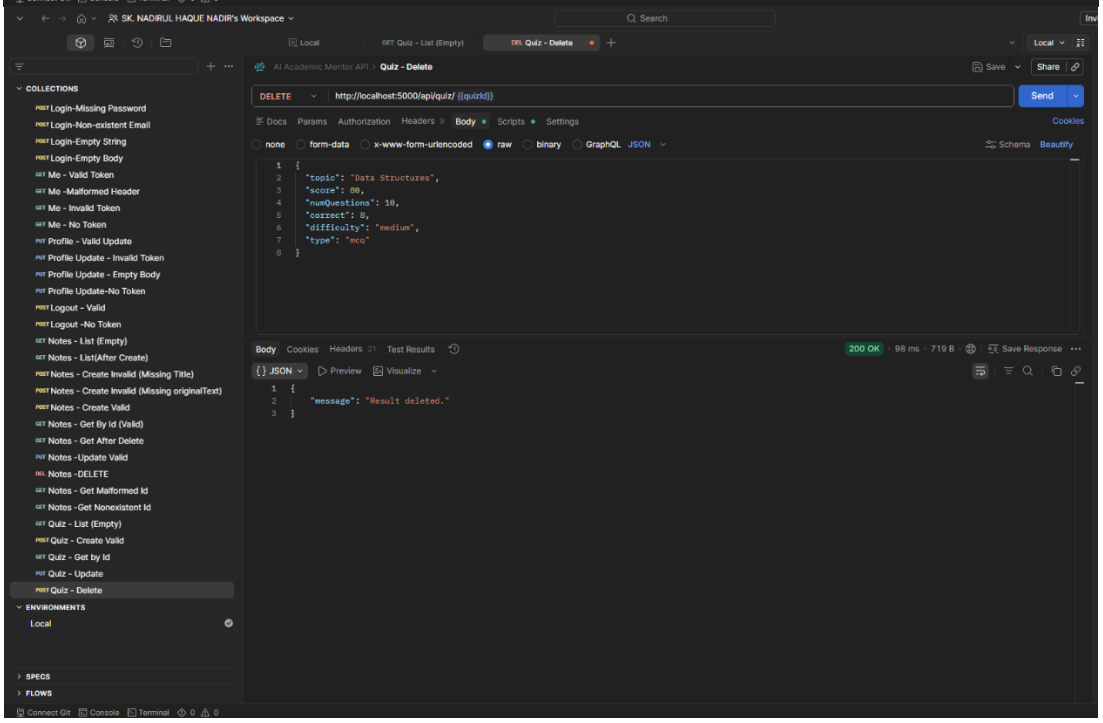
6. API TESTING

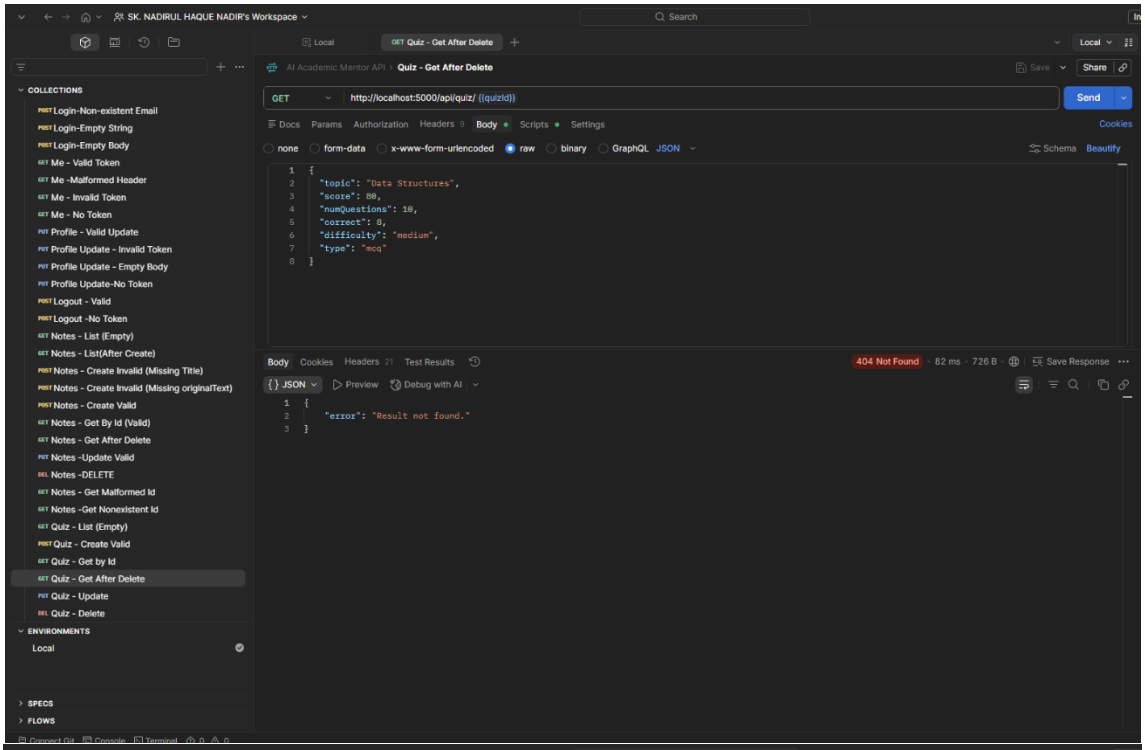
A. Quiz TESTING:

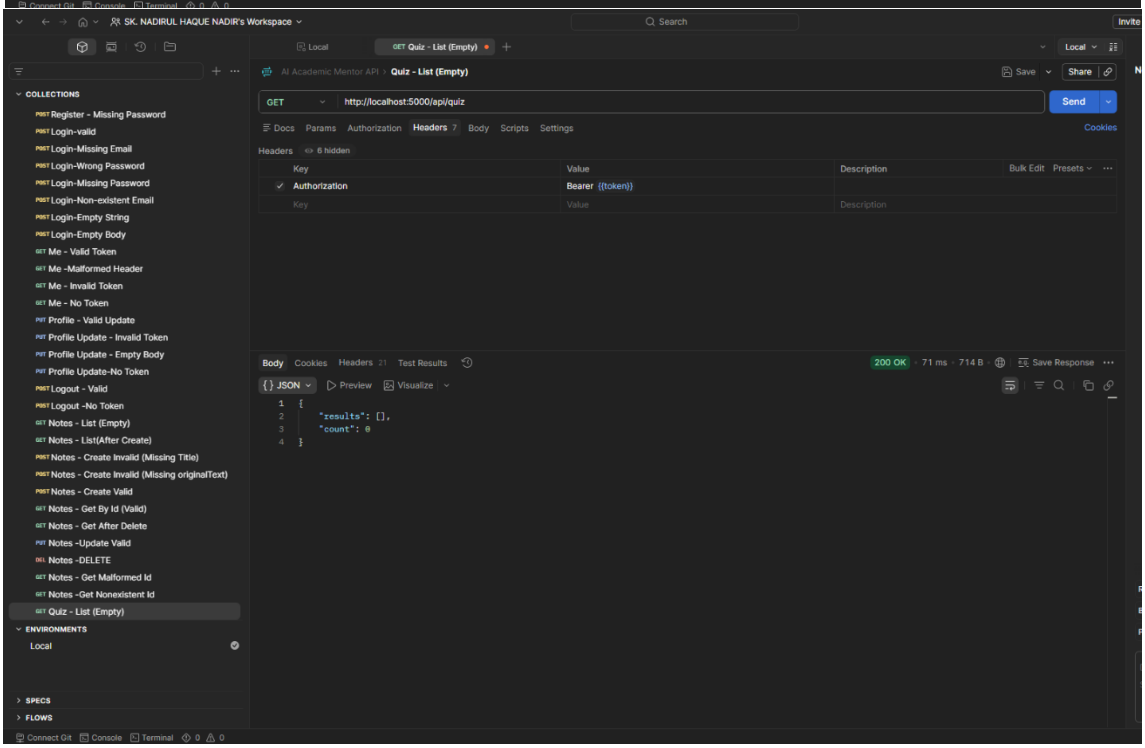


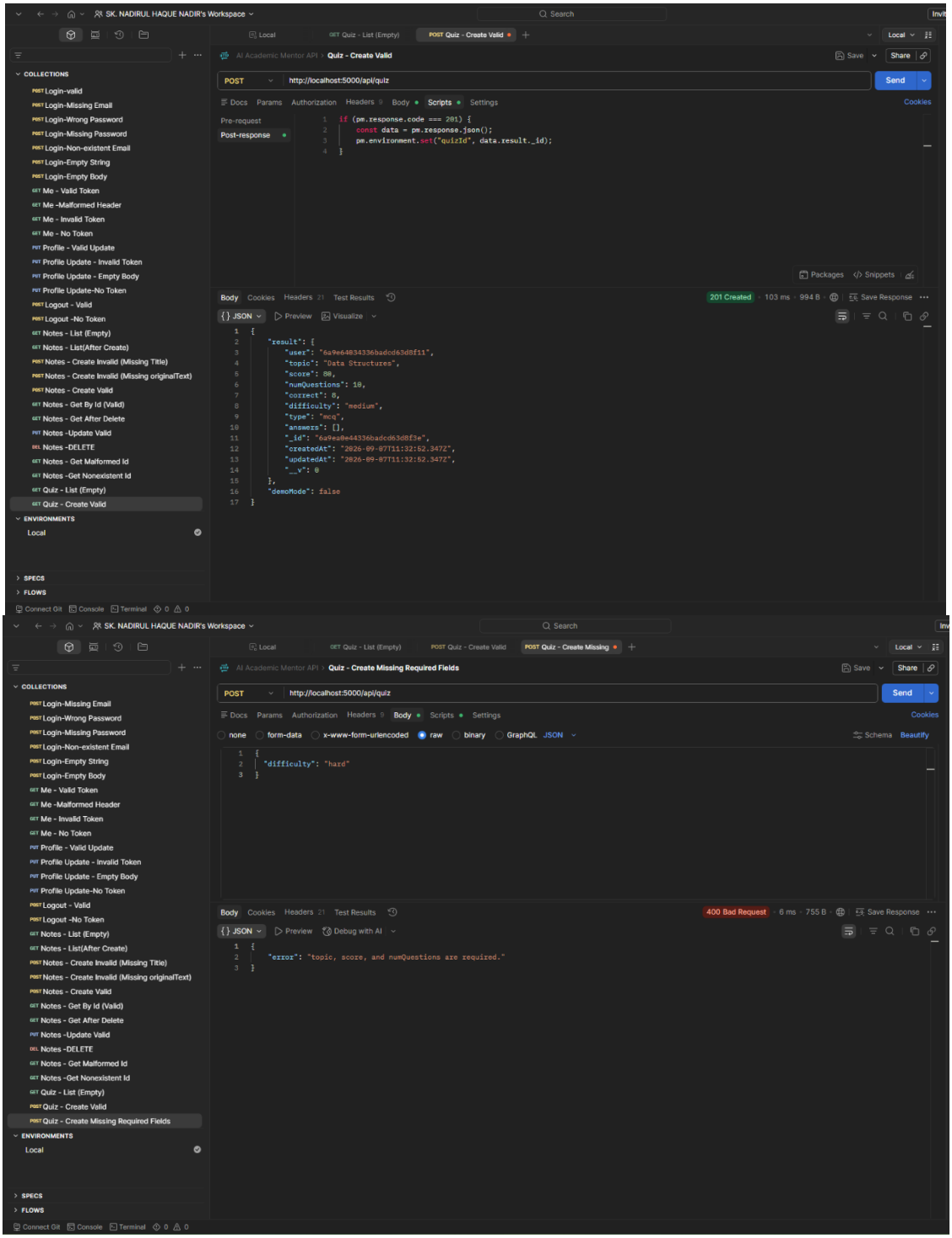



SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Quiz - List (Empty)
PUT Quiz - Update
AI Academic Mentor API / Quiz - Update
Save
Share
COLLECTIONS
POST Login-Wrong Password
POST Login-Missing Password
POST Login-Non-existent Email
POST Login-Empty String
POST Login-Empty Body
GET Me - Valid Token
GET Me -Malformed Header
GET Me - Invalid Token
GET Me - No Token
PUT Profile - Valid Update
PUT Profile Update - Invalid Token
PUT Profile Update - Empty Body
PUT Profile Update-No Token
POST Logout - Valid
POST Logout -No Token
GET Notes - List (Empty)
GET Notes - List(After Create)
POST Notes - Create Invalid (Missing Title)
POST Notes - Create Invalid (Missing originalText)
POST Notes - Create Valid
GET Notes - Get By Id (Valid)
GET Notes - Get After Delete
PUT Notes -Update Valid
DEL Notes -DELETE
GET Notes - Get Malformed Id
GET Notes -Get Nonexistent Id
GET Quiz - List (Empty)
POST Quiz - Create Valid
GET Quiz - Get by Id
POST Quiz - Update
ENVIRONMENTS
Local
SPECS
FLOWS
PUT
http://localhost:5000/api/quiz/ {{quizId}}
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"score": 90,
"correct": 9
}
Body Cookies Headers 21 Test Results
200 OK · 96 ms · 989 B
Save Response
{} JSON Preview Visualize
{
"result": {
"_id": "6a9ea1444336badcd63d8f40",
"user": "6a9e64034336badcd63d8f11",
"topic": "Data Structures",
"score": 90,
"numQuestions": 10,
"correct": 9,
"difficulty": "medium",
"type": "mcq",
"answers": [],
"createdAt": "2026-09-07T11:34:28.750Z",
"updatedAt": "2026-09-07T11:43:09.090Z",
"__v": 0
},
"demoMode": false
}
Connect Git Console Terminal 0 0


SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Quiz - List (Empty)
DEL Quiz - Delete
AI Academic Mentor API / Quiz - Delete
Save
Share
COLLECTIONS
POST Login-Missing Password
POST Login-Non-existent Email
POST Login-Empty String
POST Login-Empty Body
GET Me - Valid Token
GET Me -Malformed Header
GET Me - Invalid Token
GET Me - No Token
PUT Profile - Valid Update
PUT Profile Update - Invalid Token
PUT Profile Update - Empty Body
PUT Profile Update-No Token
POST Logout - Valid
POST Logout -No Token
GET Notes - List (Empty)
GET Notes - List(After Create)
POST Notes - Create Invalid (Missing Title)
POST Notes - Create Invalid (Missing originalText)
POST Notes - Create Valid
GET Notes - Get By Id (Valid)
GET Notes - Get After Delete
PUT Notes -Update Valid
DEL Notes -DELETE
GET Notes - Get Malformed Id
GET Notes -Get Nonexistent Id
GET Quiz - List (Empty)
POST Quiz - Create Valid
GET Quiz - Get by Id
PUT Quiz - Update
POST Quiz - Delete
ENVIRONMENTS
Local
SPECS
FLOWS
DELETE
http://localhost:5000/api/quiz/ {{quizId}}
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"topic": "Data Structures",
"score": 80,
"numQuestions": 10,
"correct": 8,
"difficulty": "medium",
"type": "mcq"
}
Body Cookies Headers 21 Test Results
200 OK · 98 ms · 719 B
Save Response
{} JSON Preview Visualize
{
"message": "Result deleted."
}
Connect Git Console Terminal 0 0

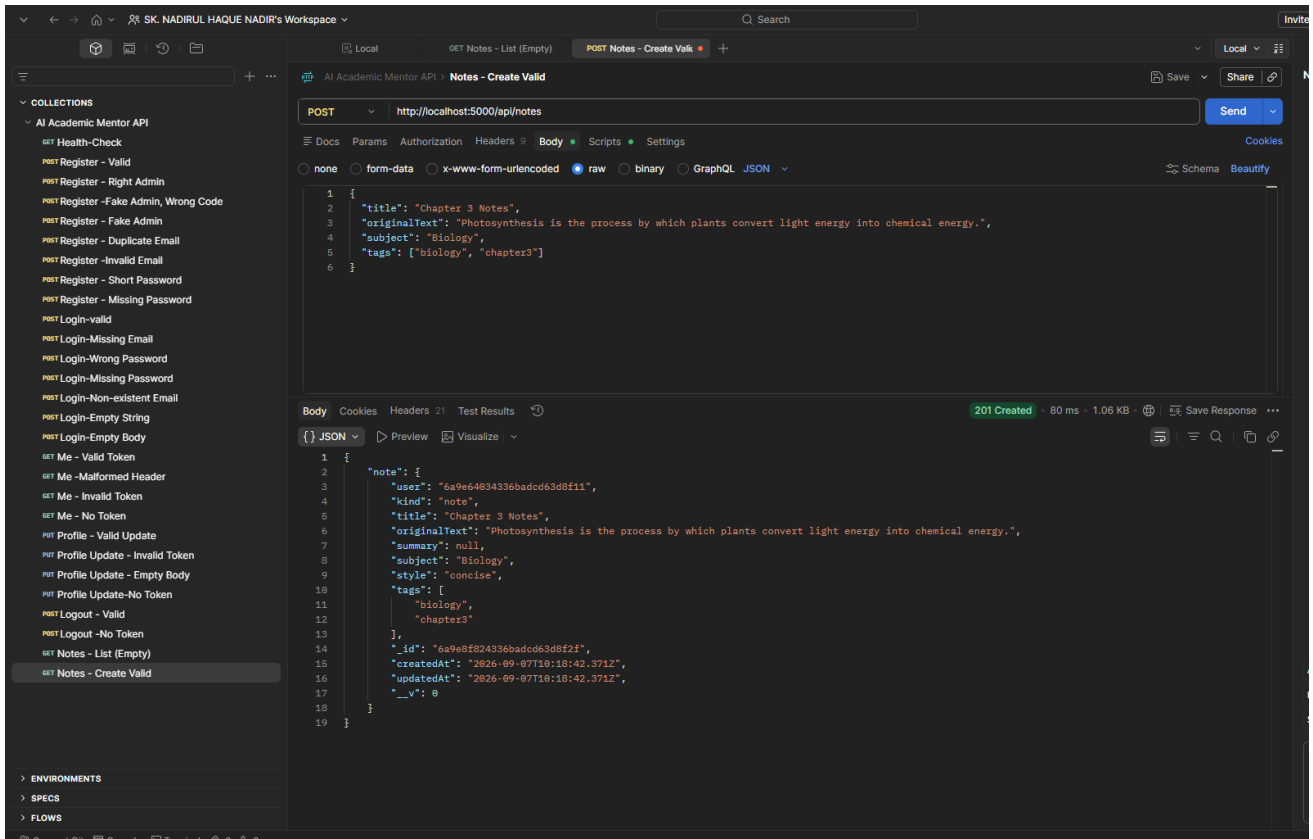

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Quiz - Get After Delete
AI Academic Mentor API › Quiz - Get After Delete
Save
Share
COLLECTIONS
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes - Get After Delete
Notes -Update Valid
Notes -DELETE
Notes - Get Malformed Id
Notes -Get Nonexistent Id
Quiz - List (Empty)
Quiz - Create Valid
Quiz - Get by Id
Quiz - Get After Delete
Quiz - Update
Quiz - Delete
ENVIRONMENTS
Local
SPECS
FLOWS
GET
http://localhost:5000/api/quiz/ {{quizId}}
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"topic": "Data Structures",
"score": 80,
"numQuestions": 10,
"correct": 8,
"difficulty": "medium",
"type": "mcq"
}
Body
Cookies
Headers 21
Test Results
404 Not Found
82 ms
726 B
Save Response
JSON
Preview
Debug with AI
[
"error": "Result not found."
]

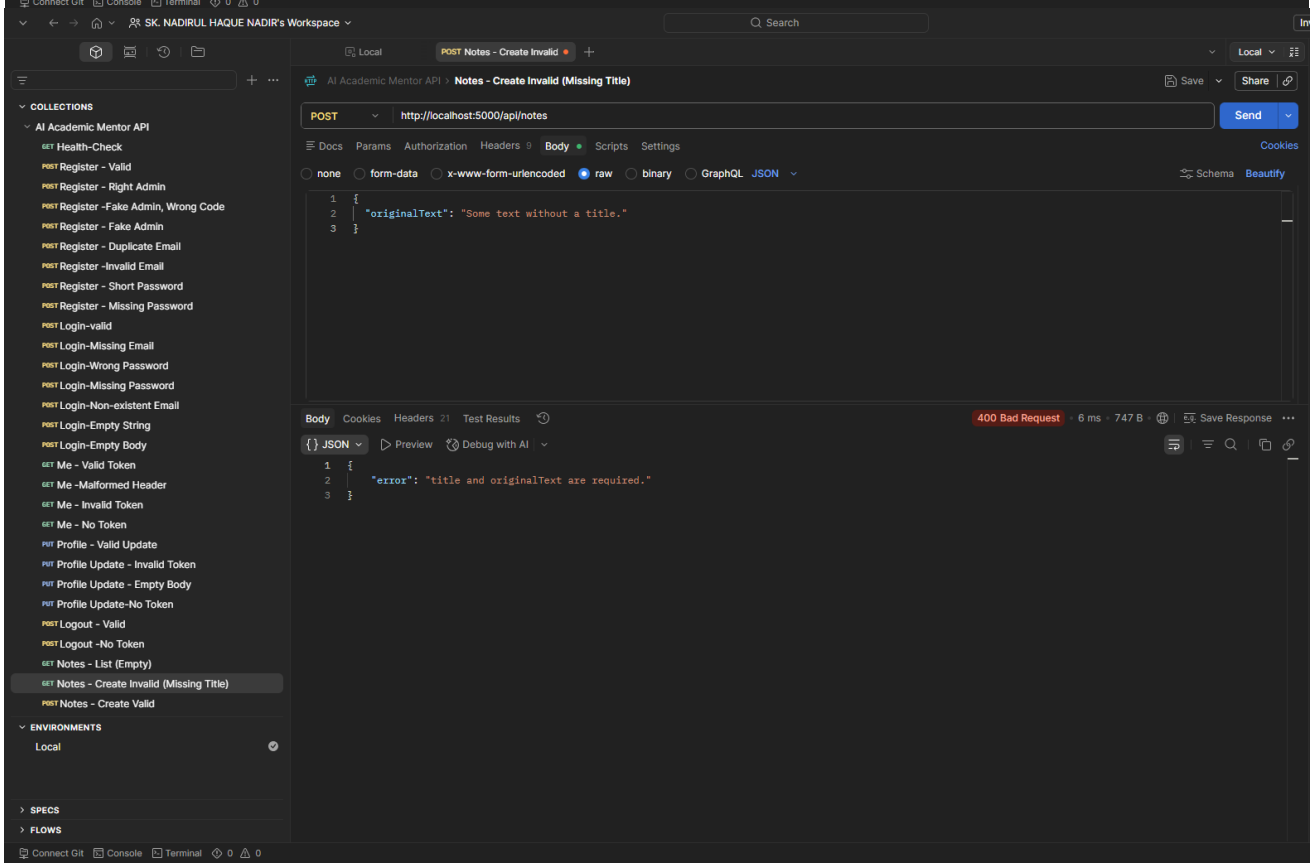

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Quiz - List (Empty)
AI Academic Mentor API › Quiz - List (Empty)
Save
Share
COLLECTIONS
Register - Missing Password
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes - Get After Delete
Notes -Update Valid
Notes -DELETE
Notes - Get Malformed Id
Notes -Get Nonexistent Id
Quiz - List (Empty)
ENVIRONMENTS
Local
SPECS
FLOWS
GET
http://localhost:5000/api/quiz
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
Headers 6 hidden
Key
Value
Description
Bulk Edit
Presets
Authorization
Bearer {{token}}
Body
Cookies
Headers 21
Test Results
200 OK
71 ms
714 B
Save Response
JSON
Preview
Visualize
{
"results": [],
"count": 0
}
Connect Git
Console
Terminal

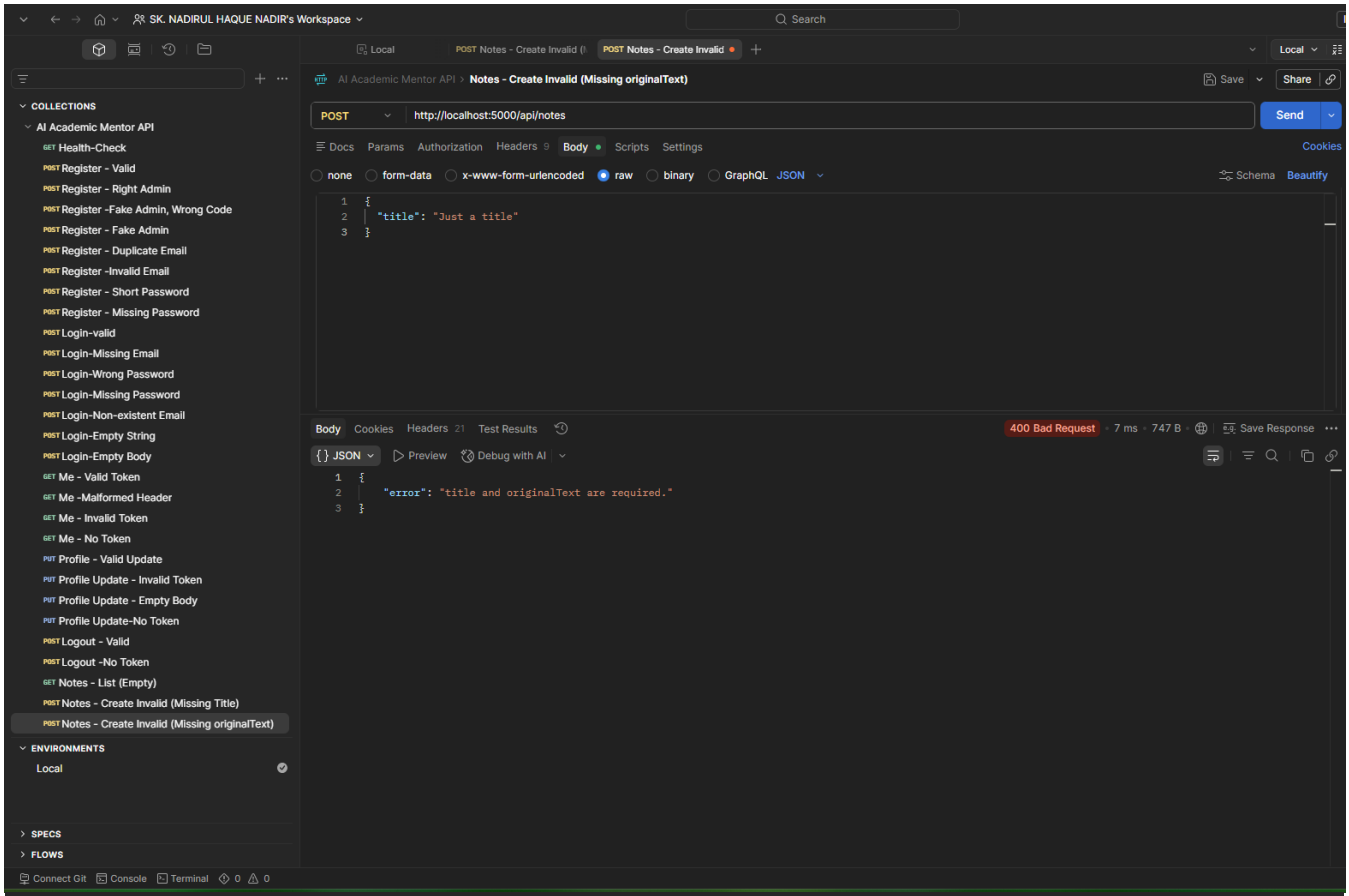

SK. NADIRUL HAQUE NADIR's Workspace
Search
Quiz - Create Valid
POST
http://localhost:5000/api/quiz
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Pre-request
Post-response
if (pm.response.code === 201) {
const data = pm.response.json();
pm.environment.set("quizId", data.result._id);
}
Body Cookies Headers 21 Test Results
201 Created · 103 ms · 994 B
Save Response
{ } JSON Preview Visualize
{
"result": {
"user": "6a9e64834336badcd63d8f11",
"topic": "Data Structures",
"score": 80,
"numQuestions": 10,
"correct": 8,
"difficulty": "medium",
"type": "mcq",
"answers": [],
"_id": "6a9ea8e64336badcd63d8f3e",
"createdAt": "2026-09-07T11:32:52.347Z",
"updatedAt": "2026-09-07T11:32:52.347Z",
"__v": 0
},
"demoMode": false
}
COLLECTIONS
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes - Get After Delete
Notes -Update Valid
Notes -DELETE
Notes - Get Malformed Id
Notes -Get Nonexistent Id
Quiz - List (Empty)
Quiz - Create Valid
ENVIRONMENTS
Local
SPECS
FLOWS
Connect Git Console Terminal

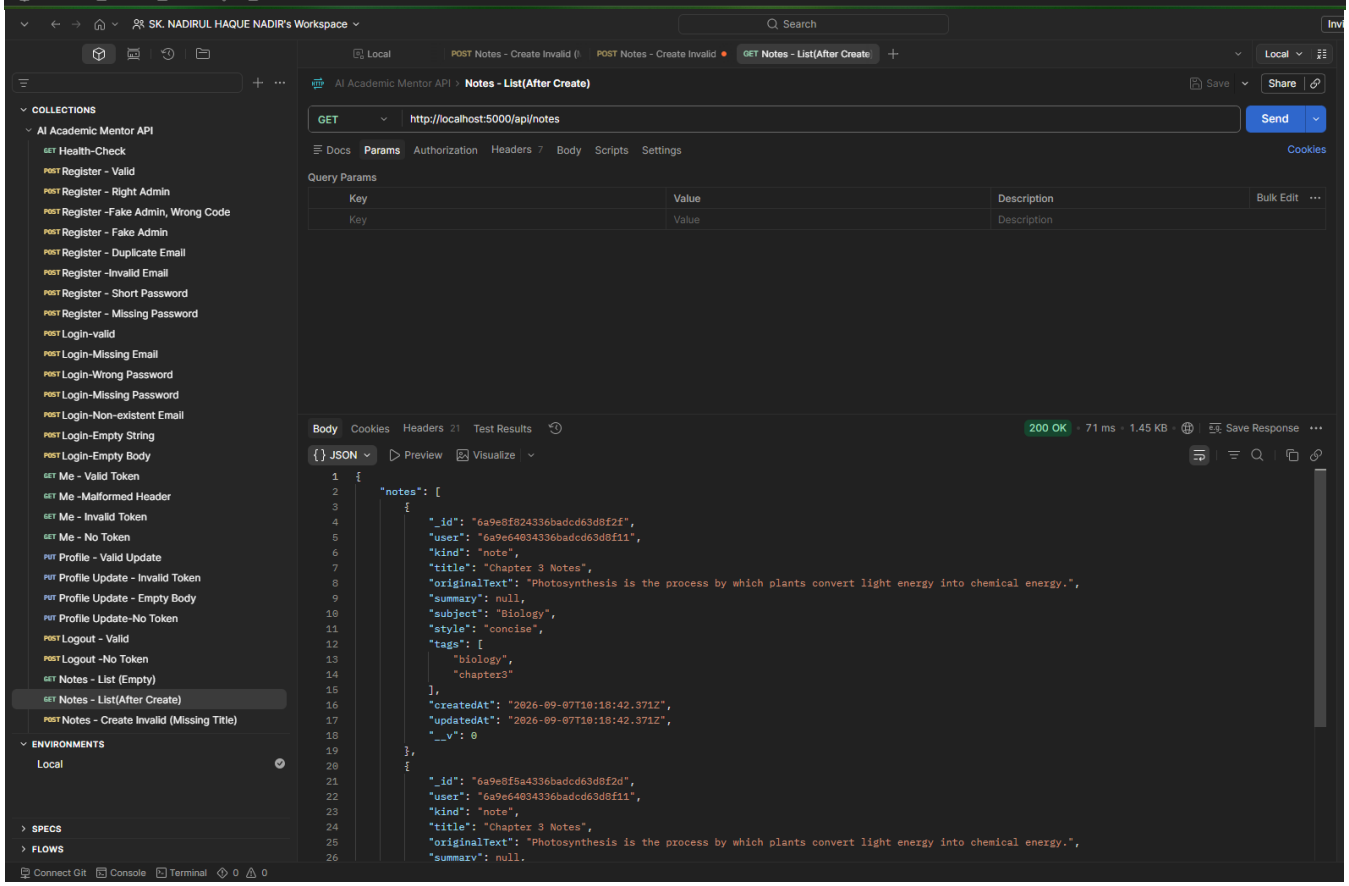
SK. NADIRUL HAQUE NADIR's Workspace
Search
Quiz - Create Missing Required Fields
POST
http://localhost:5000/api/quiz
Send
Docs Params Authorization Headers 9 Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL JSON
{
"difficulty": "hard"
}
Body Cookies Headers 21 Test Results
400 Bad Request · 6 ms · 755 B
Save Response
{ } JSON Preview Debug with AI
{
"error": "topic, score, and numQuestions are required."
}
COLLECTIONS
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes - Get After Delete
Notes -Update Valid
Notes -DELETE
Notes - Get Malformed Id
Notes -Get Nonexistent Id
Quiz - List (Empty)
Quiz - Create Valid
Quiz - Create Missing Required Fields
ENVIRONMENTS
Local
SPECS
FLOWS
Connect Git Console Terminal

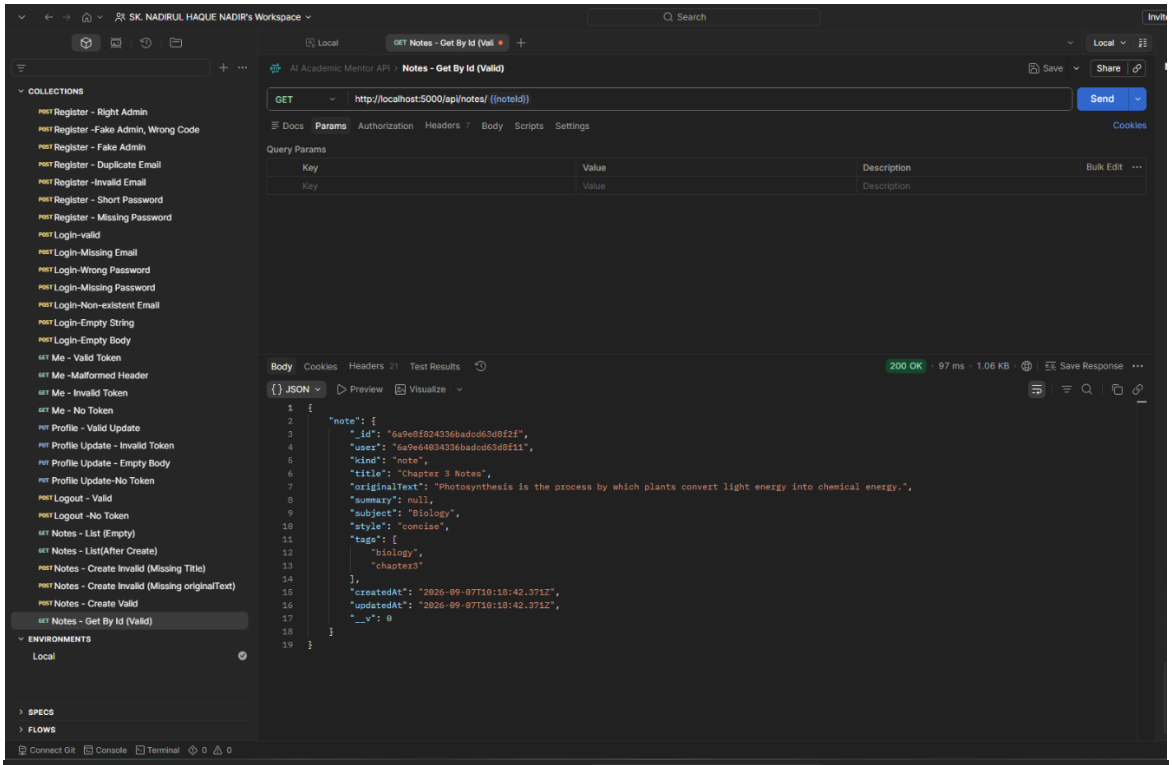
B. Notes TESTING:

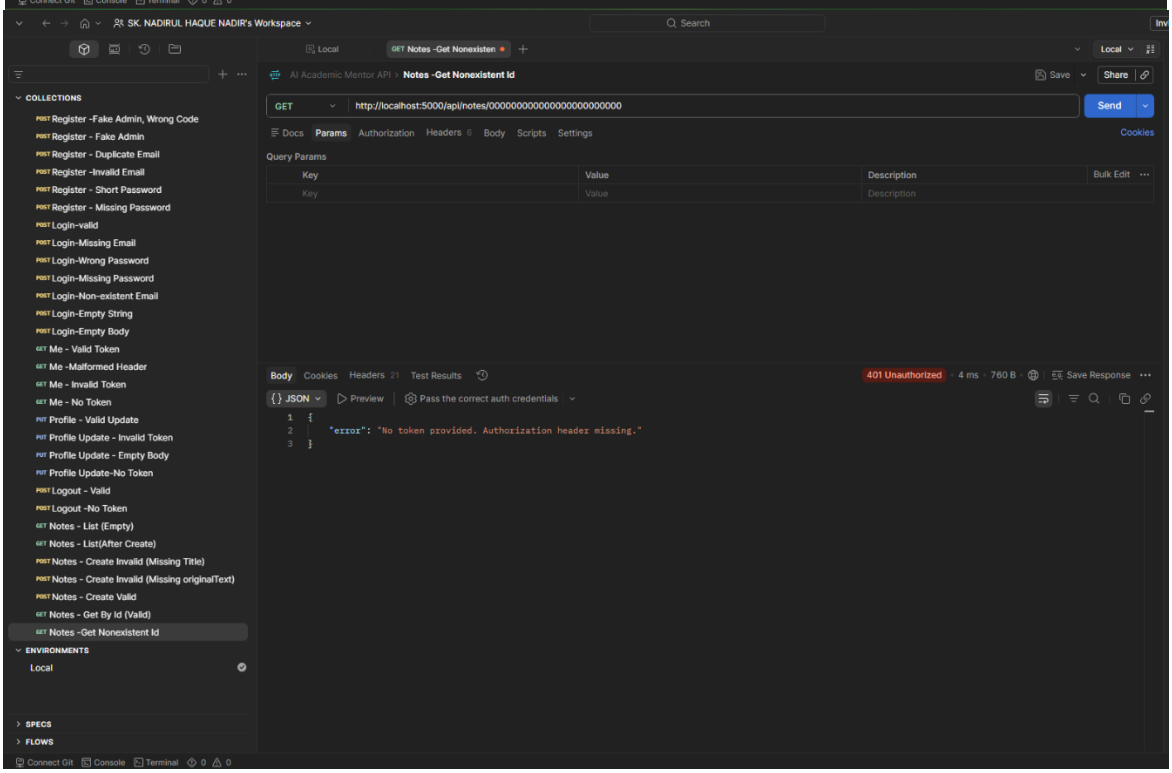


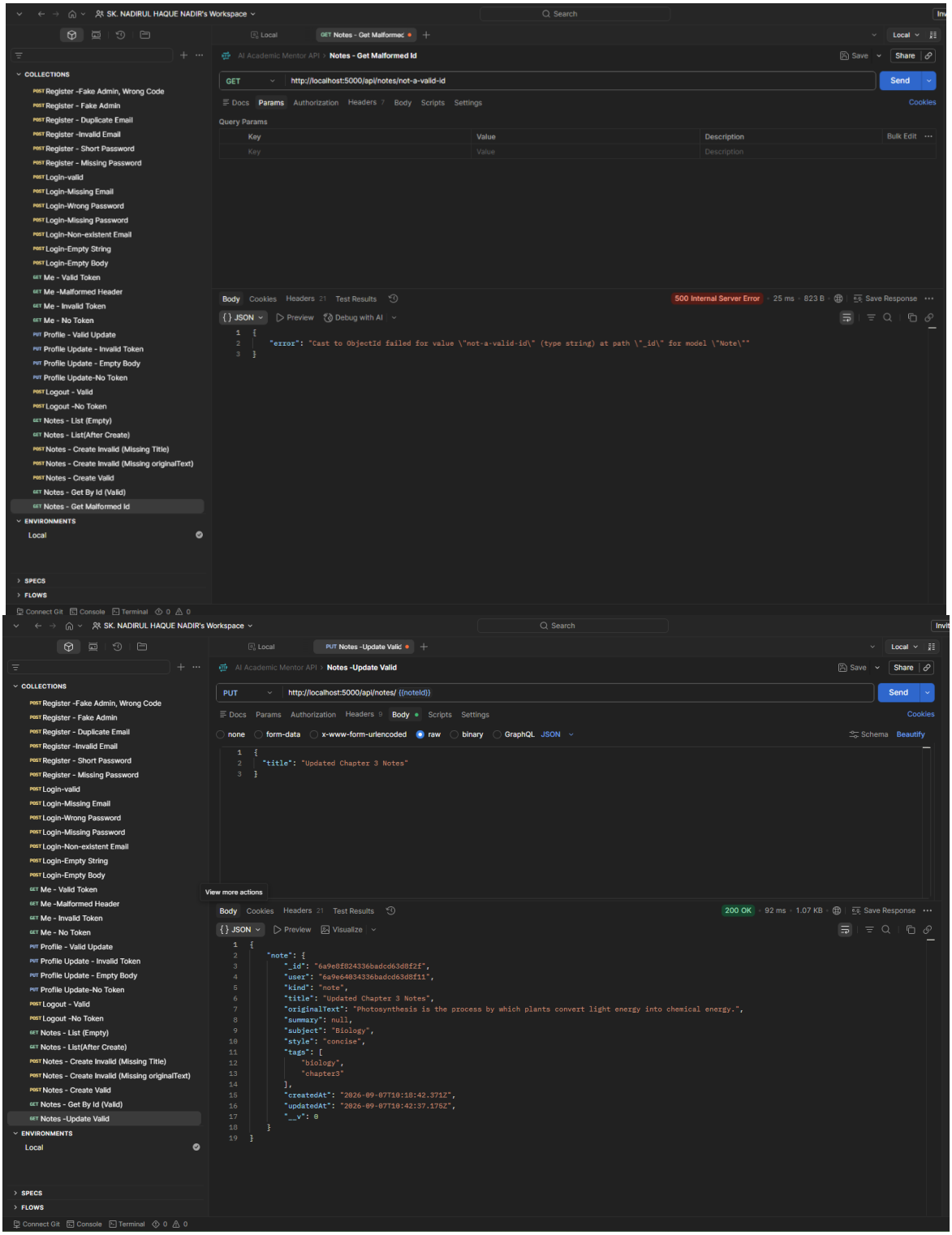


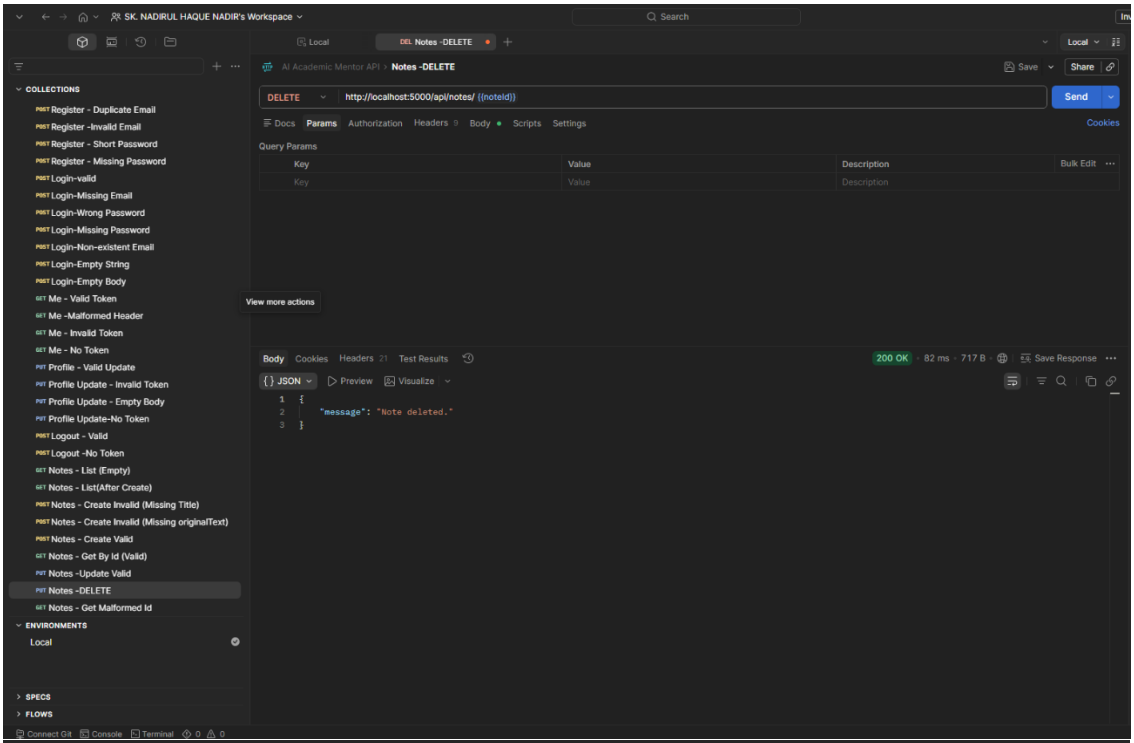

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
POST Notes - Create Invalid (
POST Notes - Create Invalid
Local
AI Academic Mentor API › Notes - Create Invalid (Missing originalText)
Save
Share
COLLECTIONS
AI Academic Mentor API
GET Health-Check
POST Register - Valid
POST Register - Right Admin
POST Register -Fake Admin, Wrong Code
POST Register - Fake Admin
POST Register - Duplicate Email
POST Register -Invalid Email
POST Register - Short Password
POST Register - Missing Password
POST Login-valid
POST Login-Missing Email
POST Login-Wrong Password
POST Login-Missing Password
POST Login-Non-existent Email
POST Login-Empty String
POST Login-Empty Body
GET Me - Valid Token
GET Me -Malformed Header
GET Me - Invalid Token
GET Me - No Token
PUT Profile - Valid Update
PUT Profile Update - Invalid Token
PUT Profile Update - Empty Body
PUT Profile Update-No Token
POST Logout - Valid
POST Logout -No Token
GET Notes - List (Empty)
POST Notes - Create Invalid (Missing Title)
POST Notes - Create Invalid (Missing originalText)
ENVIRONMENTS
Local
SPECS
FLOWS
POST
http://localhost:5000/api/notes
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"title": "Just a title"
}
Body Cookies Headers 21 Test Results
400 Bad Request 7 ms 747 B
Save Response
{ } JSON Preview Debug with AI
{
"error": "title and originalText are required."
}
Connect Git Console Terminal 0 0

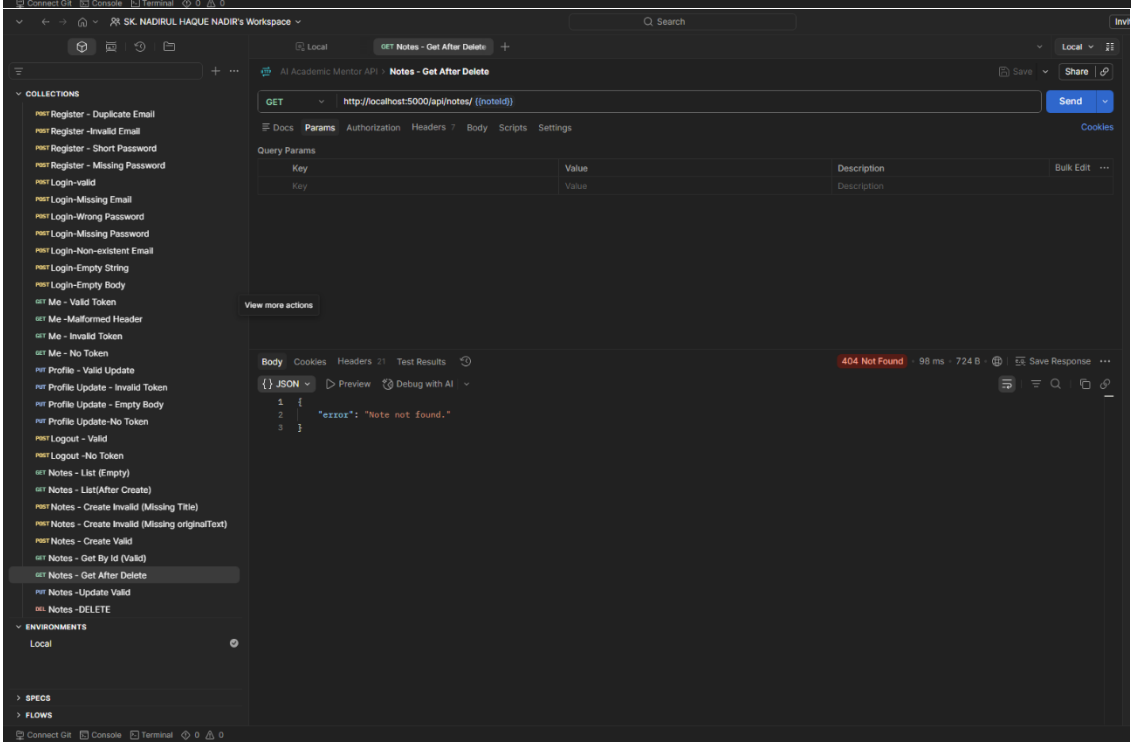

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
POST Notes - Create Invalid (
POST Notes - Create Invalid
GET Notes - List(After Create
Local
AI Academic Mentor API › Notes - List(After Create)
Save
Share
COLLECTIONS
AI Academic Mentor API
GET Health-Check
POST Register - Valid
POST Register - Right Admin
POST Register -Fake Admin, Wrong Code
POST Register - Fake Admin
POST Register - Duplicate Email
POST Register -Invalid Email
POST Register - Short Password
POST Register - Missing Password
POST Login-valid
POST Login-Missing Email
POST Login-Wrong Password
POST Login-Missing Password
POST Login-Non-existent Email
POST Login-Empty String
POST Login-Empty Body
GET Me - Valid Token
GET Me -Malformed Header
GET Me - Invalid Token
GET Me - No Token
PUT Profile - Valid Update
PUT Profile Update - Invalid Token
PUT Profile Update - Empty Body
PUT Profile Update-No Token
POST Logout - Valid
POST Logout -No Token
GET Notes - List (Empty)
GET Notes - List(After Create)
POST Notes - Create Invalid (Missing Title)
ENVIRONMENTS
Local
SPECS
FLOWS
GET
http://localhost:5000/api/notes
Send
Docs Params Authorization Headers 7 Body Scripts Settings
Cookies
Query Params
Key Value Description Bulk Edit
Key Value Description
Body Cookies Headers 21 Test Results
200 OK 71 ms 1.45 KB
Save Response
{ } JSON Preview Visualize
{
"notes": [
{
"_id": "6a9e8f824336badcd63d8f2f",
"user": "6a9e64034336badcd63d8f11",
"kind": "note",
"title": "Chapter 3 Notes",
"originalText": "Photosynthesis is the process by which plants convert light energy into chemical energy.",
"summary": null,
"subject": "Biology",
"style": "concise",
"tags": [
"biology",
"chapter3"
],
"createdAt": "2026-09-07T10:18:42.371Z",
"updatedAt": "2026-09-07T10:18:42.371Z",
"__v": 0
},
{
"_id": "6a9e8f5a4336badcd63d8f2d",
"user": "6a9e64034336badcd63d8f11",
"kind": "note",
"title": "Chapter 3 Notes",
"originalText": "Photosynthesis is the process by which plants convert light energy into chemical energy.",
"summary": null,
Connect Git Console Terminal 0 0

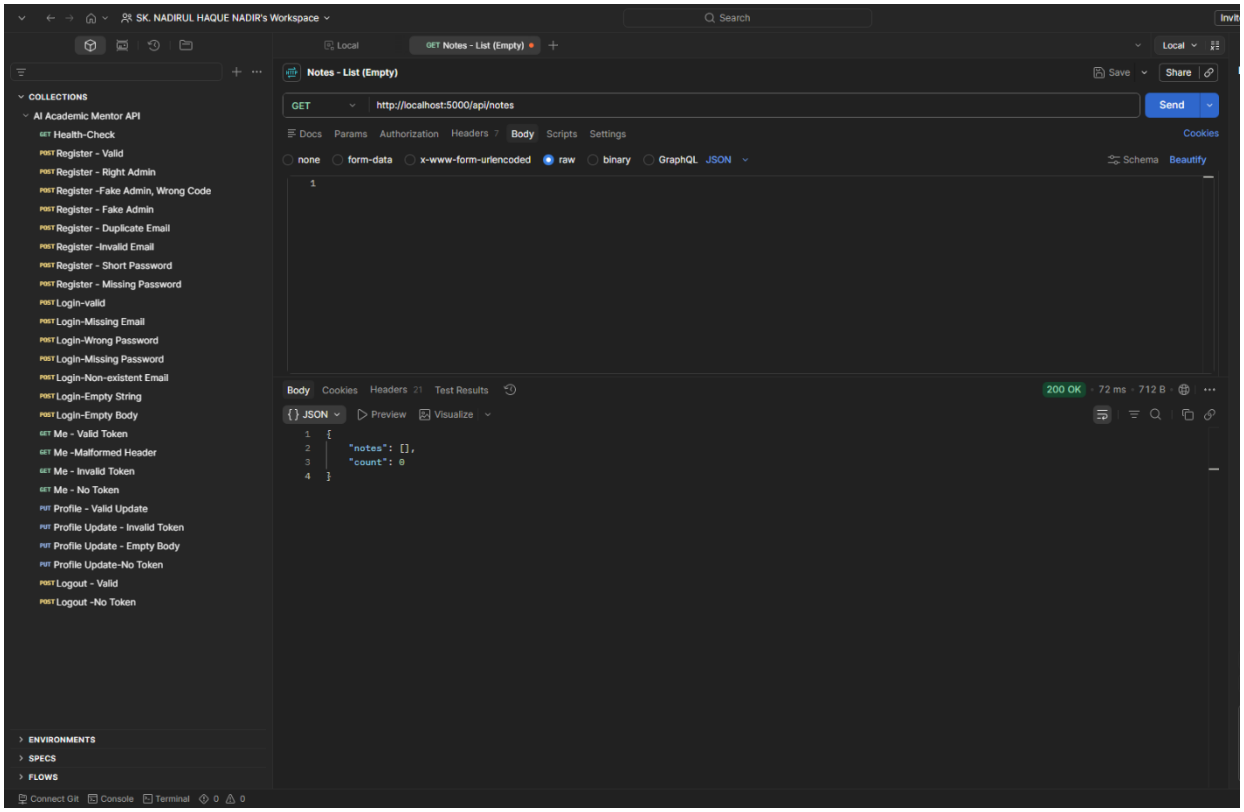

SK. NADIRUL HAQUE NADIR's Workspace
Search
Notes - Get By Id (Valid)
AI Academic Mentor API
Notes - Get By Id (Valid)
Save
Share
GET
http://localhost:5000/api/notes/ {{noteId}}
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 21
Test Results
200 OK
97 ms
1.06 KB
Save Response
JSON
Preview
Visualize
{
"note": {
"_id": "6a9e8f824336badcd63d8f2f",
"user": "6a9e64034336badcd63d8f11",
"kind": "note",
"title": "Chapter 3 Notes",
"originalText": "Photosynthesis is the process by which plants convert light energy into chemical energy.",
"summary": null,
"subject": "Biology",
"style": "concise",
"tags": [
"biology",
"chapter3"
],
"createdAt": "2026-09-07T10:18:42.371Z",
"updatedAt": "2026-09-07T10:18:42.371Z",
"__v": 0
}
}
COLLECTIONS
Register - Right Admin
Register -Fake Admin, Wrong Code
Register - Fake Admin
Register - Duplicate Email
Register -Invalid Email
Register - Short Password
Register - Missing Password
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
ENVIRONMENTS
Local
SPECS
FLOWS
Connect Git
Console
Terminal


SK. NADIRUL HAQUE NADIR's Workspace
Search
Notes -Get Nonexisten
AI Academic Mentor API
Notes -Get Nonexistent Id
Save
Share
GET
http://localhost:5000/api/notes/000000000000000000000000
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 21
Test Results
401 Unauthorized
4 ms
760 B
Save Response
JSON
Preview
Pass the correct auth credentials
{
"error": "No token provided. Authorization header missing."
}
COLLECTIONS
Register -Fake Admin, Wrong Code
Register - Fake Admin
Register - Duplicate Email
Register -Invalid Email
Register - Short Password
Register - Missing Password
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes -Get Nonexistent Id
ENVIRONMENTS
Local
SPECS
FLOWS
Connect Git
Console
Terminal

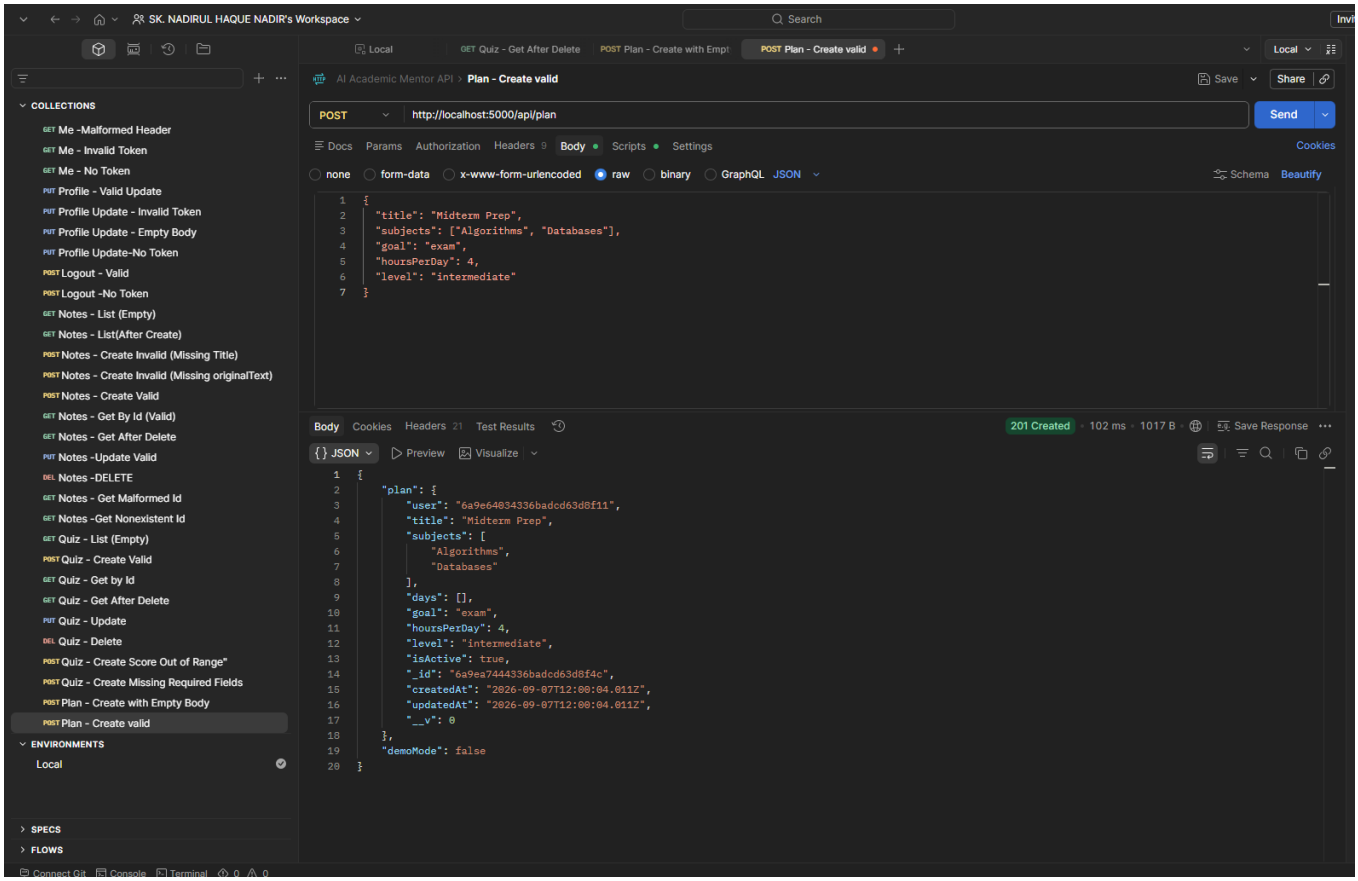

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Notes - Get Malformed Id
AI Academic Mentor API / Notes - Get Malformed Id
Save
Share
COLLECTIONS
Register -Fake Admin, Wrong Code
Register - Fake Admin
Register - Duplicate Email
Register -Invalid Email
Register - Short Password
Register - Missing Password
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes - Get Malformed Id
ENVIRONMENTS
Local
SPECS
FLOWS
GET
http://localhost:5000/api/notes/not-a-valid-id
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 21
Test Results
500 Internal Server Error
25 ms
823 B
Save Response
JSON
Preview
Debug with AI
{
"error": "Cast to ObjectId failed for value \"not-a-valid-id\" (type string) at path \"_id\" for model \"Note\""
}
Connect Git
Console
Terminal
SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
PUT Notes -Update Valid
AI Academic Mentor API / Notes -Update Valid
Save
Share
PUT
http://localhost:5000/api/notes/ {{noteId}}
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"title": "Updated Chapter 3 Notes"
}
View more actions
Body
Cookies
Headers 21
Test Results
200 OK
92 ms
1.07 KB
Save Response
JSON
Preview
Visualize
{
"note": {
"_id": "6a9e8f824336badcd63d8f2f",
"user": "6a9e6403436badcd63d8f11",
"kind": "note",
"title": "Updated Chapter 3 Notes",
"originalText": "Photosynthesis is the process by which plants convert light energy into chemical energy.",
"summary": null,
"subject": "Biology",
"style": "concise",
"tags": [
"biology",
"chapter3"
],
"createdAt": "2026-09-07T10:18:42.371Z",
"updatedAt": "2026-09-07T10:42:37.175Z",
"__v": 0
}
}
Connect Git
Console
Terminal

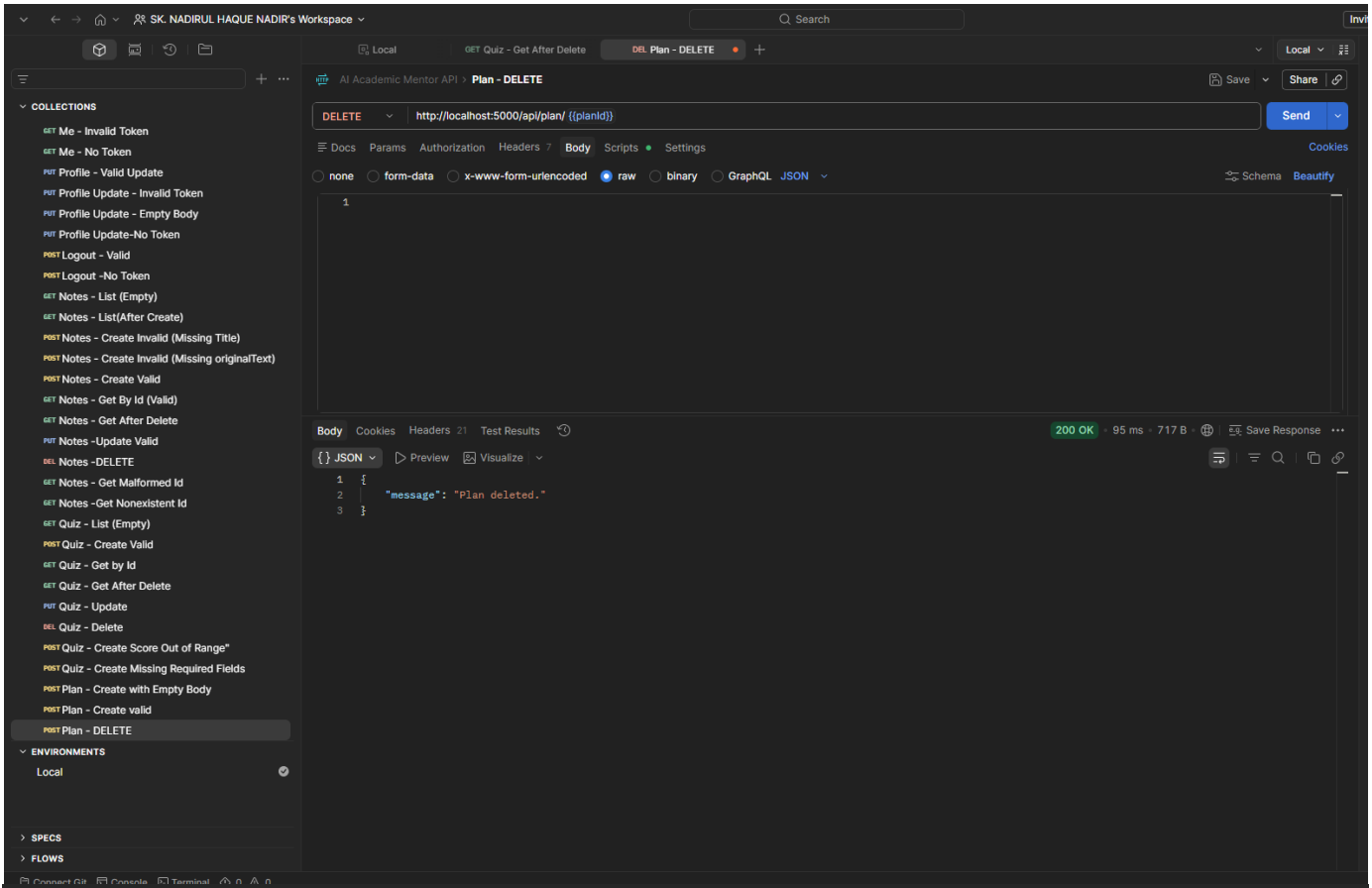

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
Notes -DELETE
AI Academic Mentor API / Notes -DELETE
Save
Share
COLLECTIONS
Register - Duplicate Email
Register -Invalid Email
Register - Short Password
Register - Missing Password
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes -Update Valid
Notes -DELETE
Notes - Get Malformed Id
ENVIRONMENTS
Local
SPECS
FLOWS
DELETE
http://localhost:5000/api/notes/ {{noteId}}
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
View more actions
Body
Cookies
Headers 21
Test Results
200 OK
82 ms
717 B
Save Response
{ } JSON
Preview
Visualize
{
"message": "Note deleted."
}

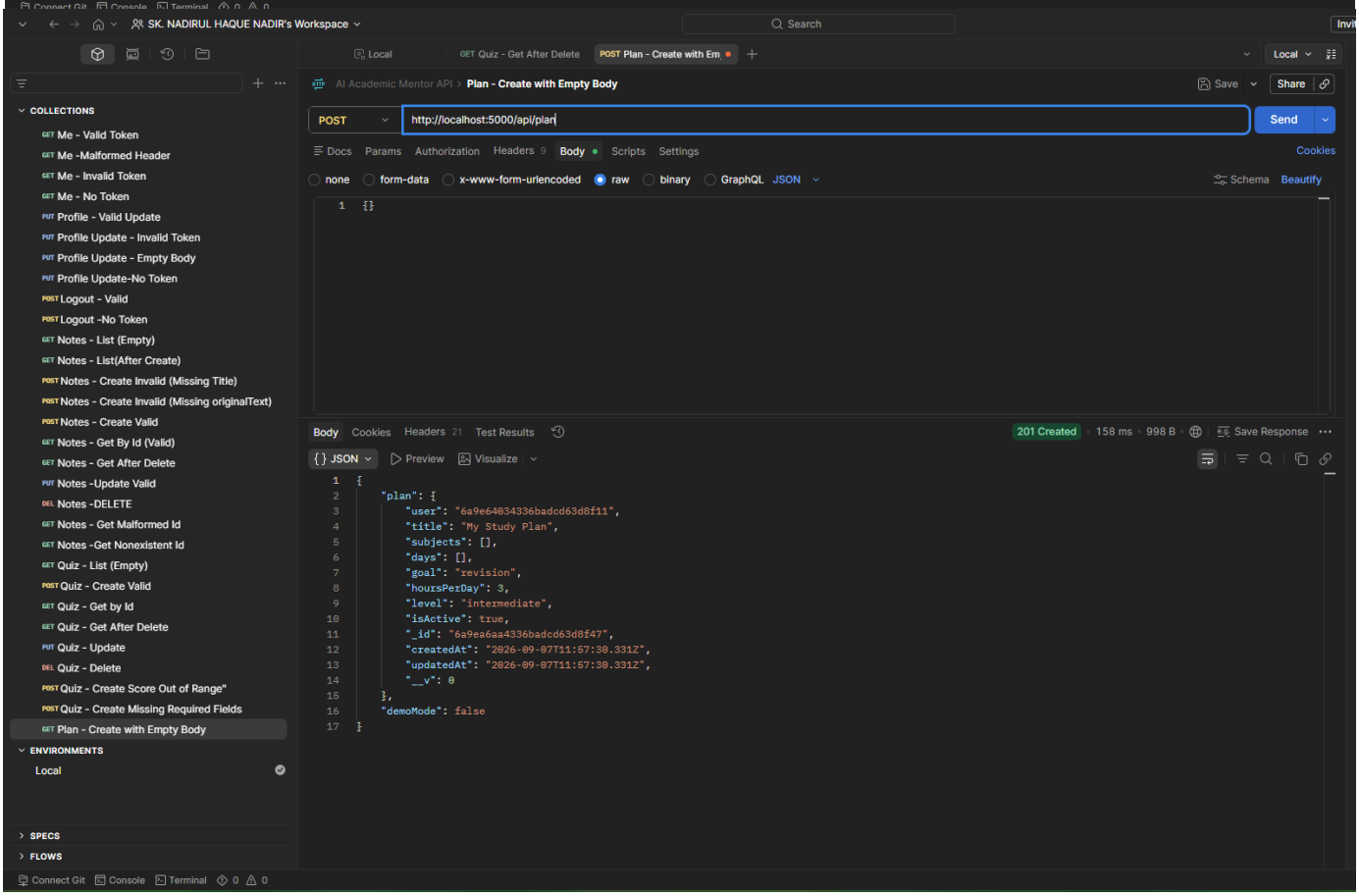

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
Notes - Get After Delete
AI Academic Mentor API / Notes - Get After Delete
Save
Share
COLLECTIONS
Register - Duplicate Email
Register -Invalid Email
Register - Short Password
Register - Missing Password
Login-valid
Login-Missing Email
Login-Wrong Password
Login-Missing Password
Login-Non-existent Email
Login-Empty String
Login-Empty Body
Me - Valid Token
Me -Malformed Header
Me - Invalid Token
Me - No Token
Profile - Valid Update
Profile Update - Invalid Token
Profile Update - Empty Body
Profile Update-No Token
Logout - Valid
Logout -No Token
Notes - List (Empty)
Notes - List(After Create)
Notes - Create Invalid (Missing Title)
Notes - Create Invalid (Missing originalText)
Notes - Create Valid
Notes - Get By Id (Valid)
Notes - Get After Delete
Notes -Update Valid
Notes -DELETE
ENVIRONMENTS
Local
SPECS
FLOWS
GET
http://localhost:5000/api/notes/ {{noteId}}
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
View more actions
Body
Cookies
Headers 21
Test Results
404 Not Found
98 ms
724 B
Save Response
{ } JSON
Preview
Debug with AI
{
"error": "Note not found."
}



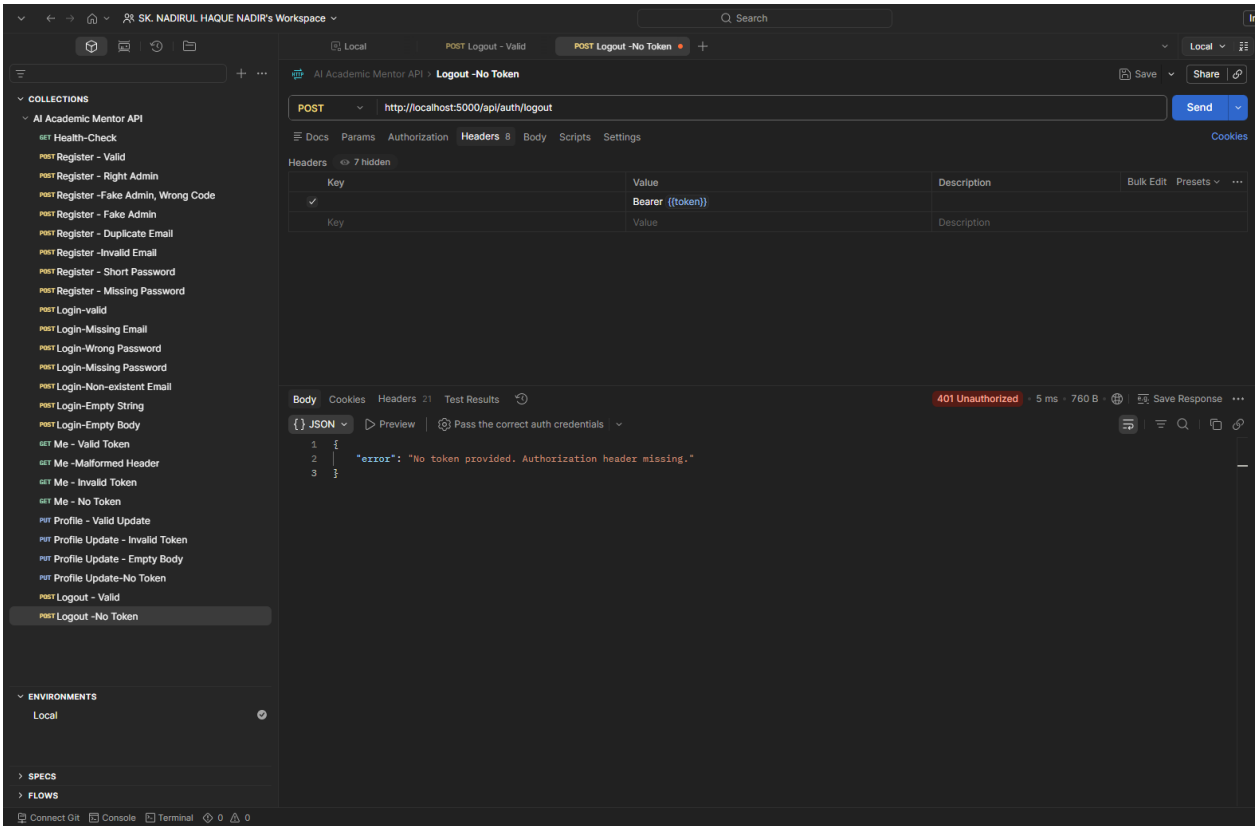
C. Study_Planner TESTING:

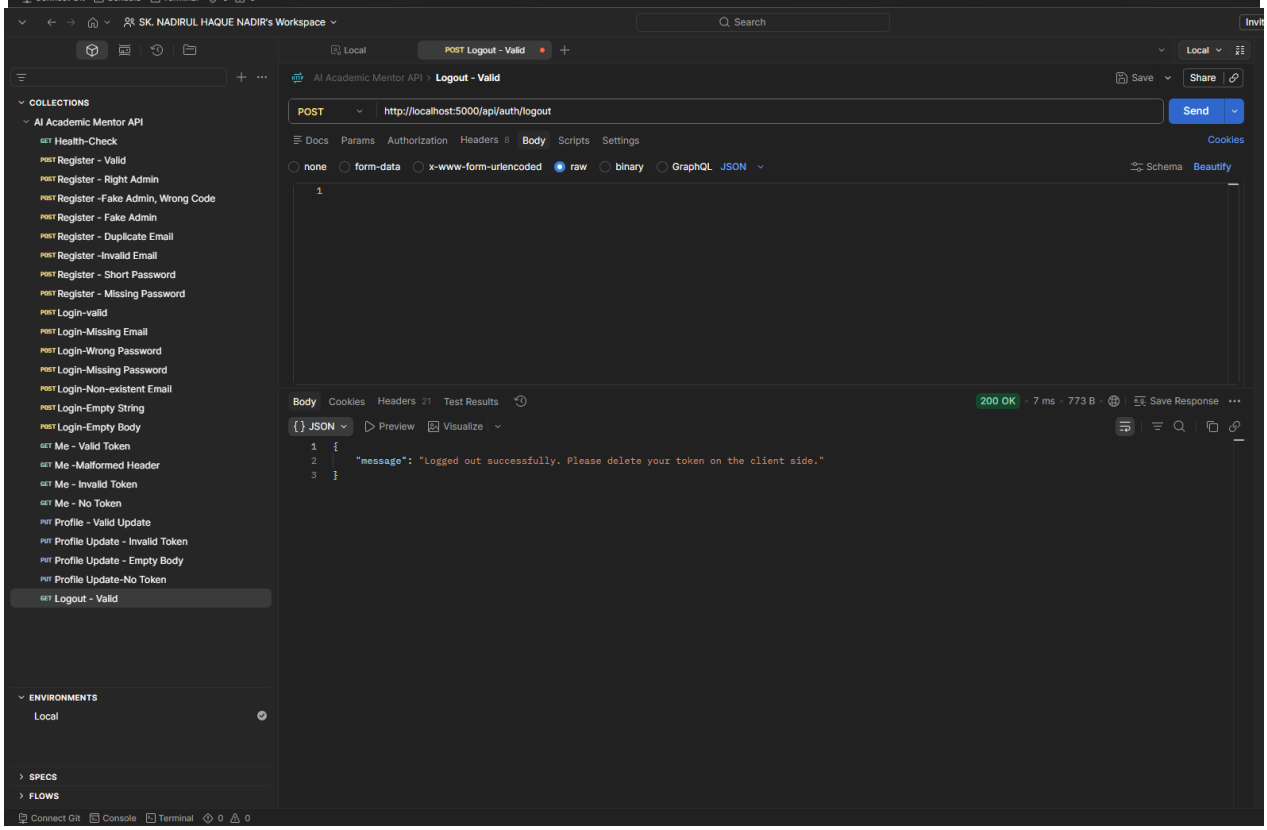



SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Quiz - Get After Delete
DEL Plan - DELETE
Local
COLLECTIONS
GET Me - Invalid Token
GET Me - No Token
PUT Profile - Valid Update
PUT Profile Update - Invalid Token
PUT Profile Update - Empty Body
PUT Profile Update-No Token
POST Logout - Valid
POST Logout -No Token
GET Notes - List (Empty)
GET Notes - List(After Create)
POST Notes - Create Invalid (Missing Title)
POST Notes - Create Invalid (Missing originalText)
POST Notes - Create Valid
GET Notes - Get By Id (Valid)
GET Notes - Get After Delete
PUT Notes -Update Valid
DEL Notes -DELETE
GET Notes - Get Malformed Id
GET Notes -Get Nonexistent Id
GET Quiz - List (Empty)
POST Quiz - Create Valid
GET Quiz - Get by Id
GET Quiz - Get After Delete
PUT Quiz - Update
DEL Quiz - Delete
POST Quiz - Create Score Out of Range"
POST Quiz - Create Missing Required Fields
POST Plan - Create with Empty Body
POST Plan - Create valid
POST Plan - DELETE
ENVIRONMENTS
Local
SPECS
FLOWS
AI Academic Mentor API › Plan - DELETE
Save
Share
DELETE
http://localhost:5000/api/plan/ {{planId}}
Send
Docs Params Authorization Headers 7 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
Body Cookies Headers 21 Test Results
200 OK 95 ms 717 B
Save Response
{ } JSON Preview Visualize
{
"message": "Plan deleted."
}

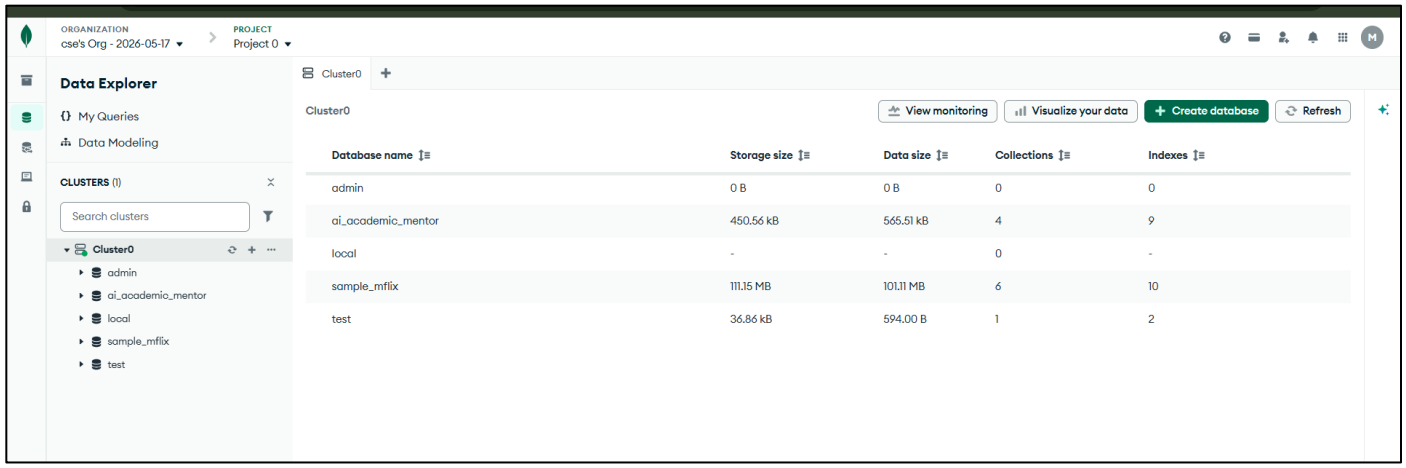

SK. NADIRUL HAQUE NADIR's Workspace
Search
Local
GET Quiz - Get After Delete
POST Plan - Create with Em
Local
COLLECTIONS
GET Me - Valid Token
GET Me -Malformed Header
GET Me - Invalid Token
GET Me - No Token
PUT Profile - Valid Update
PUT Profile Update - Invalid Token
PUT Profile Update - Empty Body
PUT Profile Update-No Token
POST Logout - Valid
POST Logout -No Token
GET Notes - List (Empty)
GET Notes - List(After Create)
POST Notes - Create Invalid (Missing Title)
POST Notes - Create Invalid (Missing originalText)
POST Notes - Create Valid
GET Notes - Get By Id (Valid)
GET Notes - Get After Delete
PUT Notes -Update Valid
DEL Notes -DELETE
GET Notes - Get Malformed Id
GET Notes -Get Nonexistent Id
GET Quiz - List (Empty)
POST Quiz - Create Valid
GET Quiz - Get by Id
GET Quiz - Get After Delete
PUT Quiz - Update
DEL Quiz - Delete
POST Quiz - Create Score Out of Range"
POST Quiz - Create Missing Required Fields
GET Plan - Create with Empty Body
ENVIRONMENTS
Local
SPECS
FLOWS
AI Academic Mentor API › Plan - Create with Empty Body
Save
Share
POST
http://localhost:5000/api/plan
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
Body Cookies Headers 21 Test Results
201 Created 158 ms 998 B
Save Response
{ } JSON Preview Visualize
{
"plan": {
"user": "6a9e64034336badcd63d8f11",
"title": "My Study Plan",
"subjects": [],
"days": [],
"goal": "revision",
"hoursPerDay": 3,
"level": "intermediate",
"isActive": true,
"_id": "6a9ea6aa4336badcd63d8f47",
"createdAt": "2026-09-07T11:57:30.331Z",
"updatedAt": "2026-09-07T11:57:30.331Z",
"__v": 0
},
"demoMode": false
}
Connect Git Console Terminal 0 0

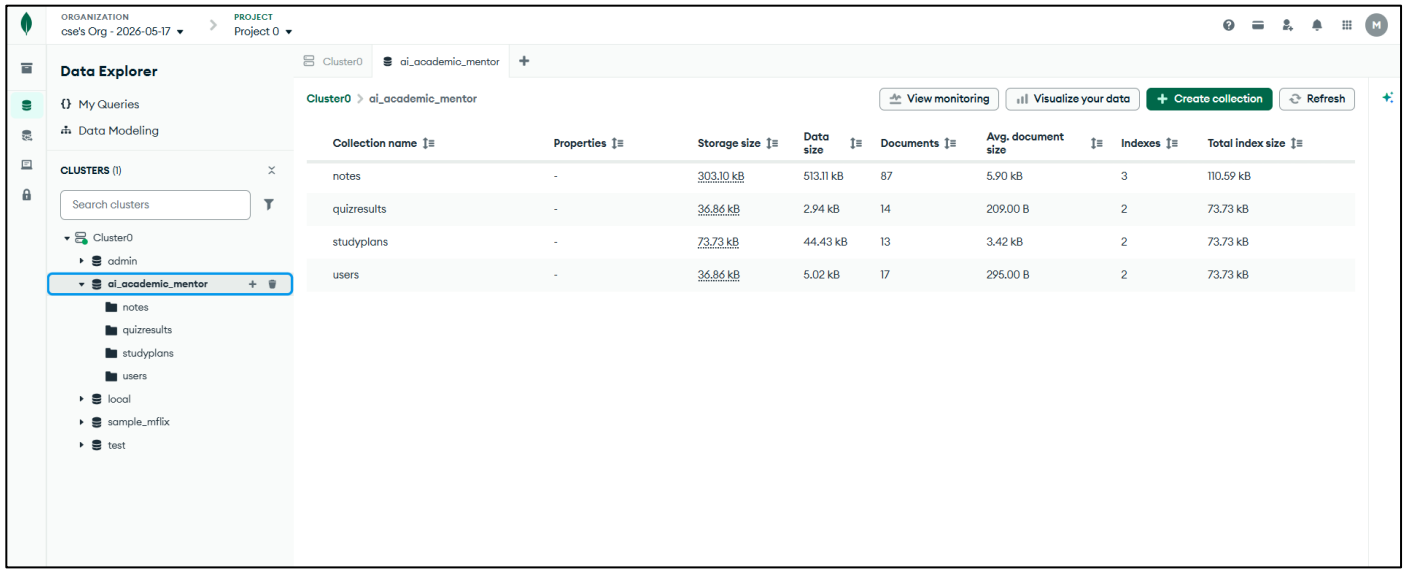
D. Logout TESTING:

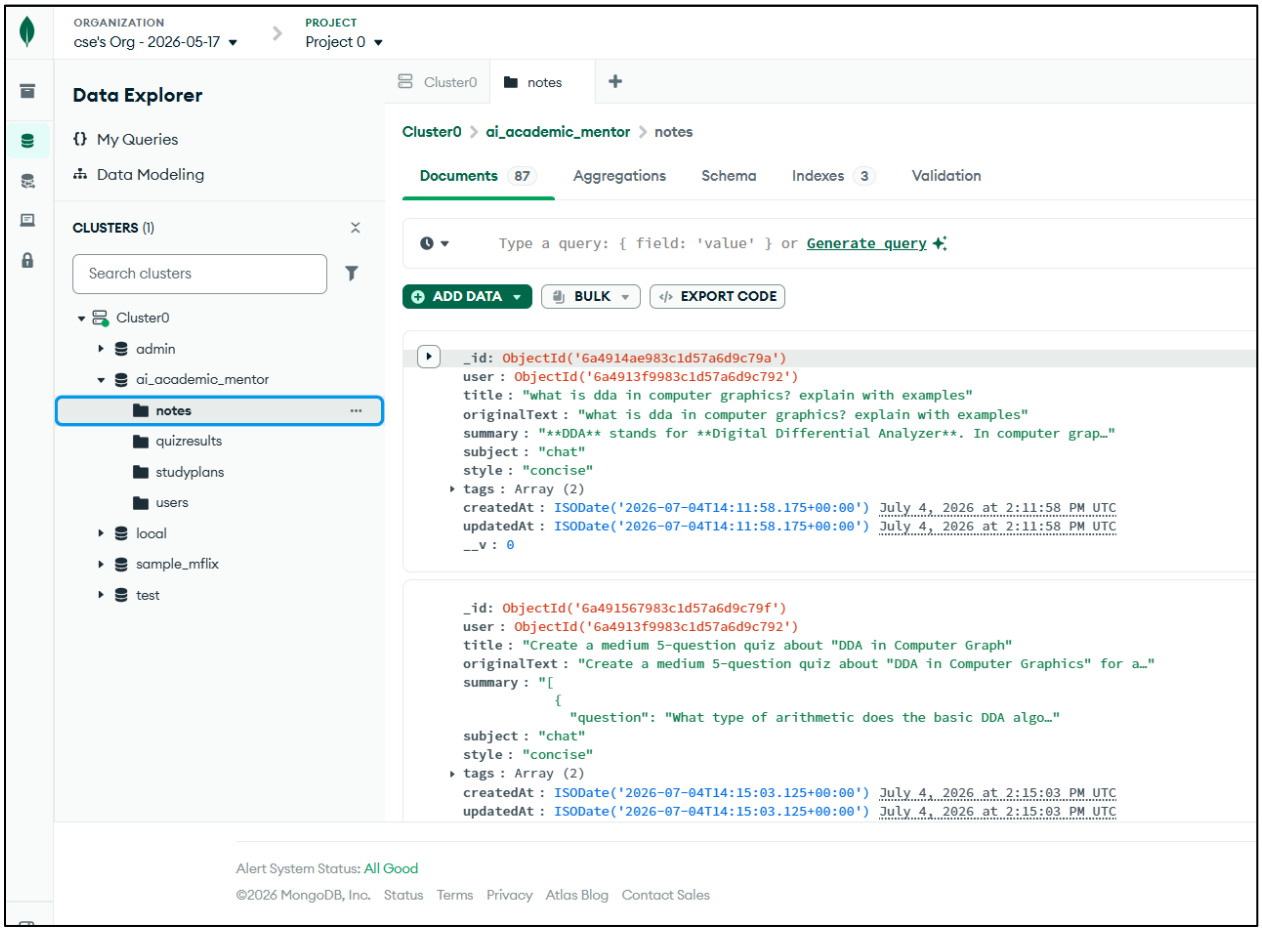


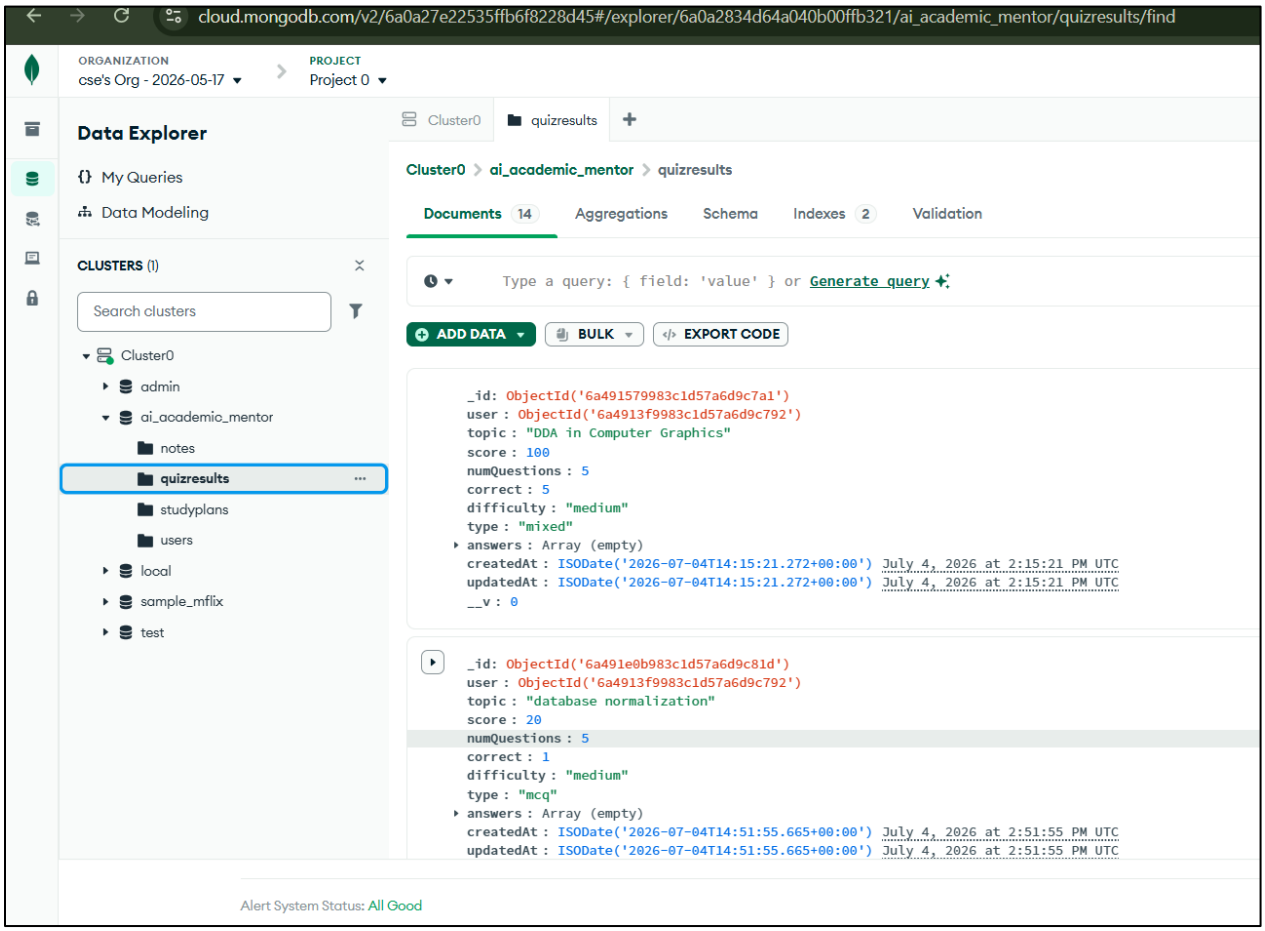


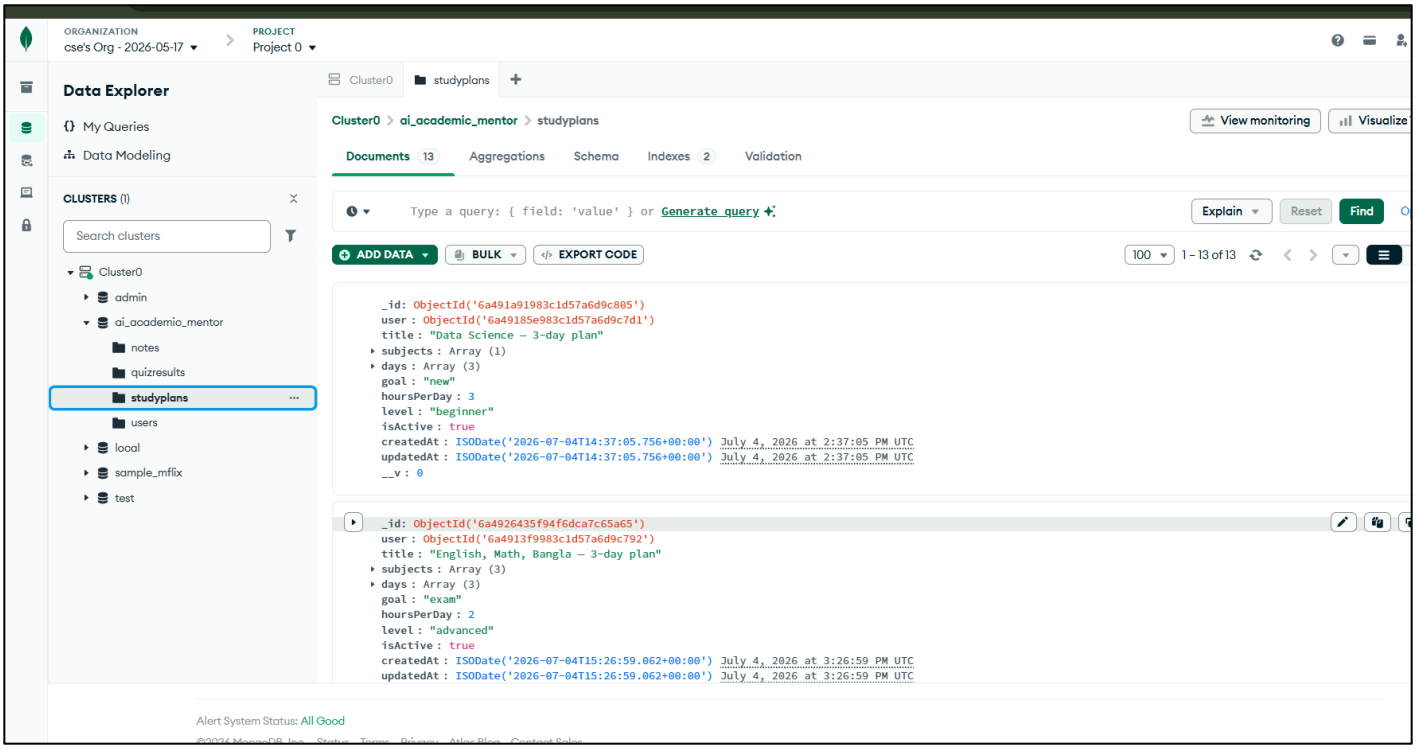
7. DATABASE TESTING

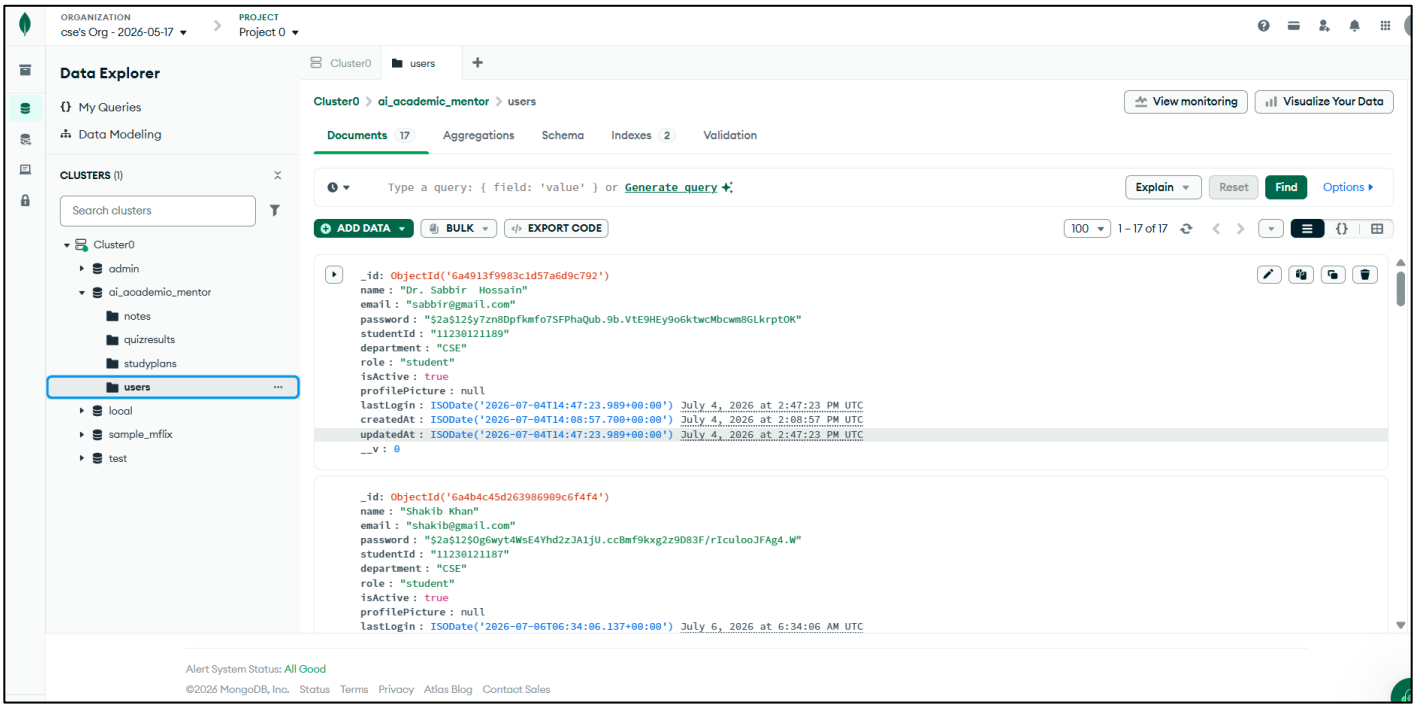





ORGANIZATION
cse's Org - 2026-05-17
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
ai_academic_mentor
notes
quizresults
studyplans
users
local
sample_mflix
test
Cluster0
notes
Cluster0 > ai_academic_mentor > notes
Documents 87
Aggregations
Schema
Indexes 3
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
BULK
EXPORT CODE
_id: ObjectId('6a4914ae983c1d57a6d9c79a')
user : ObjectId('6a4913f9983c1d57a6d9c792')
title : "what is dda in computer graphics? explain with examples"
originalText : "what is dda in computer graphics? explain with examples"
summary : "**DDA** stands for **Digital Differential Analyzer**. In computer grap…"
subject : "chat"
style : "concise"
tags : Array (2)
createdAt : ISODate('2026-07-04T14:11:58.175+00:00') July 4, 2026 at 2:11:58 PM UTC
updatedAt : ISODate('2026-07-04T14:11:58.175+00:00') July 4, 2026 at 2:11:58 PM UTC
__v : 0
_id: ObjectId('6a491567983c1d57a6d9c79f')
user : ObjectId('6a4913f9983c1d57a6d9c792')
title : "Create a medium 5-question quiz about "DDA in Computer Graph"
originalText : "Create a medium 5-question quiz about "DDA in Computer Graphics" for a…"
summary : "[
{
"question": "What type of arithmetic does the basic DDA algo…"
subject : "chat"
style : "concise"
tags : Array (2)
createdAt : ISODate('2026-07-04T14:15:03.125+00:00') July 4, 2026 at 2:15:03 PM UTC
updatedAt : ISODate('2026-07-04T14:15:03.125+00:00') July 4, 2026 at 2:15:03 PM UTC
Alert System Status: All Good
©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales

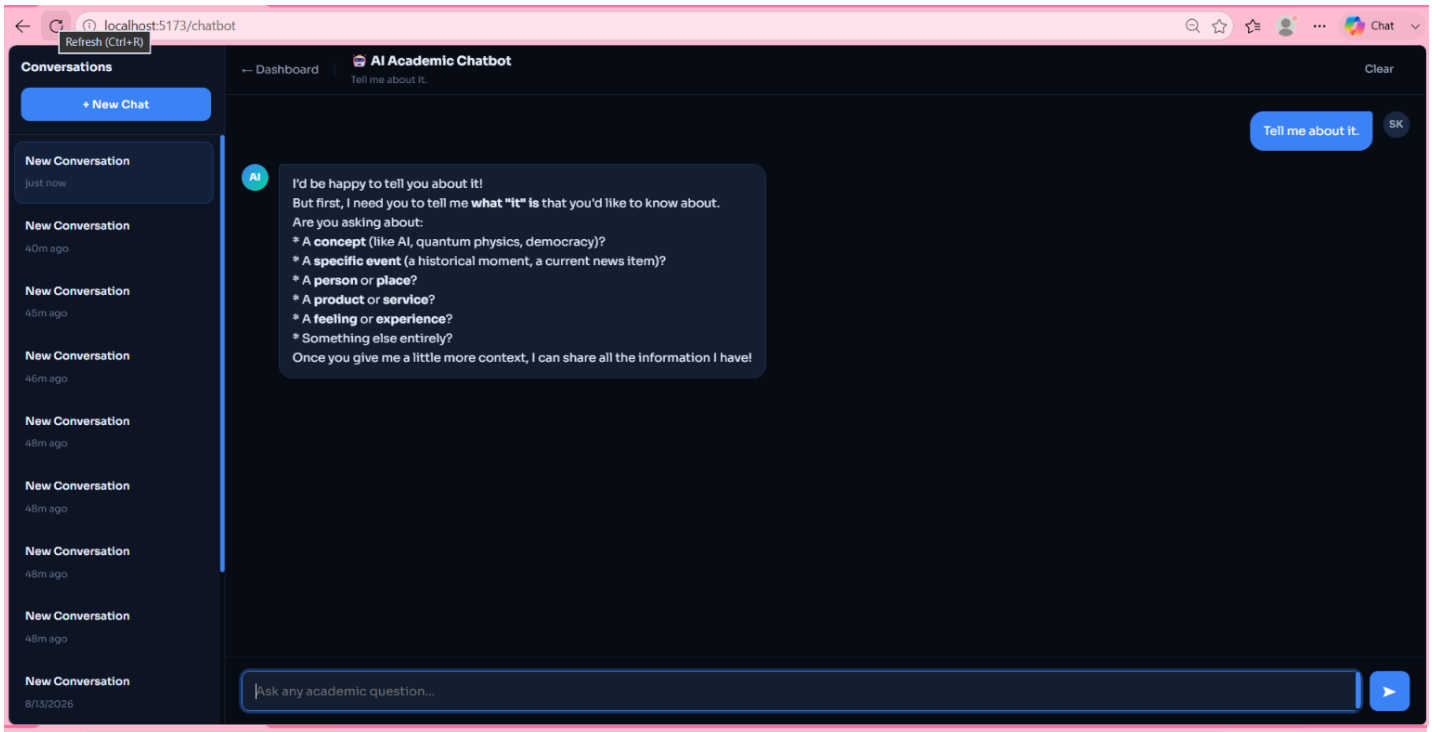

cloud.mongodb.com/v2/6a0a27e22535ffb6f8228d45#/explorer/6a0a2834d64a040b00ffb321/ai_academic_mentor/quizresults/find
ORGANIZATION
cse's Org - 2026-05-17
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
ai_academic_mentor
notes
quizresults
studyplans
users
local
sample_mflix
test
Cluster0
quizresults
Cluster0 > ai_academic_mentor > quizresults
Documents 14
Aggregations
Schema
Indexes 2
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
BULK
EXPORT CODE
_id: ObjectId('6a491579983c1d57a6d9c7a1')
user : ObjectId('6a4913f9983c1d57a6d9c792')
topic : "DDA in Computer Graphics"
score : 100
numQuestions : 5
correct : 5
difficulty : "medium"
type : "mixed"
answers : Array (empty)
createdAt : ISODate('2026-07-04T14:15:21.272+00:00') July 4, 2026 at 2:15:21 PM UTC
updatedAt : ISODate('2026-07-04T14:15:21.272+00:00') July 4, 2026 at 2:15:21 PM UTC
__v : 0
_id: ObjectId('6a491e0b983c1d57a6d9c81d')
user : ObjectId('6a4913f9983c1d57a6d9c792')
topic : "database normalization"
score : 20
numQuestions : 5
correct : 1
difficulty : "medium"
type : "mcq"
answers : Array (empty)
createdAt : ISODate('2026-07-04T14:51:55.665+00:00') July 4, 2026 at 2:51:55 PM UTC
updatedAt : ISODate('2026-07-04T14:51:55.665+00:00') July 4, 2026 at 2:51:55 PM UTC
Alert System Status: All Good

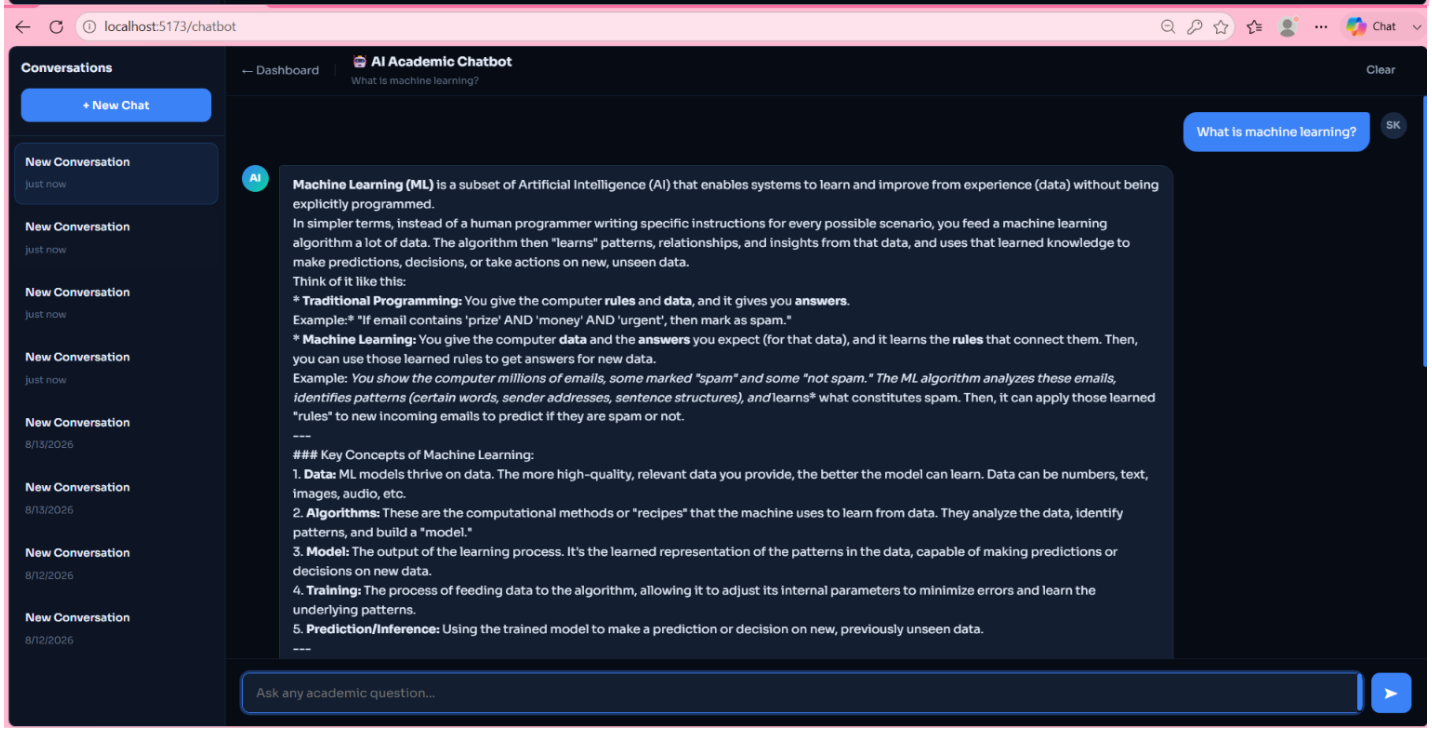

ORGANIZATION
cse's Org - 2026-05-17
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
ai_academic_mentor
notes
quizresults
studyplans
users
local
sample_mflix
test
Cluster0 > ai_academic_mentor > studyplans
View monitoring
Visualize
Documents 13
Aggregations
Schema
Indexes 2
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
ADD DATA
BULK
EXPORT CODE
100
1 - 13 of 13
_id: ObjectId('6a491a91983c1d57a6d9c805')
user : ObjectId('6a49185e983c1d57a6d9c7d1')
title : "Data Science — 3-day plan"
subjects : Array (1)
days : Array (3)
goal : "new"
hoursPerDay : 3
level : "beginner"
isActive : true
createdAt : ISODate('2026-07-04T14:37:05.756+00:00') July 4, 2026 at 2:37:05 PM UTC
updatedAt : ISODate('2026-07-04T14:37:05.756+00:00') July 4, 2026 at 2:37:05 PM UTC
__v : 0
_id: ObjectId('6a4926435f94f6dca7c65a65')
user : ObjectId('6a4913f9983c1d57a6d9c792')
title : "English, Math, Bangla — 3-day plan"
subjects : Array (3)
days : Array (3)
goal : "exam"
hoursPerDay : 2
level : "advanced"
isActive : true
createdAt : ISODate('2026-07-04T15:26:59.062+00:00') July 4, 2026 at 3:26:59 PM UTC
updatedAt : ISODate('2026-07-04T15:26:59.062+00:00') July 4, 2026 at 3:26:59 PM UTC
Alert System Status: All Good

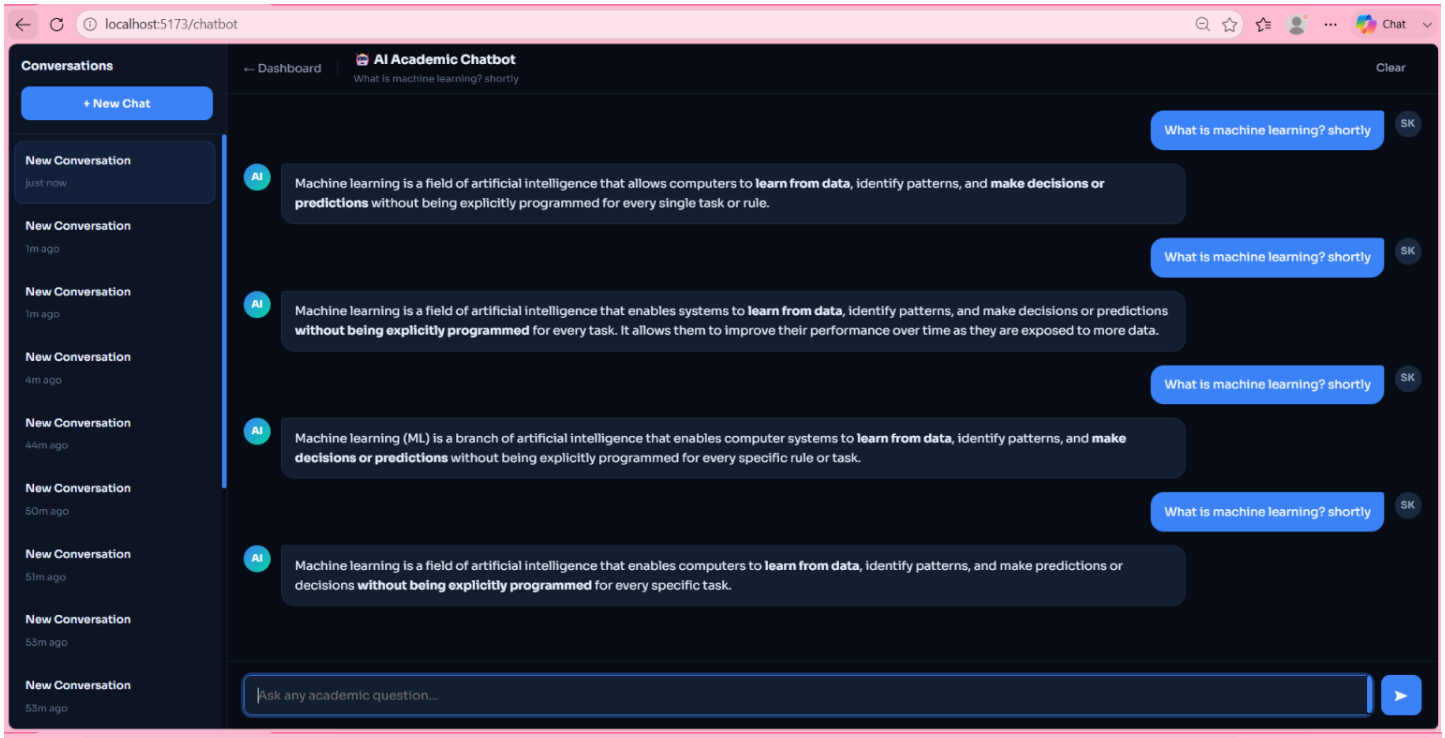

ORGANIZATION
cse's Org - 2026-05-17
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
ai_academic_mentor
notes
quizresults
studyplans
users
local
sample_mflix
test
Cluster0 > ai_academic_mentor > users
View monitoring
Visualize Your Data
Documents 17
Aggregations
Schema
Indexes 2
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
100
1 - 17 of 17
_id: ObjectId('6a4913f9983c1d57a6d9c792')
name : "Dr. Sabbir Hossain"
email : "sabbir@gmail.com"
password : "$2a$12$y7zn8DpfkmfO7SFPhaQub.9b.VtE9HEy9o6ktwcMbcwm8GLkrptOK"
studentId : "11230121189"
department : "CSE"
role : "student"
isActive : true
profilePicture : null
lastLogin : ISODate('2026-07-04T14:47:23.989+00:00') July 4, 2026 at 2:47:23 PM UTC
createdAt : ISODate('2026-07-04T14:08:57.700+00:00') July 4, 2026 at 2:08:57 PM UTC
updatedAt : ISODate('2026-07-04T14:47:23.989+00:00') July 4, 2026 at 2:47:23 PM UTC
__v : 0
_id: ObjectId('6a4b4c45d263986909c6f4f4')
name : "Shakib Khan"
email : "shakib@gmail.com"
password : "$2a$12$Og6wyt4WsE4Yhd2zJA1jU.ccBmf9kxg2z9D83F/rIculooJFAg4.W"
studentId : "11230121187"
department : "CSE"
role : "student"
isActive : true
profilePicture : null
lastLogin : ISODate('2026-07-06T06:34:06.137+00:00') July 6, 2026 at 6:34:06 AM UTC
Alert System Status: All Good
©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales

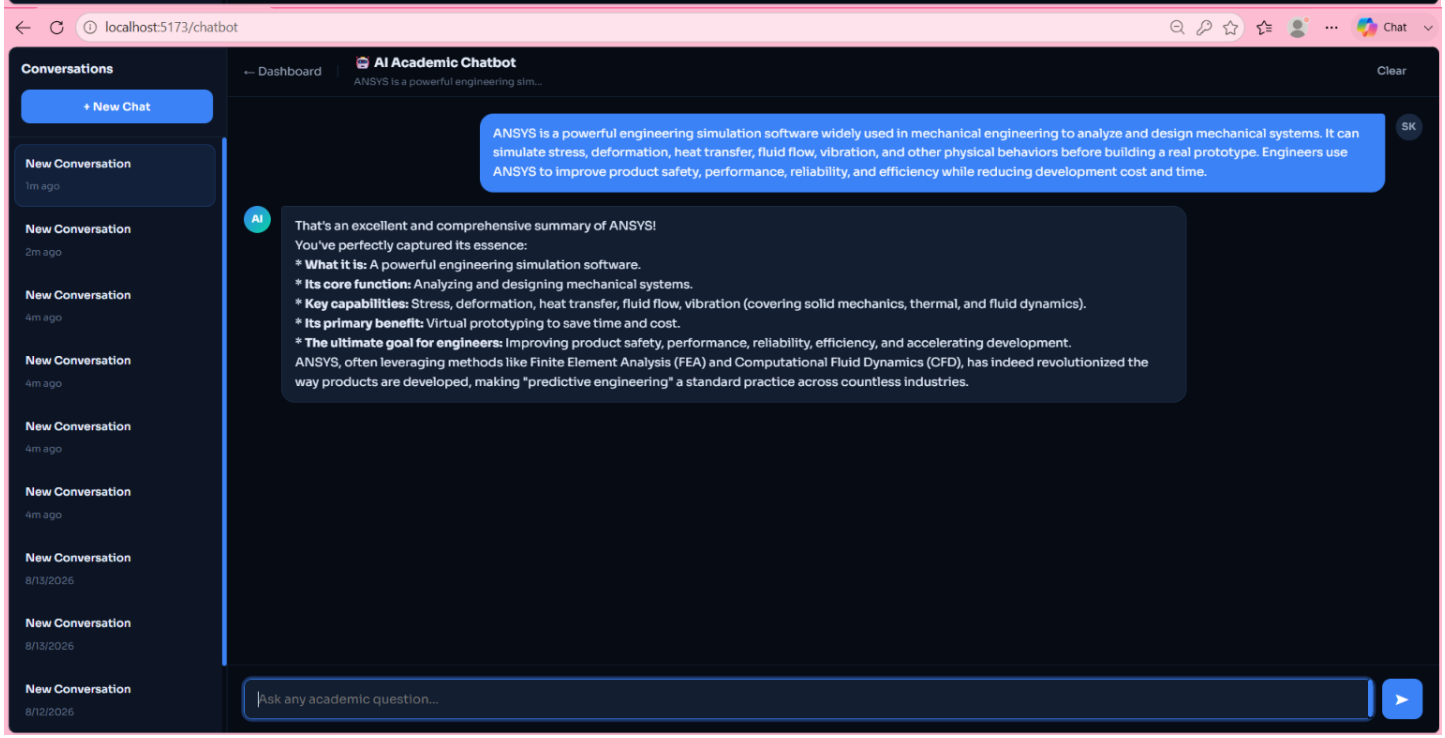
8. AI FEATURE TESTING

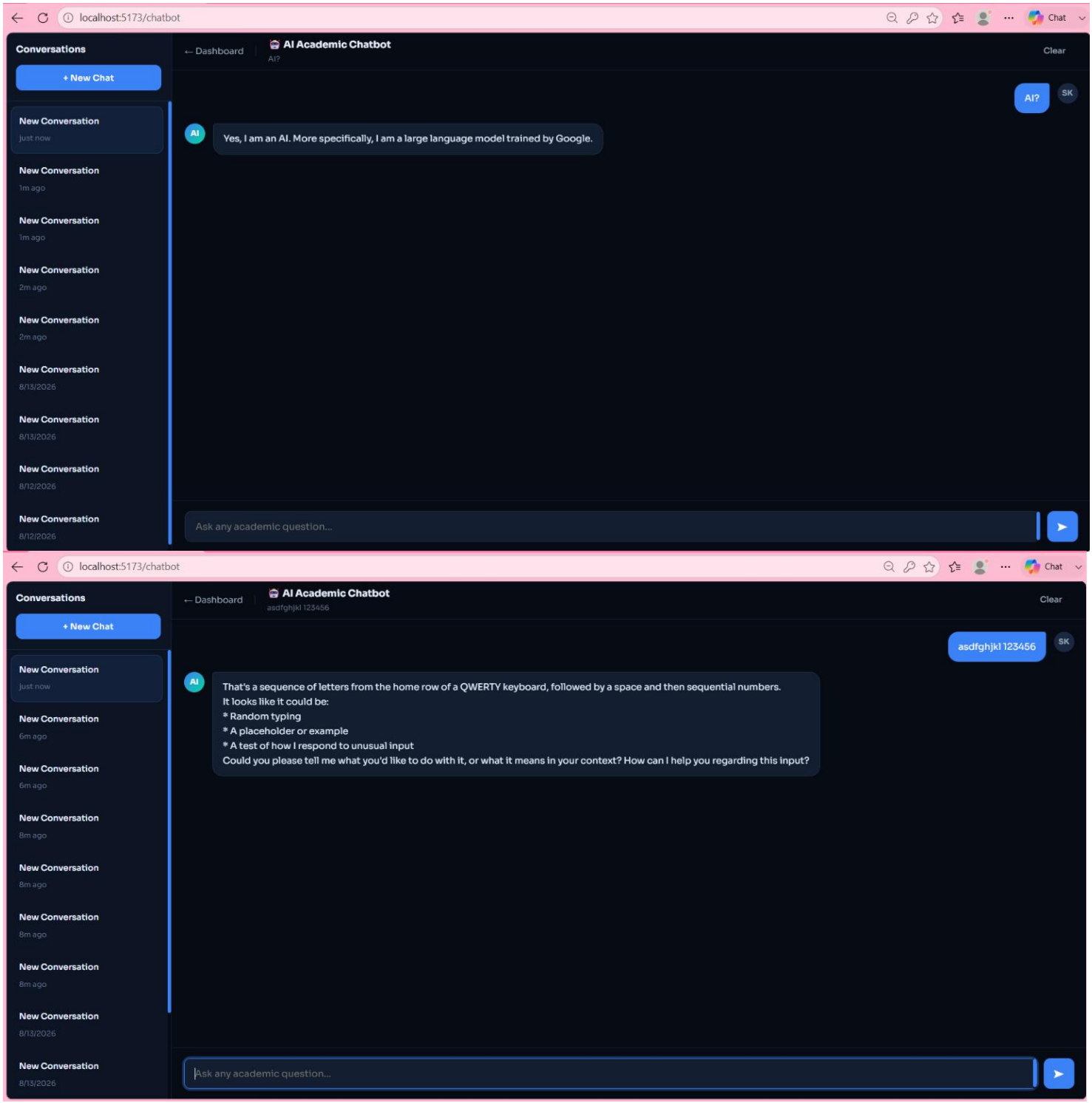
A. Chatbot TESTING:





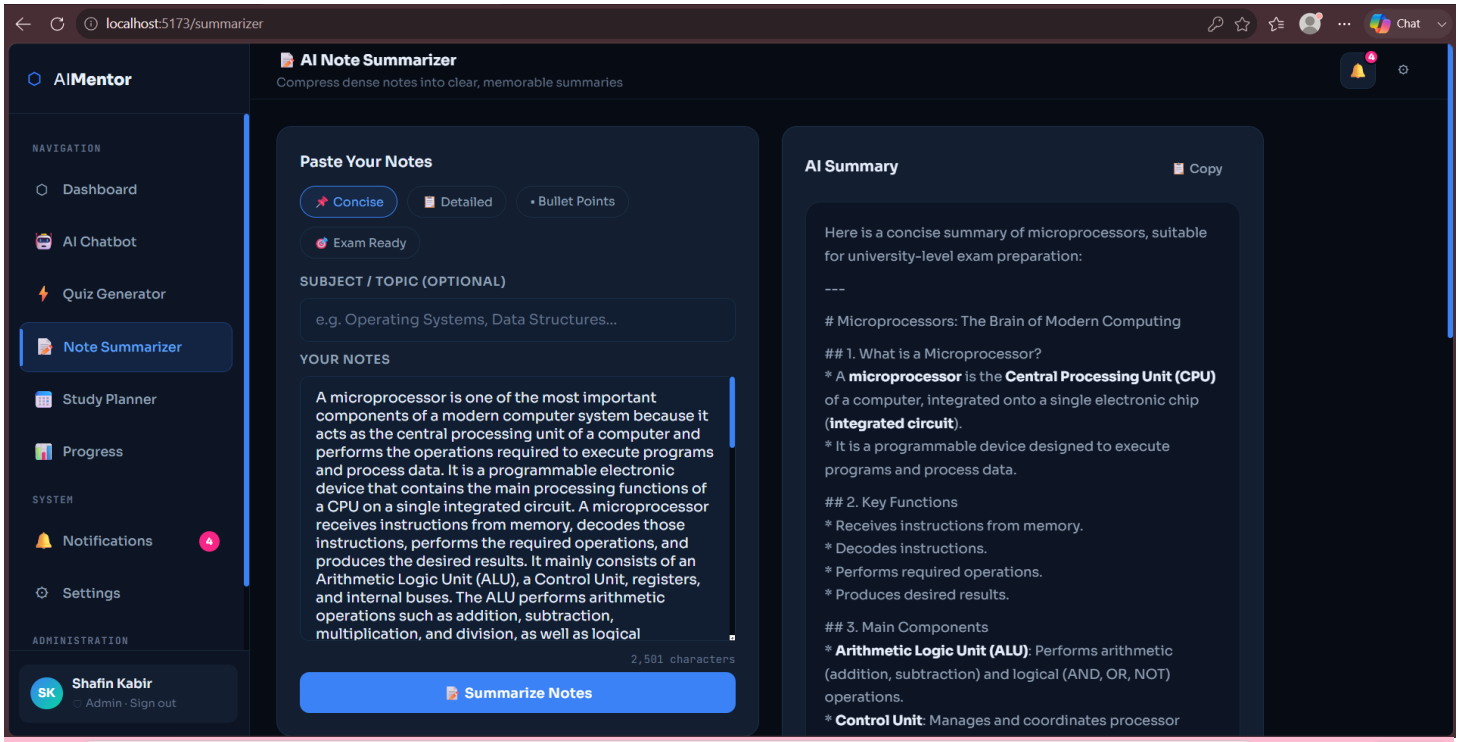

localhost:5173/chatbot
Chat
Conversations
+ New Chat
New Conversation
just now
New Conversation
1m ago
New Conversation
1m ago
New Conversation
4m ago
New Conversation
44m ago
New Conversation
50m ago
New Conversation
51m ago
New Conversation
53m ago
New Conversation
53m ago
← Dashboard
AI Academic Chatbot
What is machine learning? shortly
Clear
What is machine learning? shortly
SK
AI
Machine learning is a field of artificial intelligence that allows computers to learn from data, identify patterns, and make decisions or predictions without being explicitly programmed for every single task or rule.
What is machine learning? shortly
SK
AI
Machine learning is a field of artificial intelligence that enables systems to learn from data, identify patterns, and make decisions or predictions without being explicitly programmed for every task. It allows them to improve their performance over time as they are exposed to more data.
What is machine learning? shortly
SK
AI
Machine learning (ML) is a branch of artificial intelligence that enables computer systems to learn from data, identify patterns, and make decisions or predictions without being explicitly programmed for every specific rule or task.
What is machine learning? shortly
SK
AI
Machine learning is a field of artificial intelligence that enables computers to learn from data, identify patterns, and make predictions or decisions without being explicitly programmed for every specific task.
Ask any academic question...

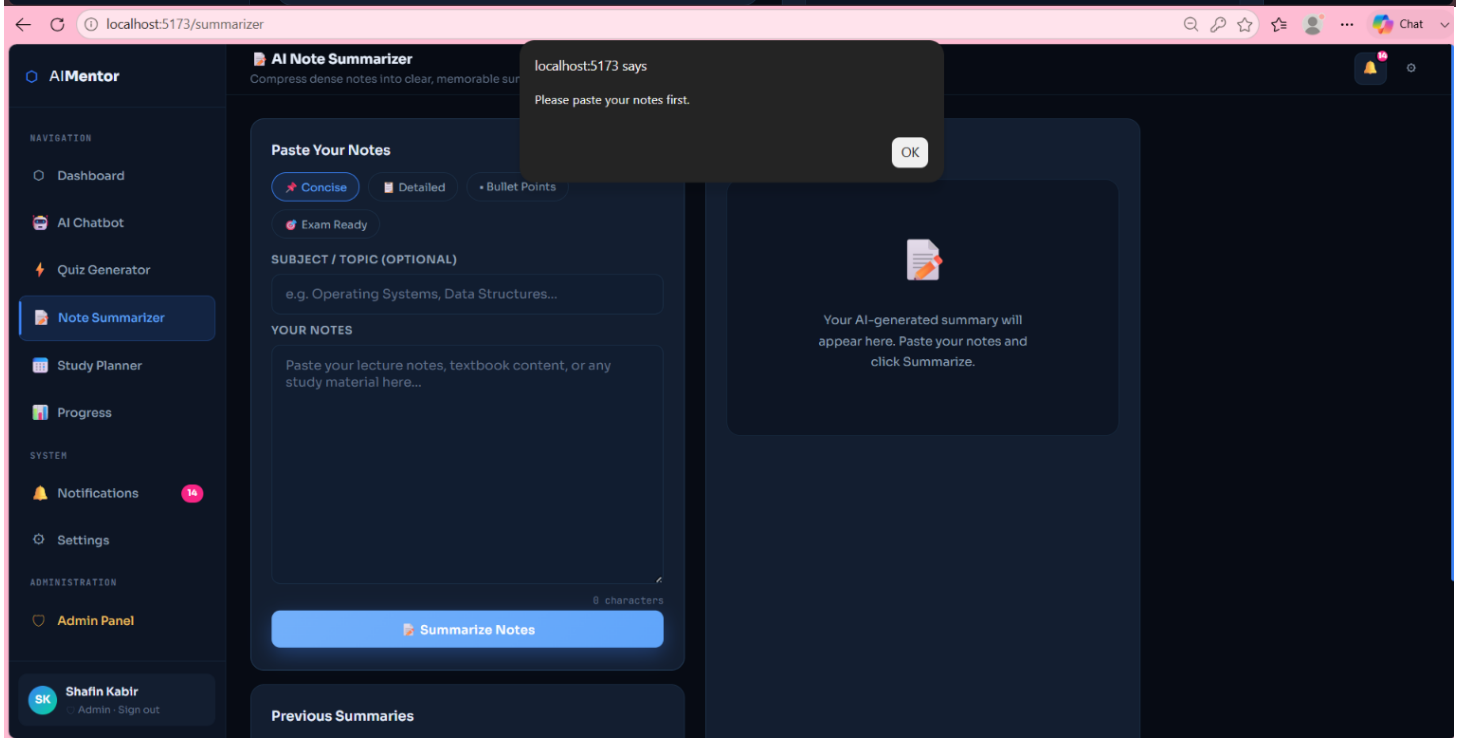

localhost:5173/chatbot
Chat
Conversations
+ New Chat
New Conversation
1m ago
New Conversation
2m ago
New Conversation
4m ago
New Conversation
4m ago
New Conversation
4m ago
New Conversation
4m ago
New Conversation
8/13/2026
New Conversation
8/13/2026
New Conversation
8/12/2026
← Dashboard
AI Academic Chatbot
ANSYS is a powerful engineering sim...
Clear
ANSYS is a powerful engineering simulation software widely used in mechanical engineering to analyze and design mechanical systems. It can simulate stress, deformation, heat transfer, fluid flow, vibration, and other physical behaviors before building a real prototype. Engineers use ANSYS to improve product safety, performance, reliability, and efficiency while reducing development cost and time.
SK
AI
That's an excellent and comprehensive summary of ANSYS!
You've perfectly captured its essence:
* What it is: A powerful engineering simulation software.
* Its core function: Analyzing and designing mechanical systems.
* Key capabilities: Stress, deformation, heat transfer, fluid flow, vibration (covering solid mechanics, thermal, and fluid dynamics).
* Its primary benefit: Virtual prototyping to save time and cost.
* The ultimate goal for engineers: Improving product safety, performance, reliability, efficiency, and accelerating development.
ANSYS, often leveraging methods like Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD), has indeed revolutionized the way products are developed, making "predictive engineering" a standard practice across countless industries.
Ask any academic question...


localhost:5173/chatbot
Chat
Conversations
+ New Chat
New Conversation
just now
New Conversation
1m ago
New Conversation
1m ago
New Conversation
2m ago
New Conversation
2m ago
New Conversation
8/13/2026
New Conversation
8/13/2026
New Conversation
8/12/2026
New Conversation
8/12/2026
← Dashboard
AI Academic Chatbot
AI?
Clear
AI?
SK
AI
Yes, I am an AI. More specifically, I am a large language model trained by Google.
Ask any academic question...

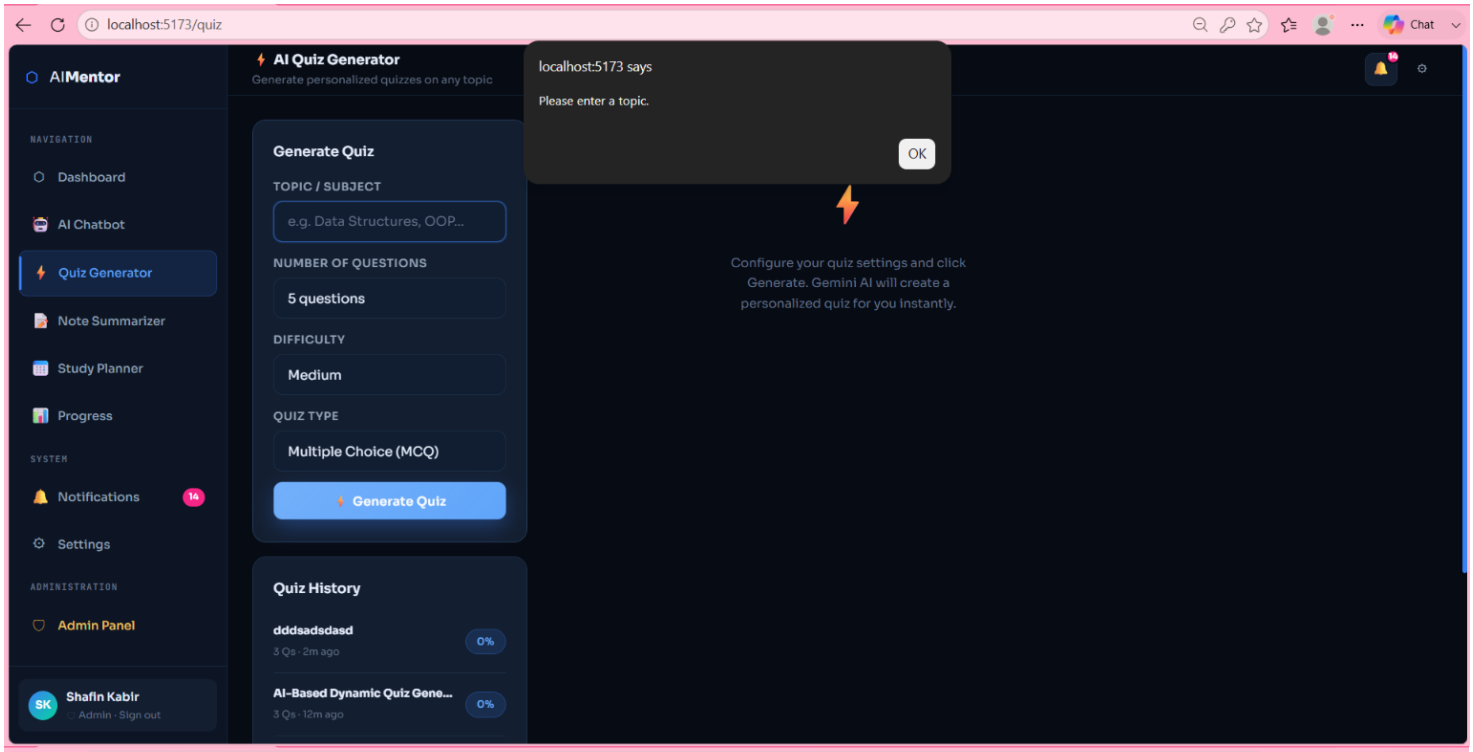
localhost:5173/chatbot
Chat
Conversations
+ New Chat
New Conversation
just now
New Conversation
6m ago
New Conversation
6m ago
New Conversation
8m ago
New Conversation
8m ago
New Conversation
8m ago
New Conversation
8m ago
New Conversation
8/13/2026
New Conversation
8/13/2026
← Dashboard
AI Academic Chatbot
asdfghjkl 123456
Clear
asdfghjkl 123456
SK
AI
That's a sequence of letters from the home row of a QWERTY keyboard, followed by a space and then sequential numbers.
It looks like it could be:
* Random typing
* A placeholder or example
* A test of how I respond to unusual input
Could you please tell me what you'd like to do with it, or what it means in your context? How can I help you regarding this input?
Ask any academic question...

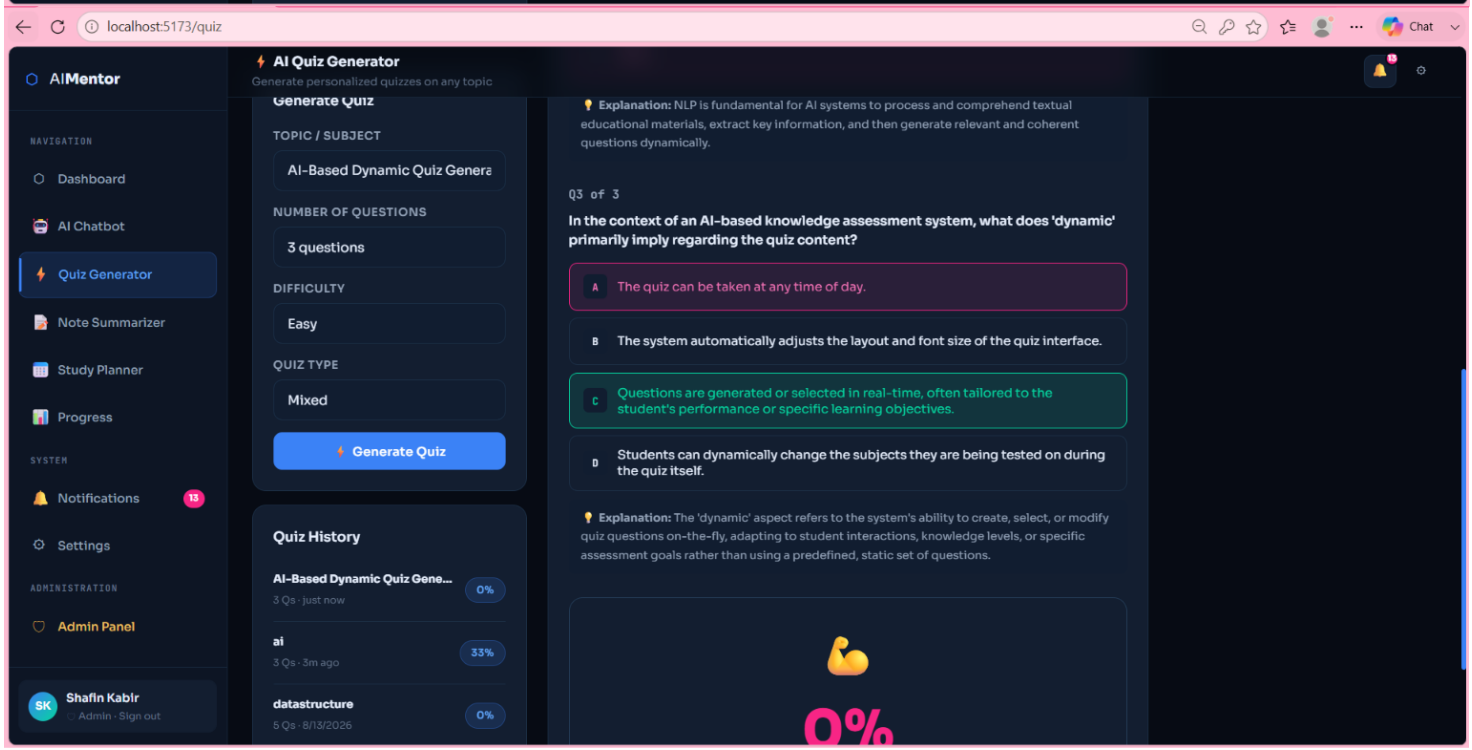
B. Note_Summarizer TESTING:

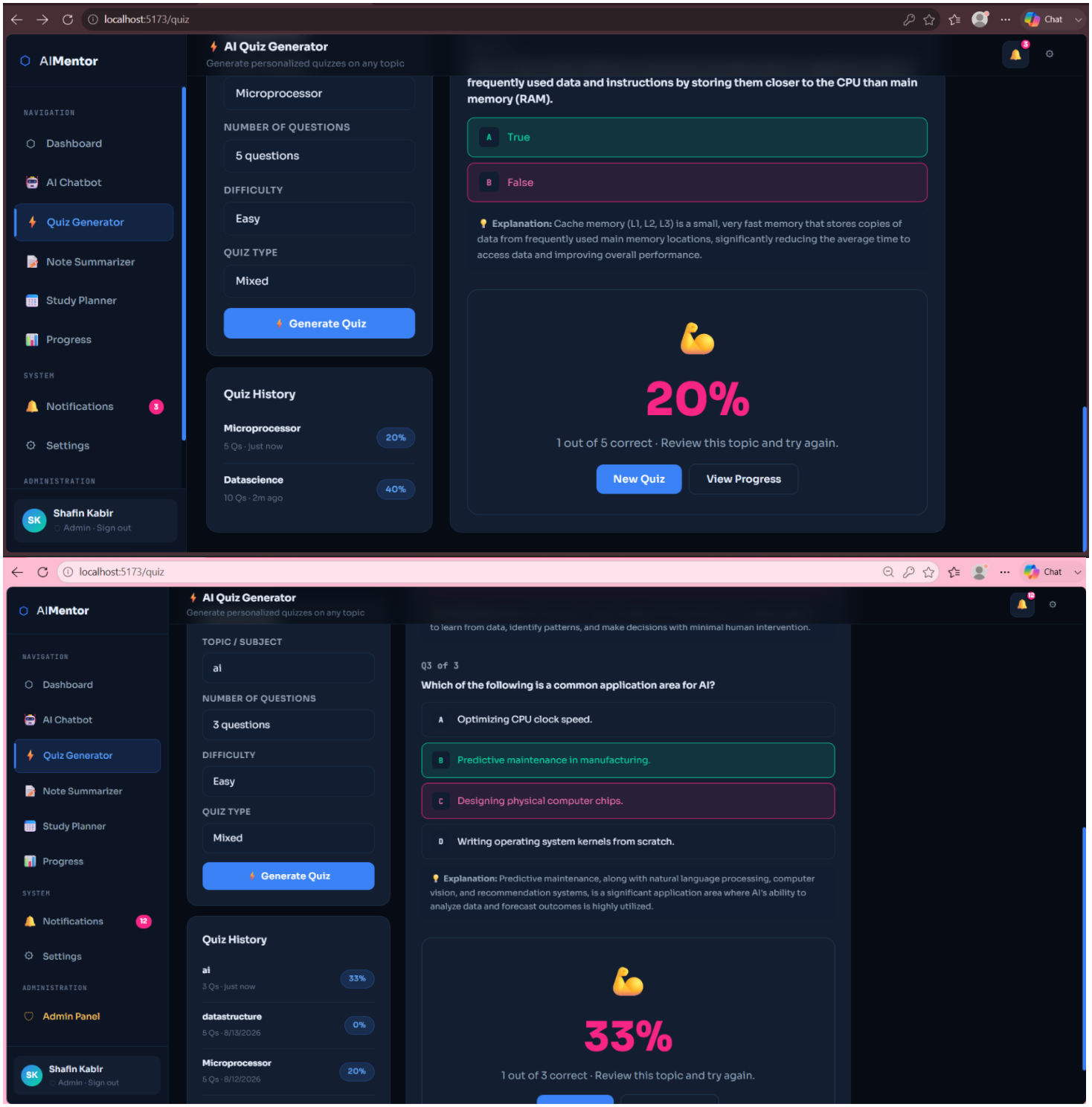




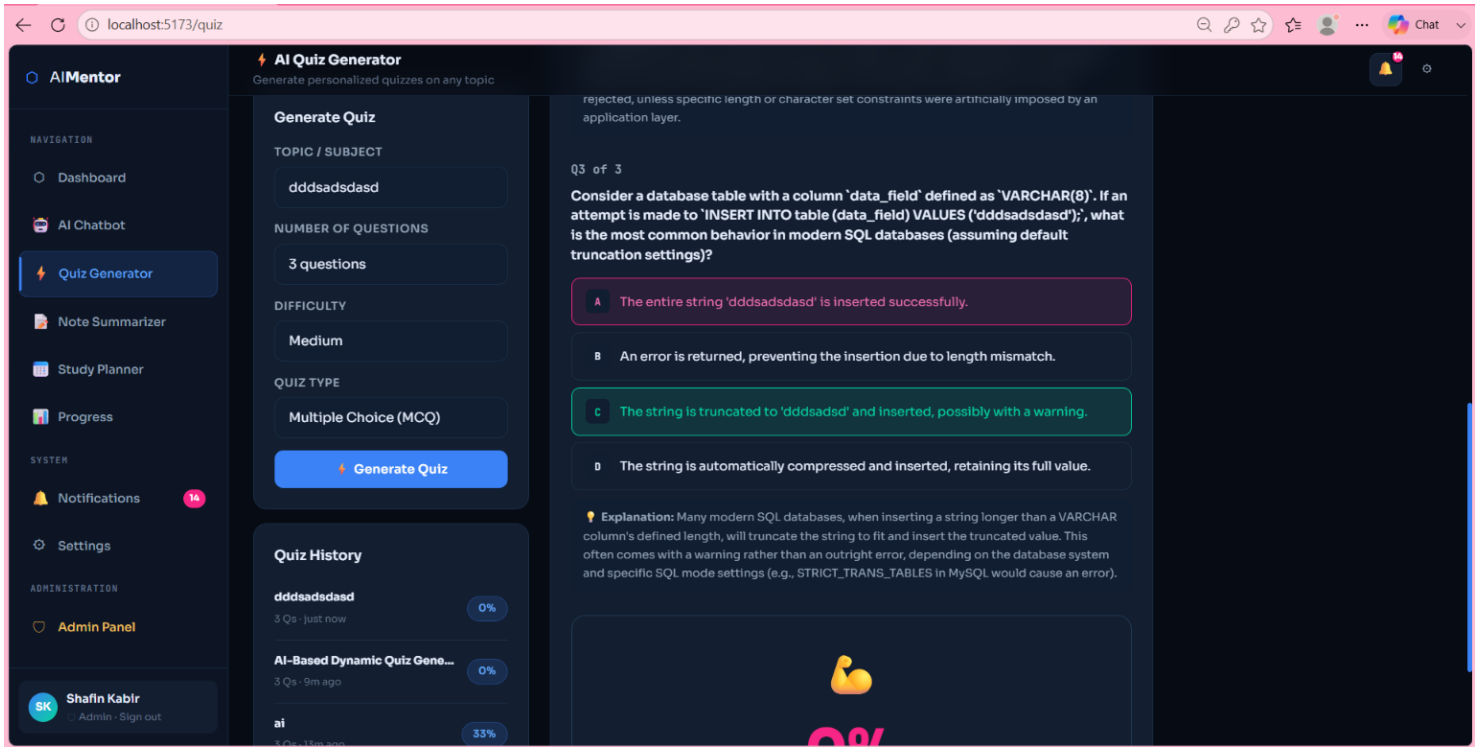
C. Quiz_Generator TESTING:



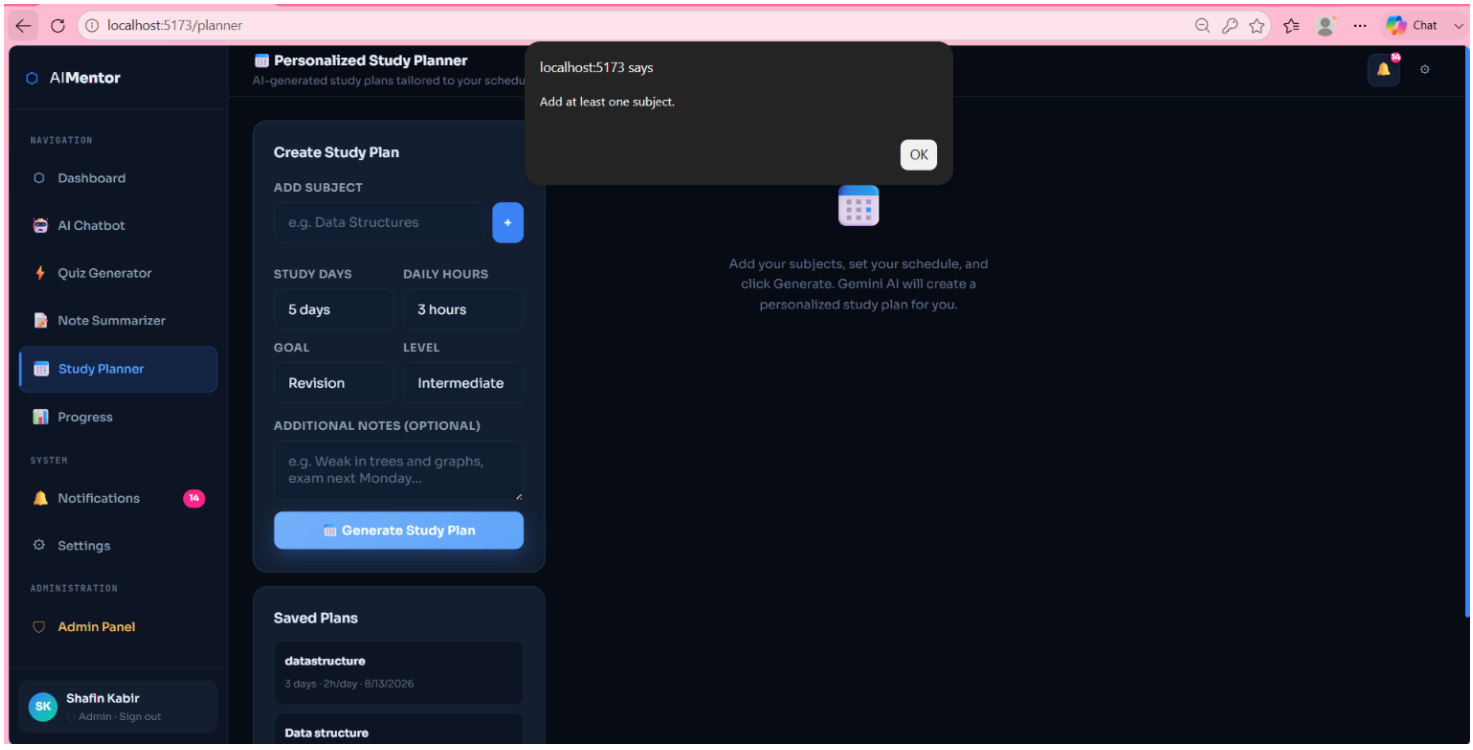


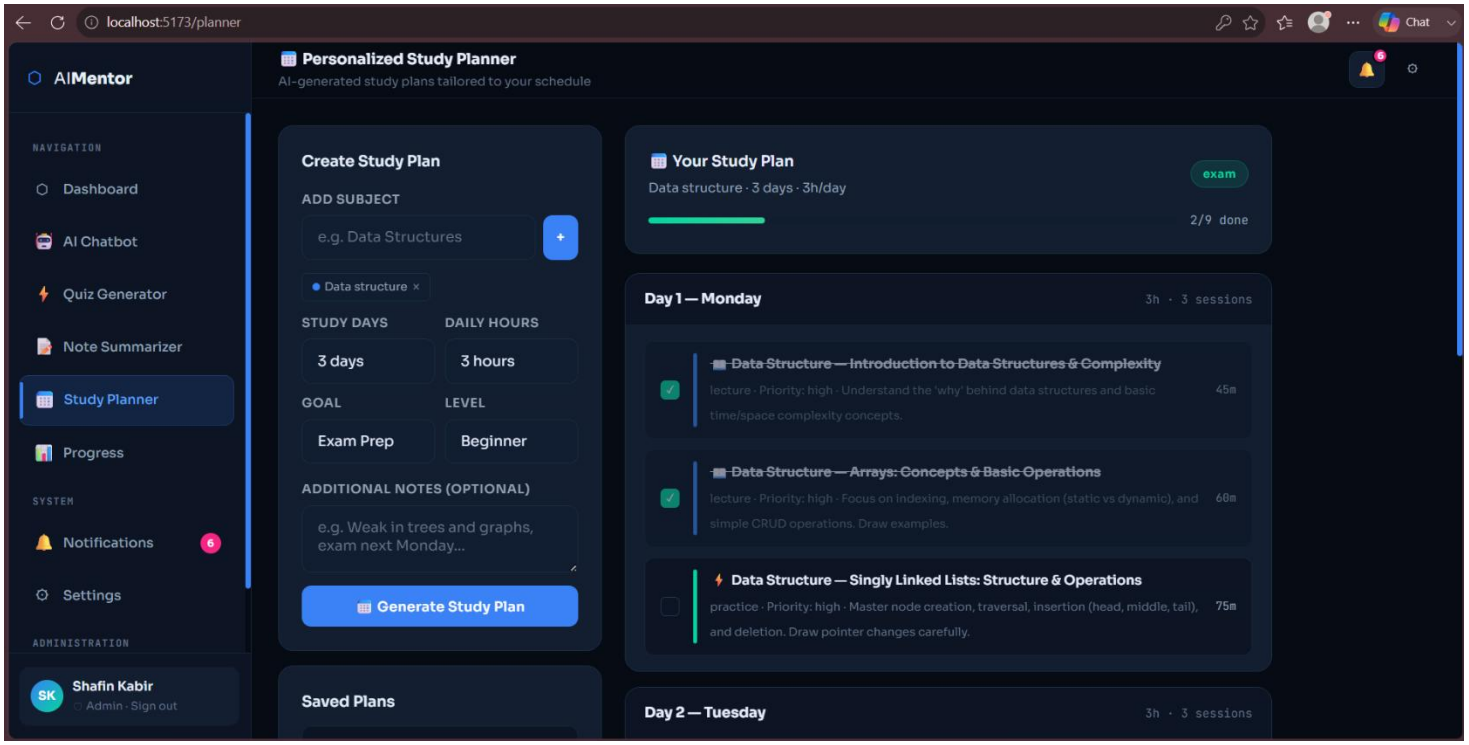

localhost:5173/quiz
AIMentor
AI Quiz Generator
Generate personalized quizzes on any topic
NAVIGATION
Dashboard
AI Chatbot
Quiz Generator
Note Summarizer
Study Planner
Progress
SYSTEM
Notifications
Settings
ADMINISTRATION
Shafin Kabir
Admin · Sign out
Microprocessor
NUMBER OF QUESTIONS
5 questions
DIFFICULTY
Easy
QUIZ TYPE
Mixed
Generate Quiz
Quiz History
Microprocessor
5 Qs · just now
20%
Datascience
10 Qs · 2m ago
40%
frequently used data and instructions by storing them closer to the CPU than main memory (RAM).
A True
B False
Explanation: Cache memory (L1, L2, L3) is a small, very fast memory that stores copies of data from frequently used main memory locations, significantly reducing the average time to access data and improving overall performance.
20%
1 out of 5 correct · Review this topic and try again.
New Quiz
View Progress

localhost:5173/quiz
AIMentor
AI Quiz Generator
Generate personalized quizzes on any topic
NAVIGATION
Dashboard
AI Chatbot
Quiz Generator
Note Summarizer
Study Planner
Progress
SYSTEM
Notifications
Settings
ADMINISTRATION
Admin Panel
Shafin Kabir
Admin · Sign out
TOPIC / SUBJECT
ai
NUMBER OF QUESTIONS
3 questions
DIFFICULTY
Easy
QUIZ TYPE
Mixed
Generate Quiz
Quiz History
ai
3 Qs · just now
33%
datastructure
5 Qs · 8/13/2026
0%
Microprocessor
5 Qs · 8/12/2026
20%
to learn from data, identify patterns, and make decisions with minimal human intervention.
Q3 of 3
Which of the following is a common application area for AI?
A Optimizing CPU clock speed.
B Predictive maintenance in manufacturing.
C Designing physical computer chips.
D Writing operating system kernels from scratch.
Explanation: Predictive maintenance, along with natural language processing, computer vision, and recommendation systems, is a significant application area where AI's ability to analyze data and forecast outcomes is highly utilized.
33%
1 out of 3 correct · Review this topic and try again.



D. Study_Planner TESTING:





9. UI/UX & RESPONSIVE TESTING

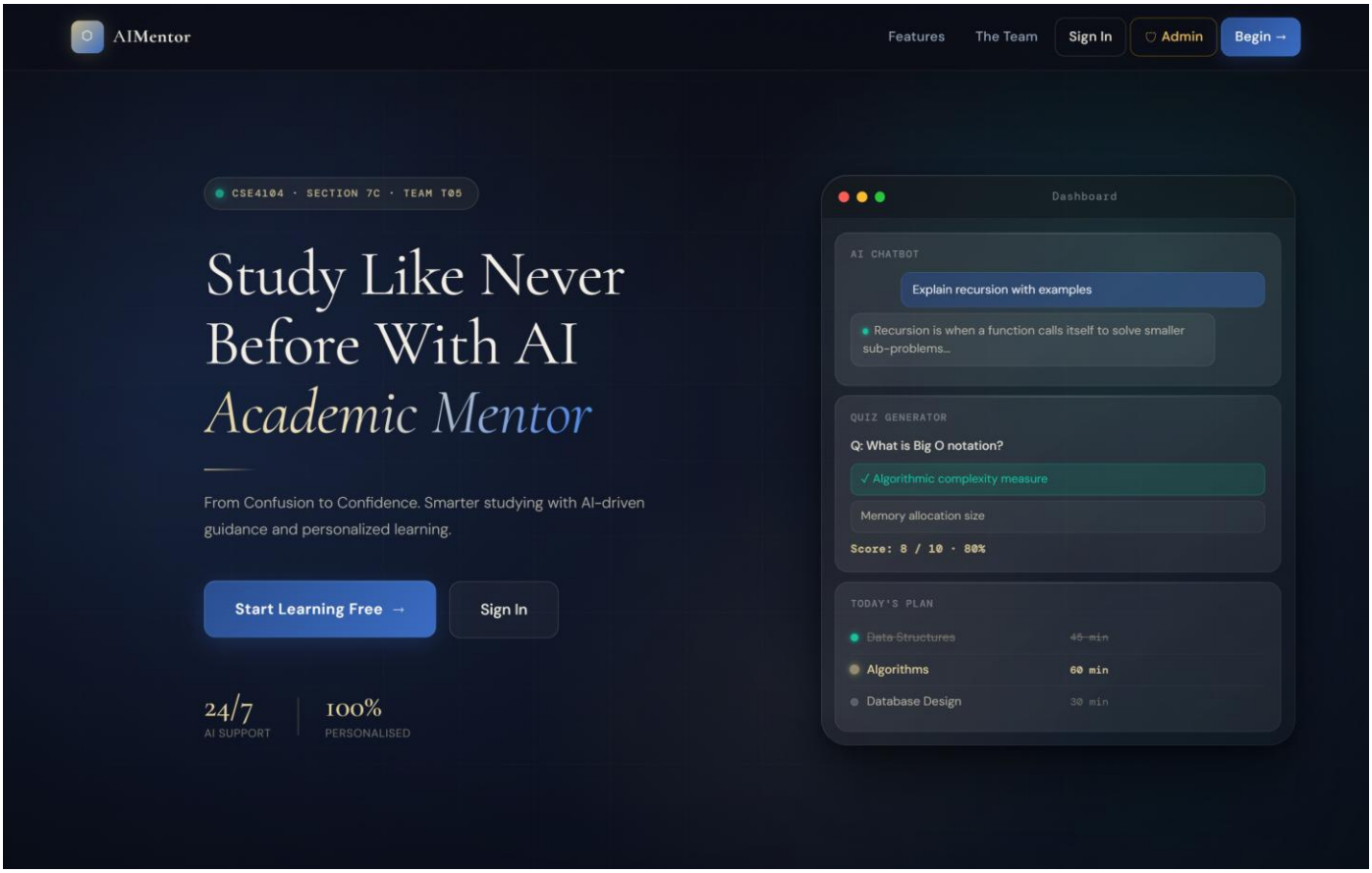
A. UI/UX TESTING:

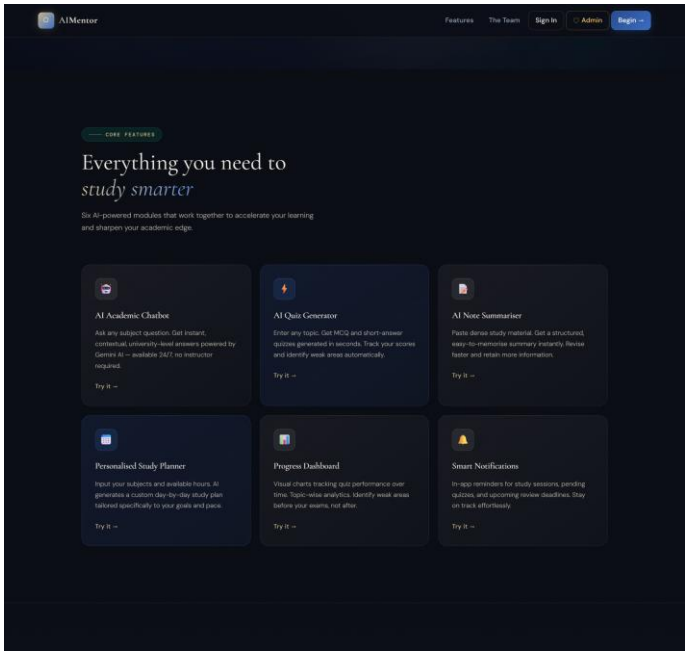
Element	Status
Navigation	TESTED
Buttons	TESTED
Forms	TESTED
Menus	TESTED
Modals	TESTED
Notifications	TESTED
Error messages	PARTIAL — API-level only
Loading states	TESTED
Empty states	TESTED
Page layouts	TESTED
Text readability	TESTED

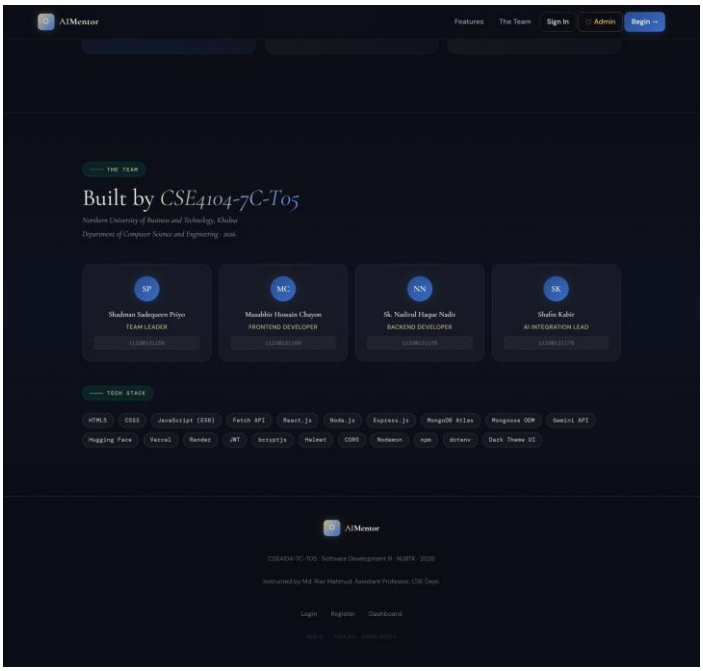
B. RESPONSIVE TESTING:

DESKTOP – 1440PX

Home

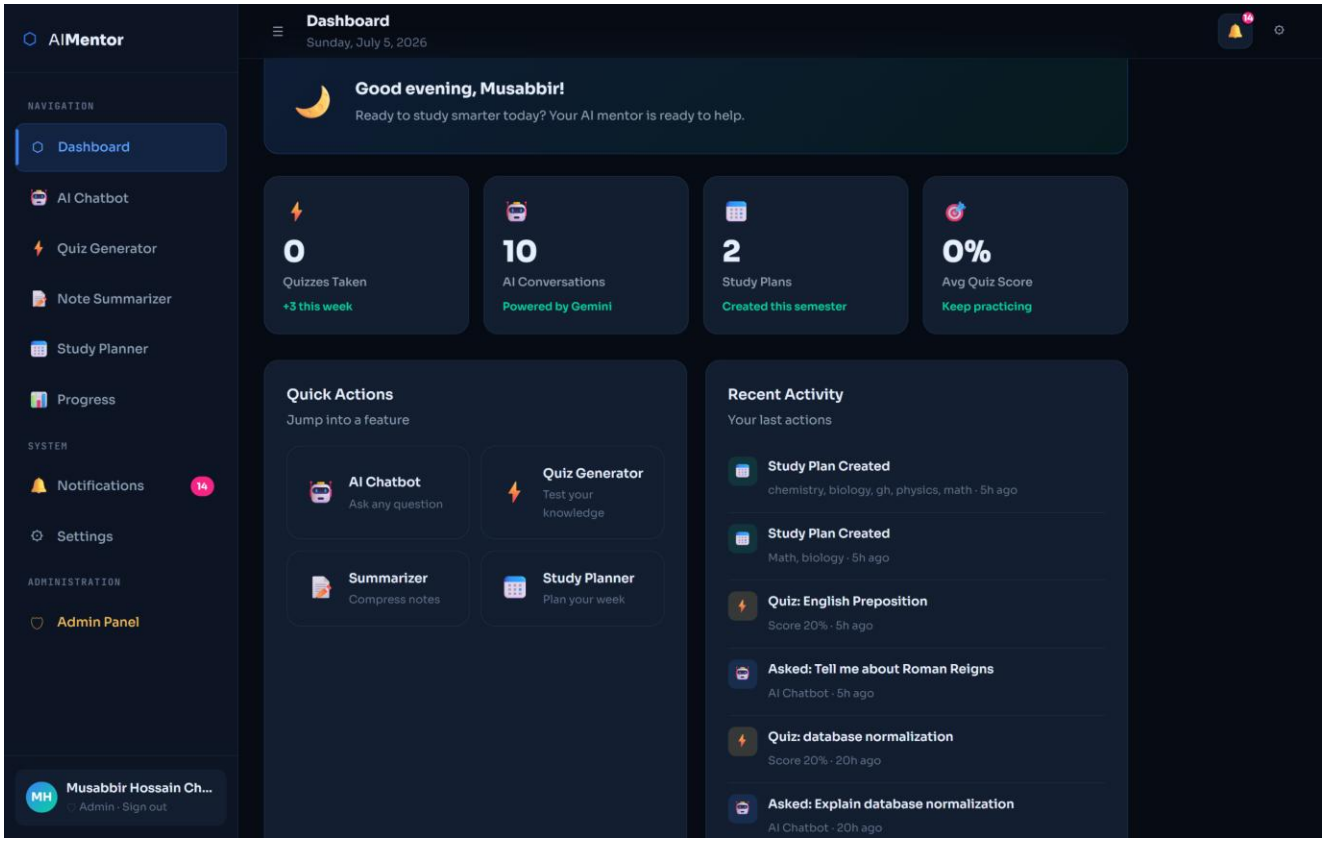


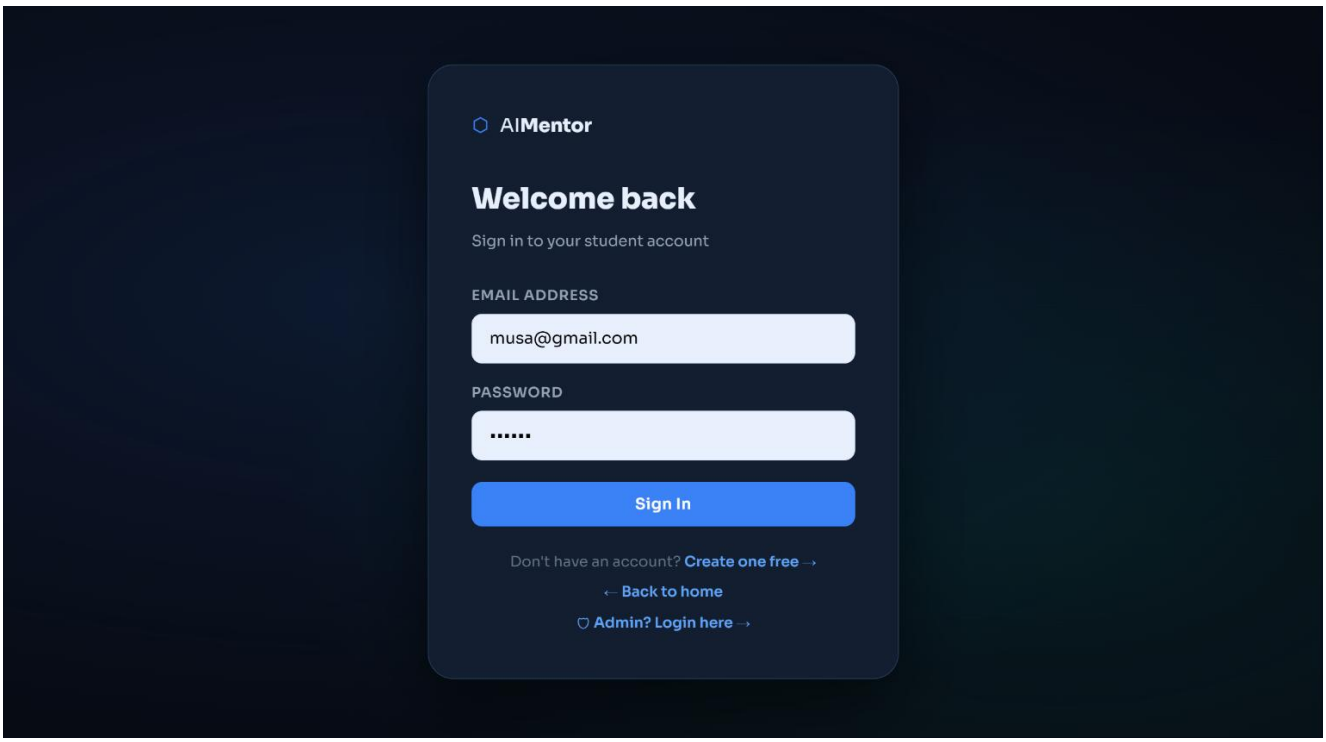




DESKTOP — 1440PX

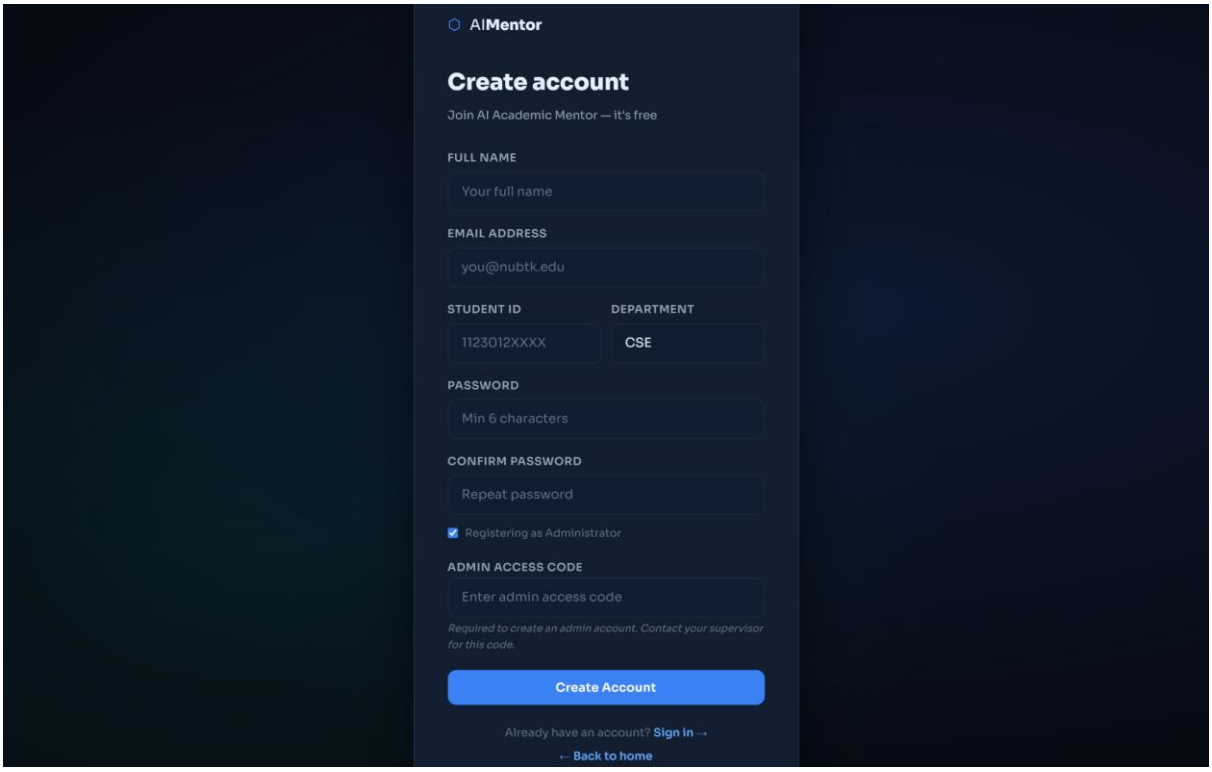
Dashboard & Login

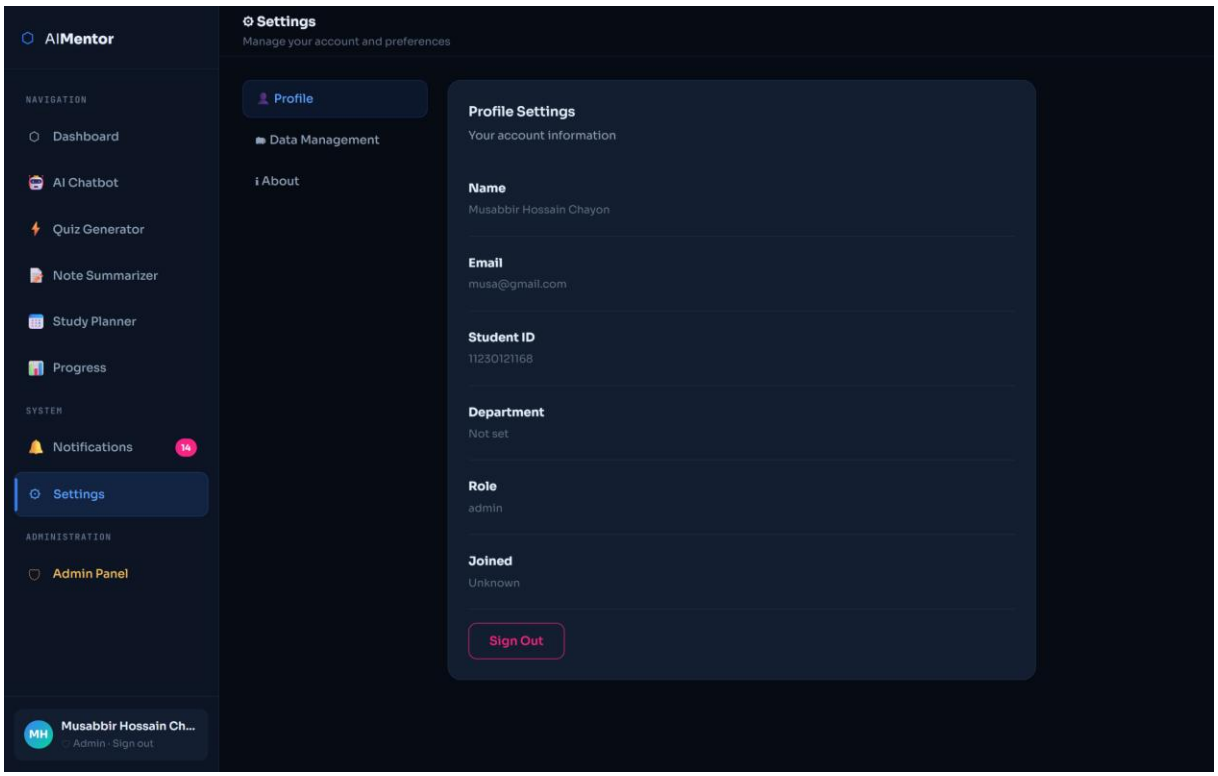




DESKTOP — 1440PX

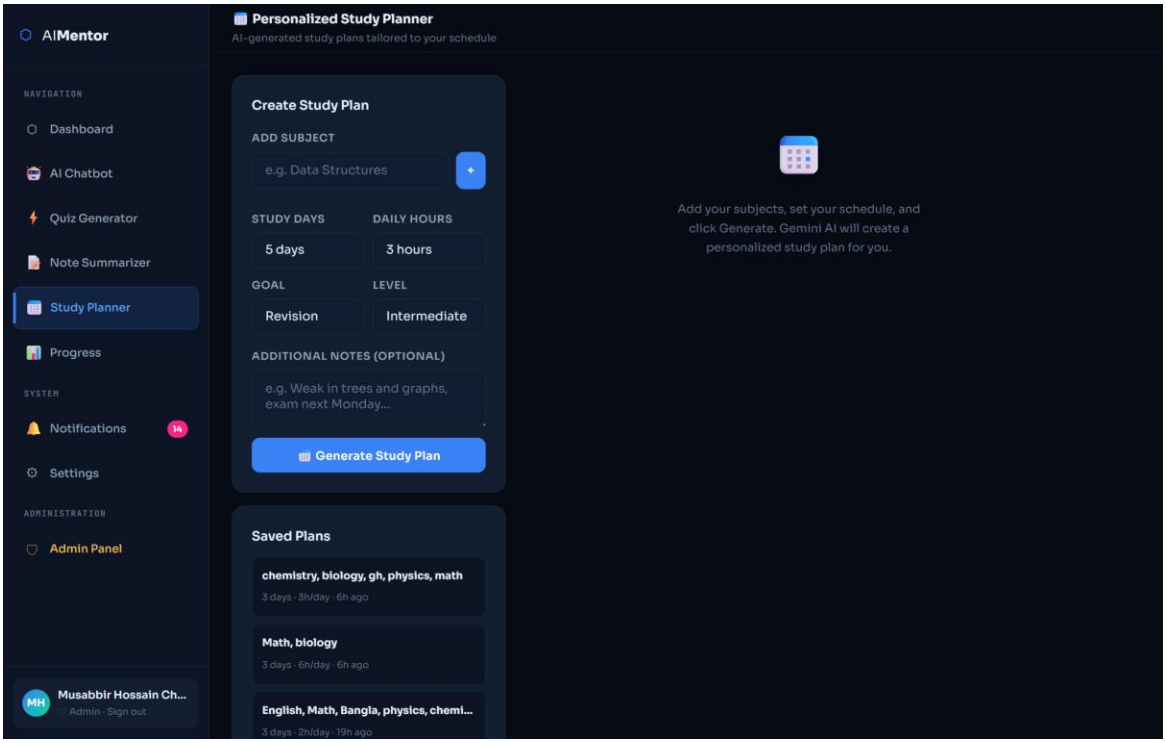
Register & Settings

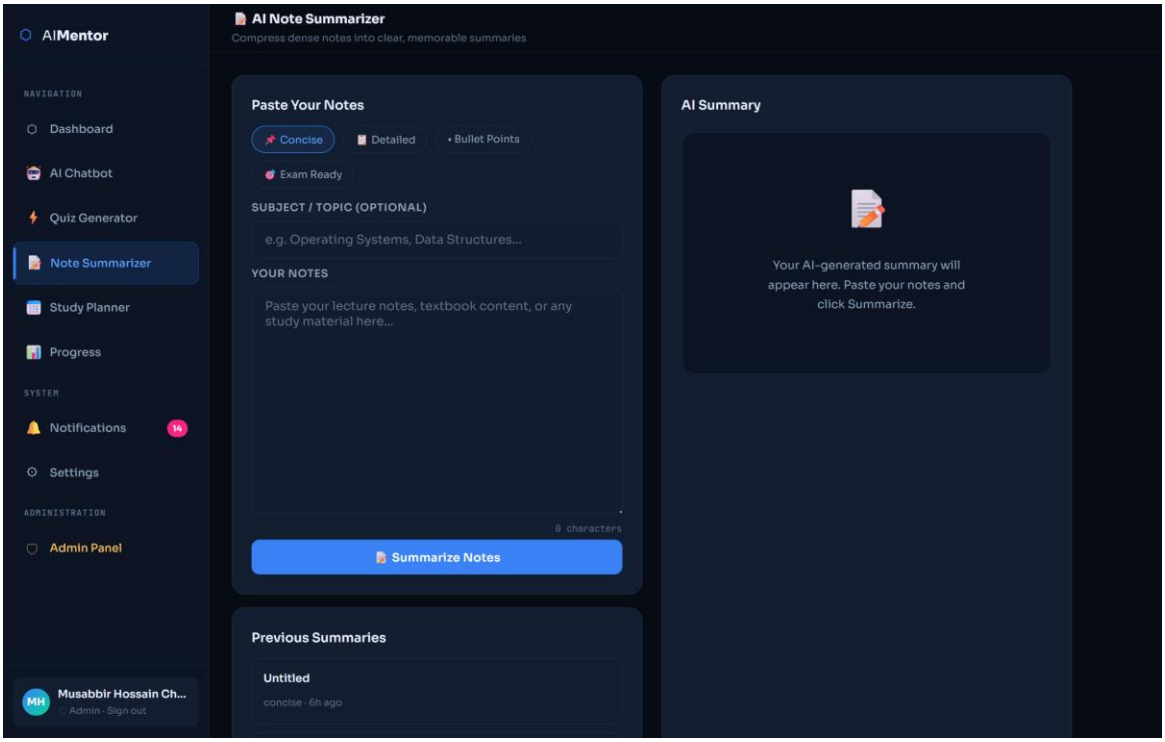




DESKTOP — 1440PX

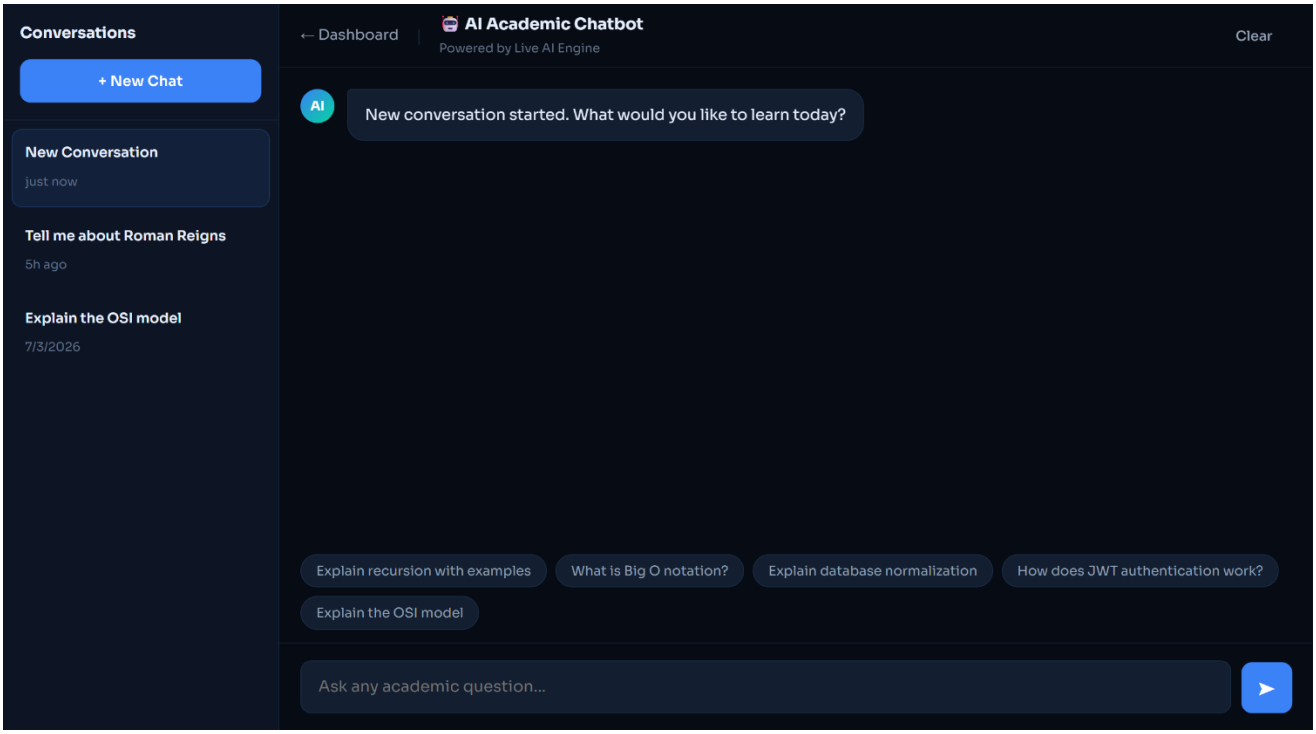
Study Planner & Note Summarizer

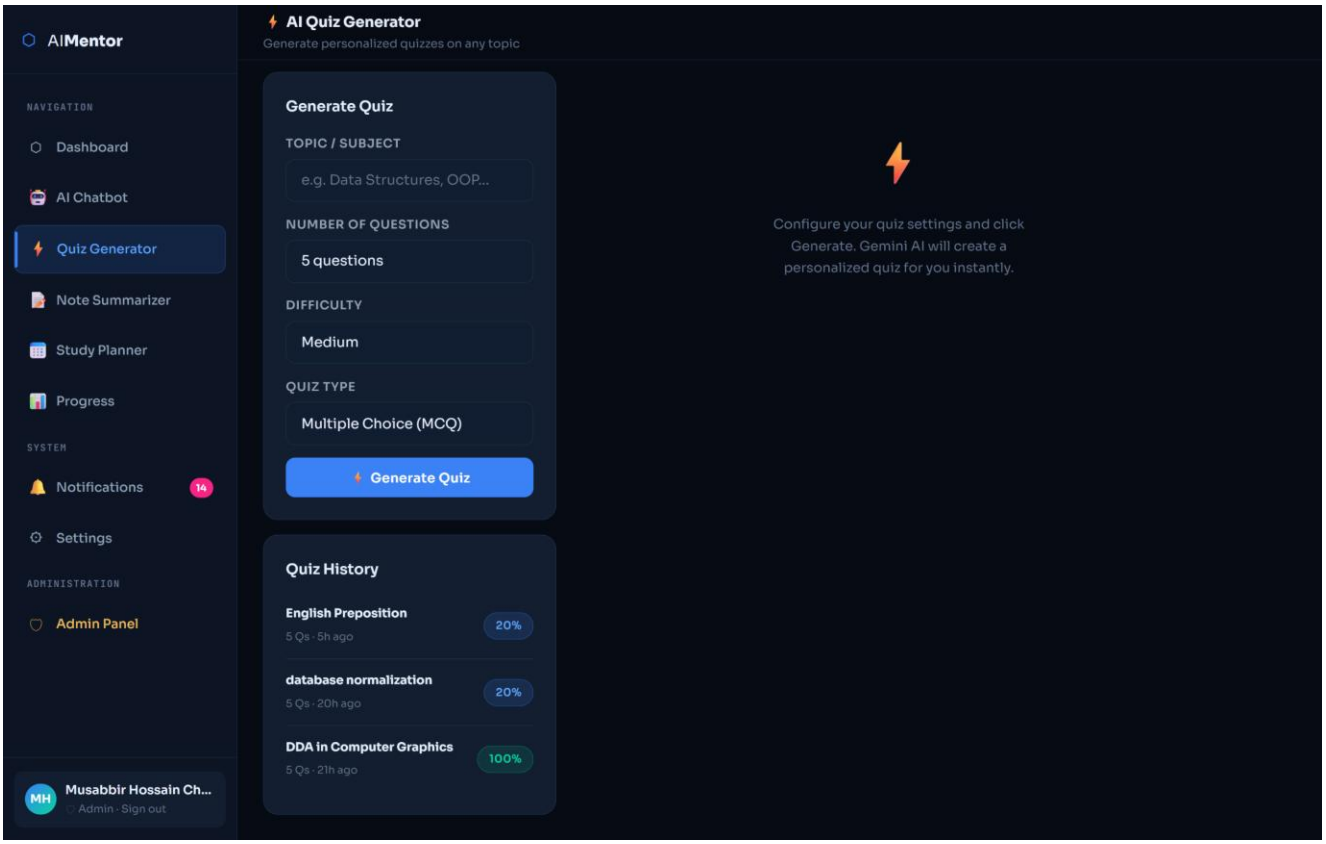




DESKTOP — 1440PX

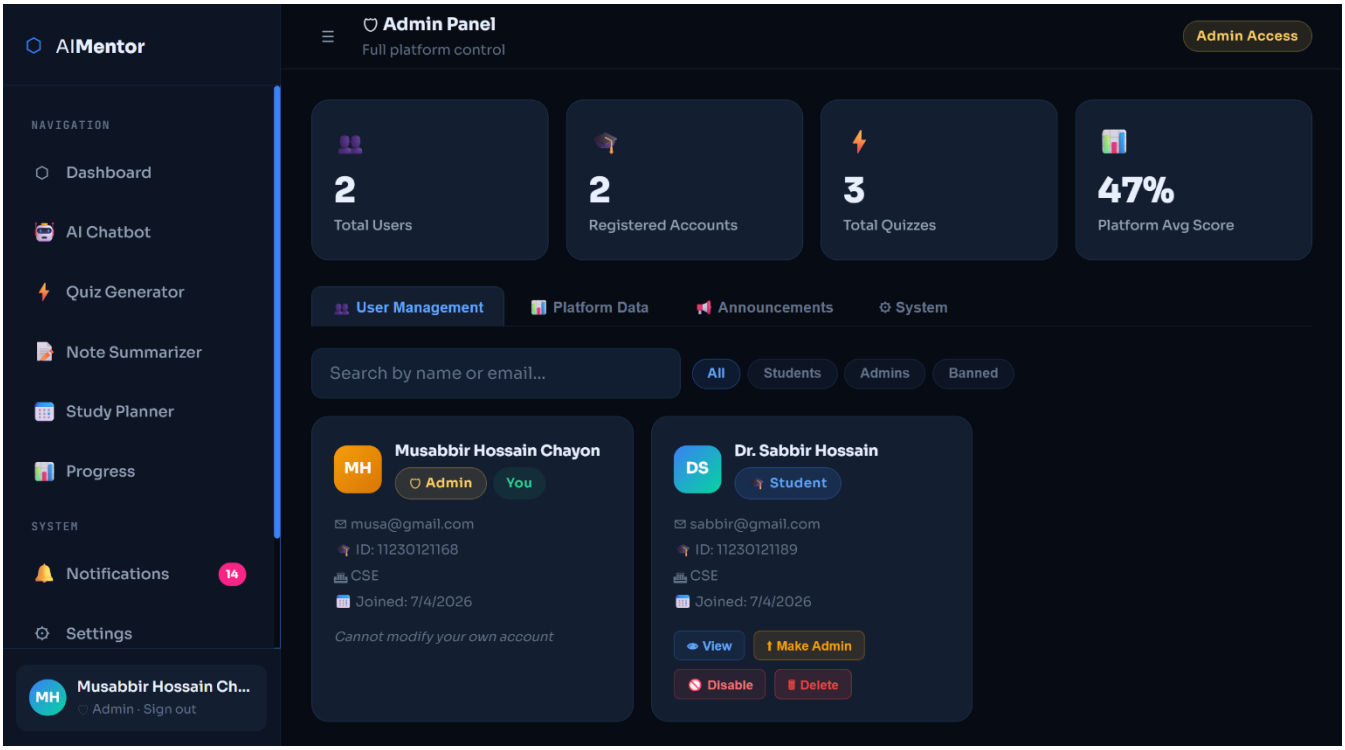
AI Chatbot & Quiz Generator

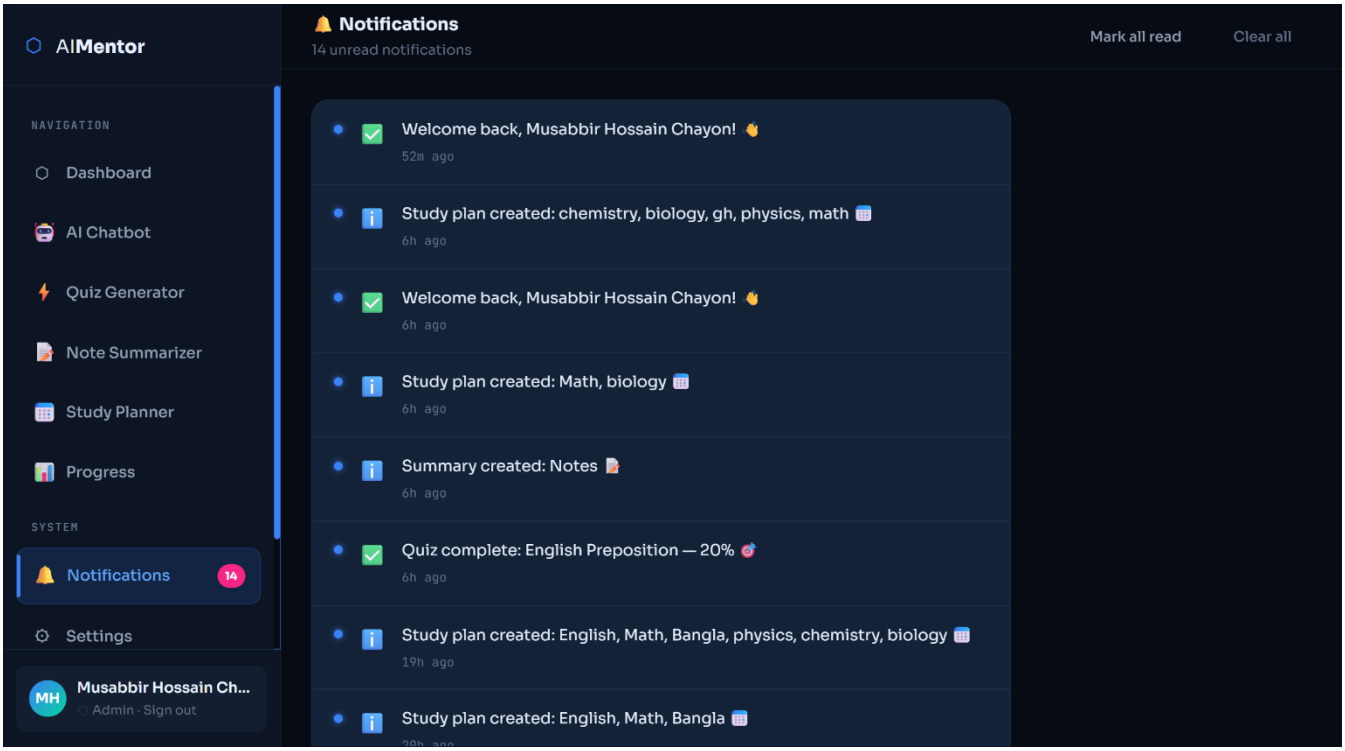




DESKTOP — 1440PX

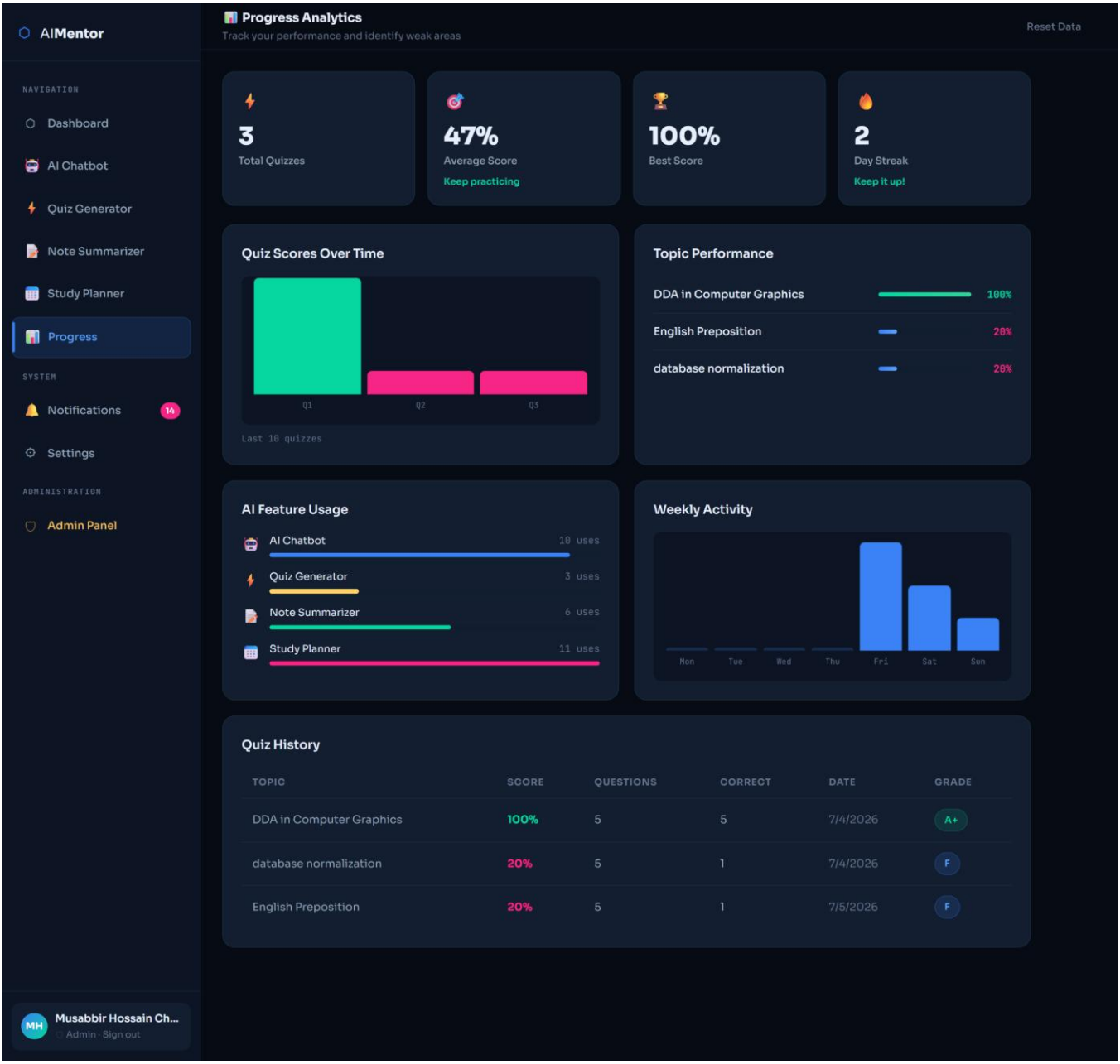
Admin Panel & Notifications





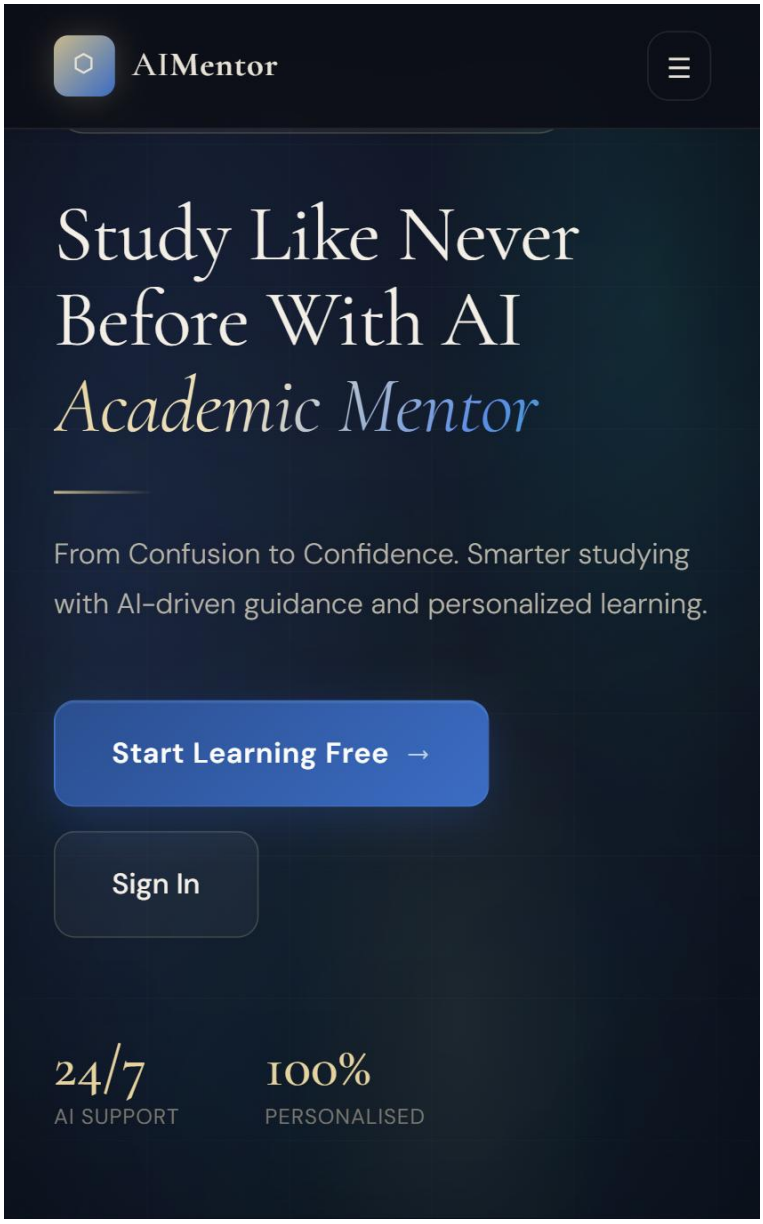
DESKTOP — 1440PX

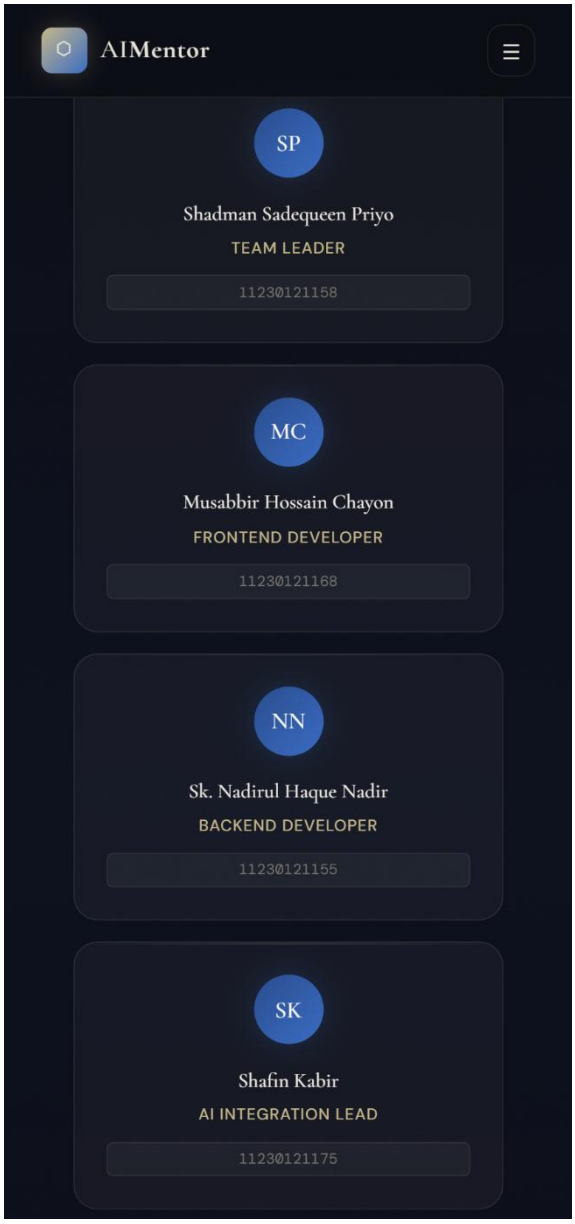
Progress Analytics



MOBILE — 425PX (RESPONSIVE VIEW)

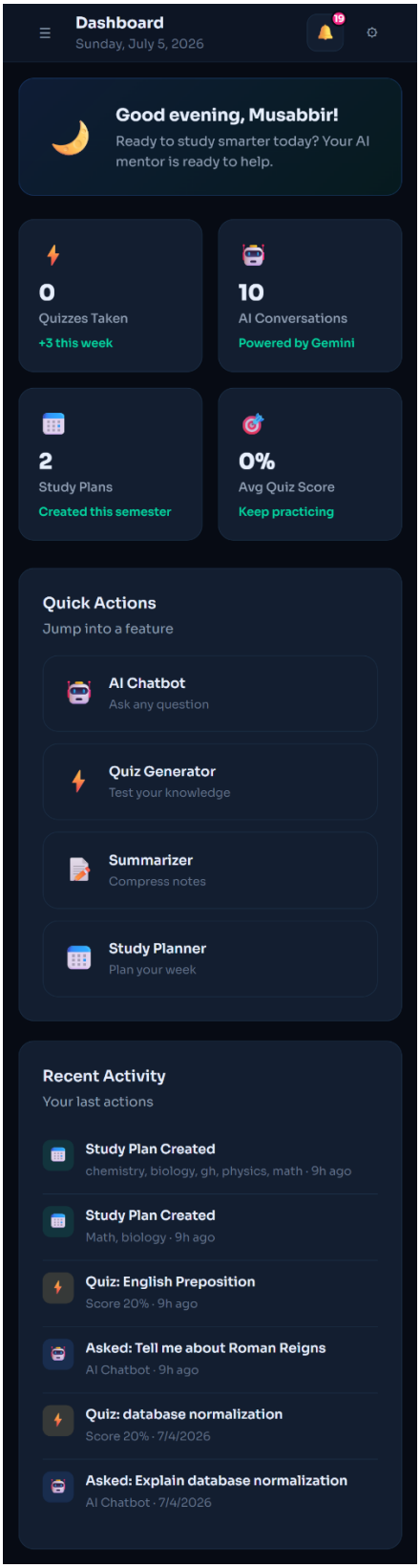
Home

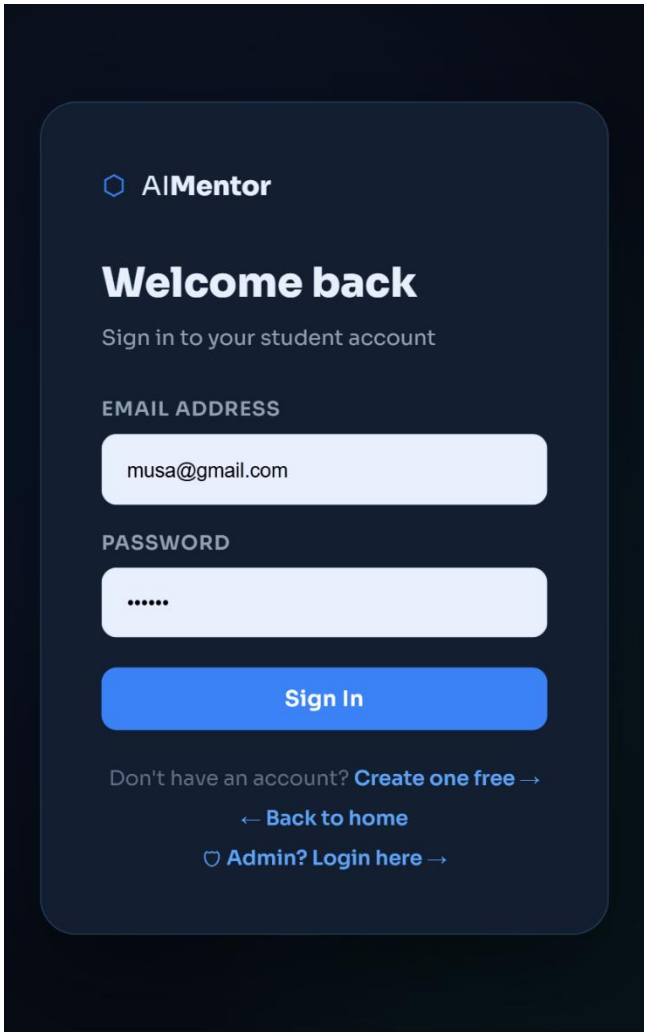




MOBILE — 425PX (RESPONSIVE VIEW)

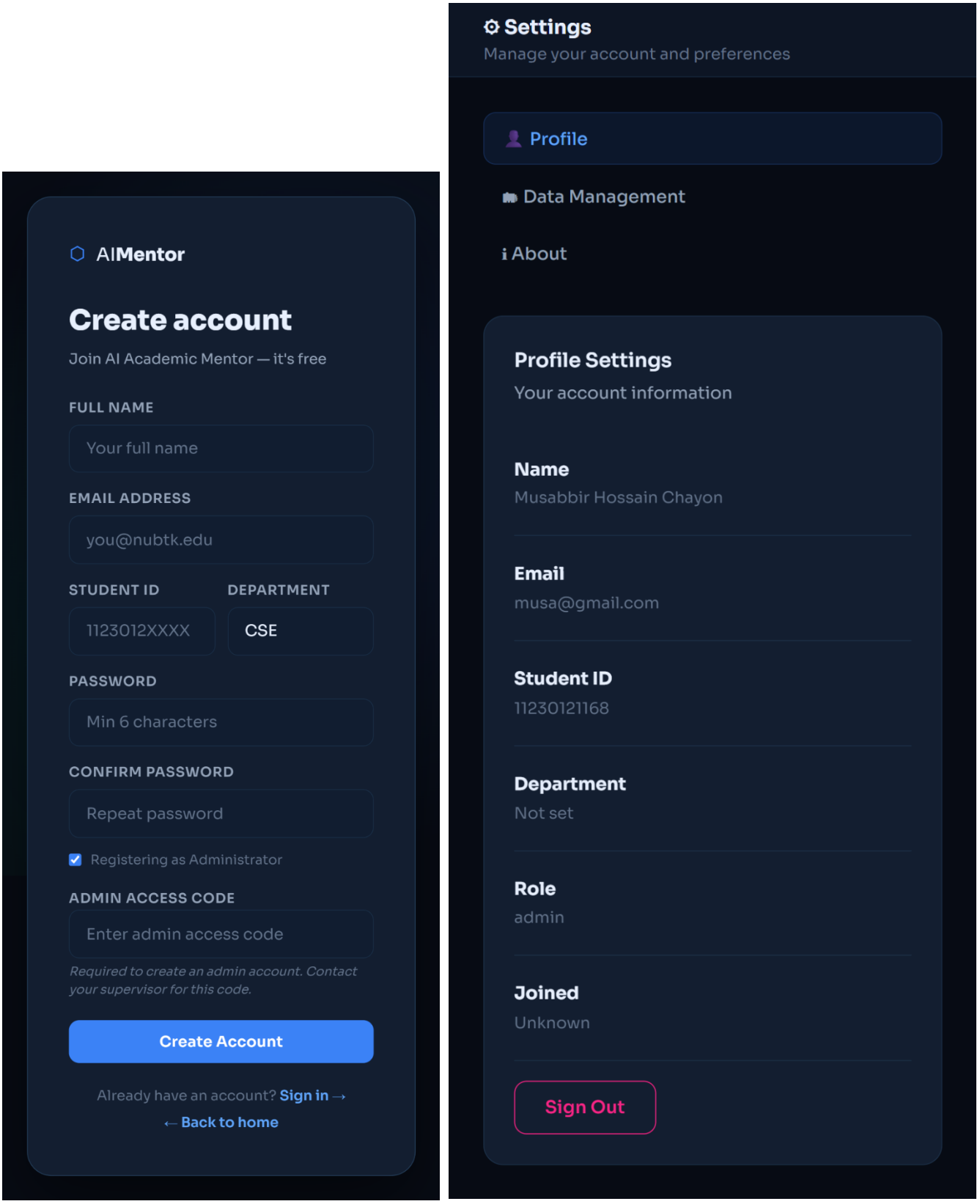
Dashboard & Login





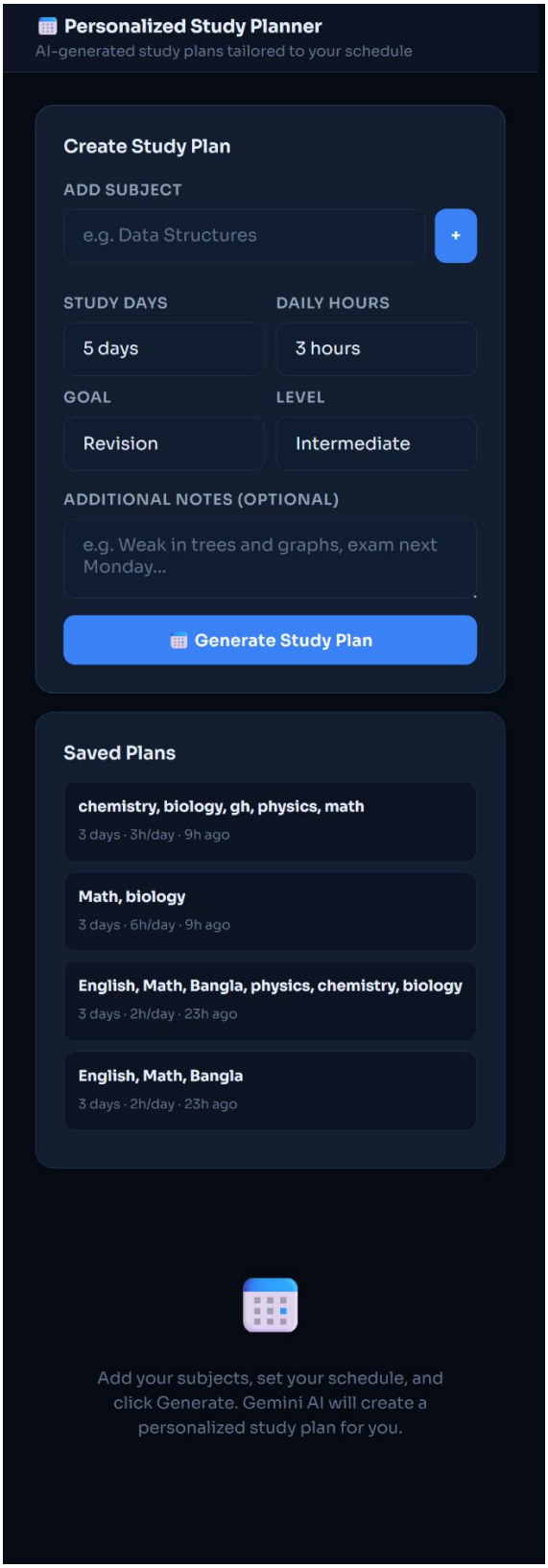
MOBILE — 425PX (RESPONSIVE VIEW)

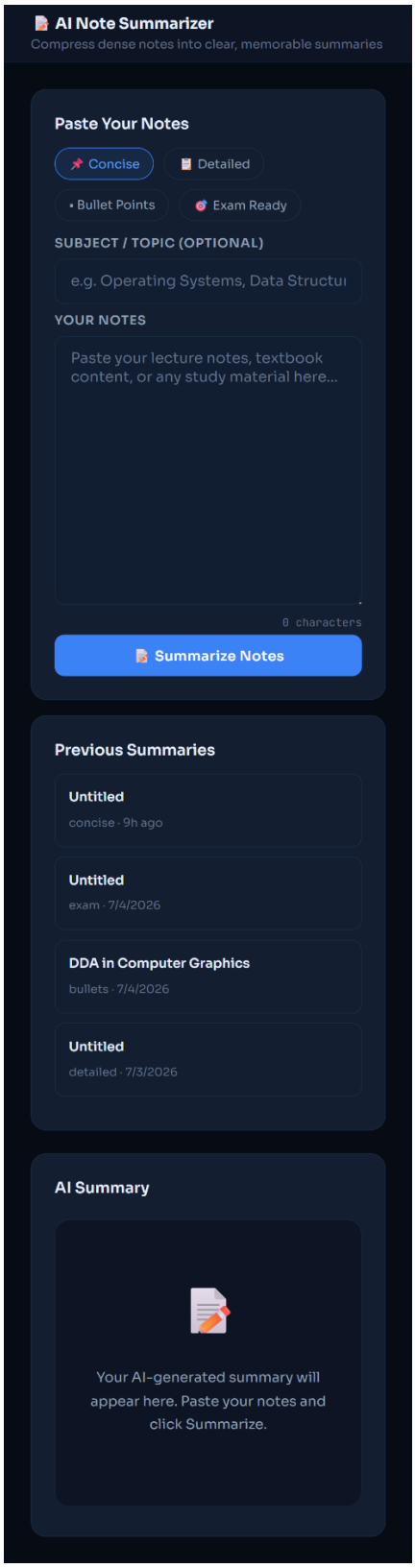
Register & Settings



MOBILE — 425PX (RESPONSIVE VIEW)

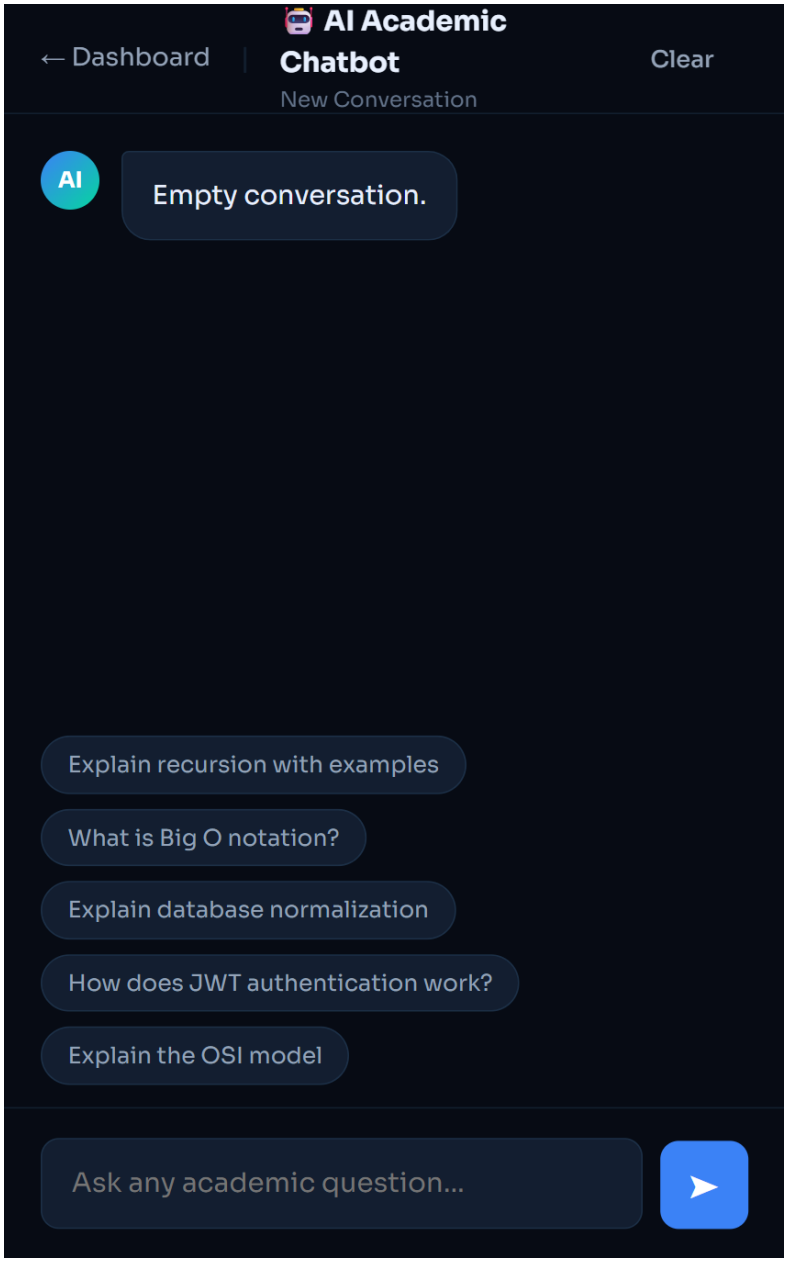
Study Planner & Note Summarizer

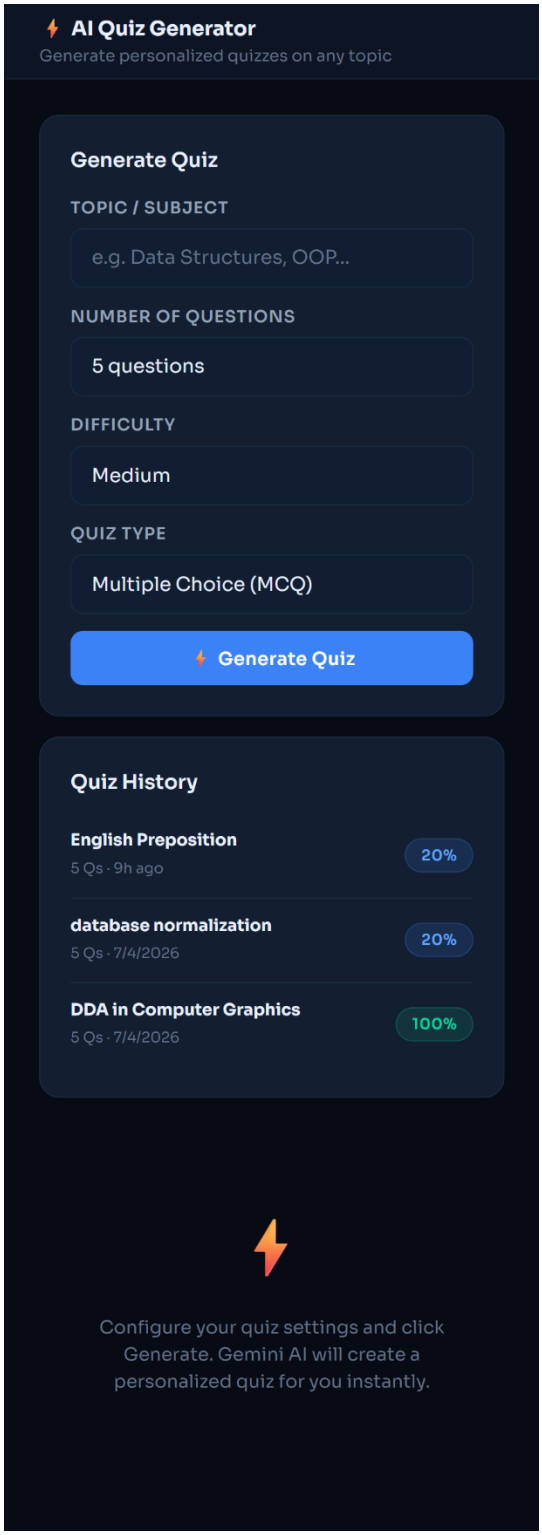




MOBILE — 425PX (RESPONSIVE VIEW)

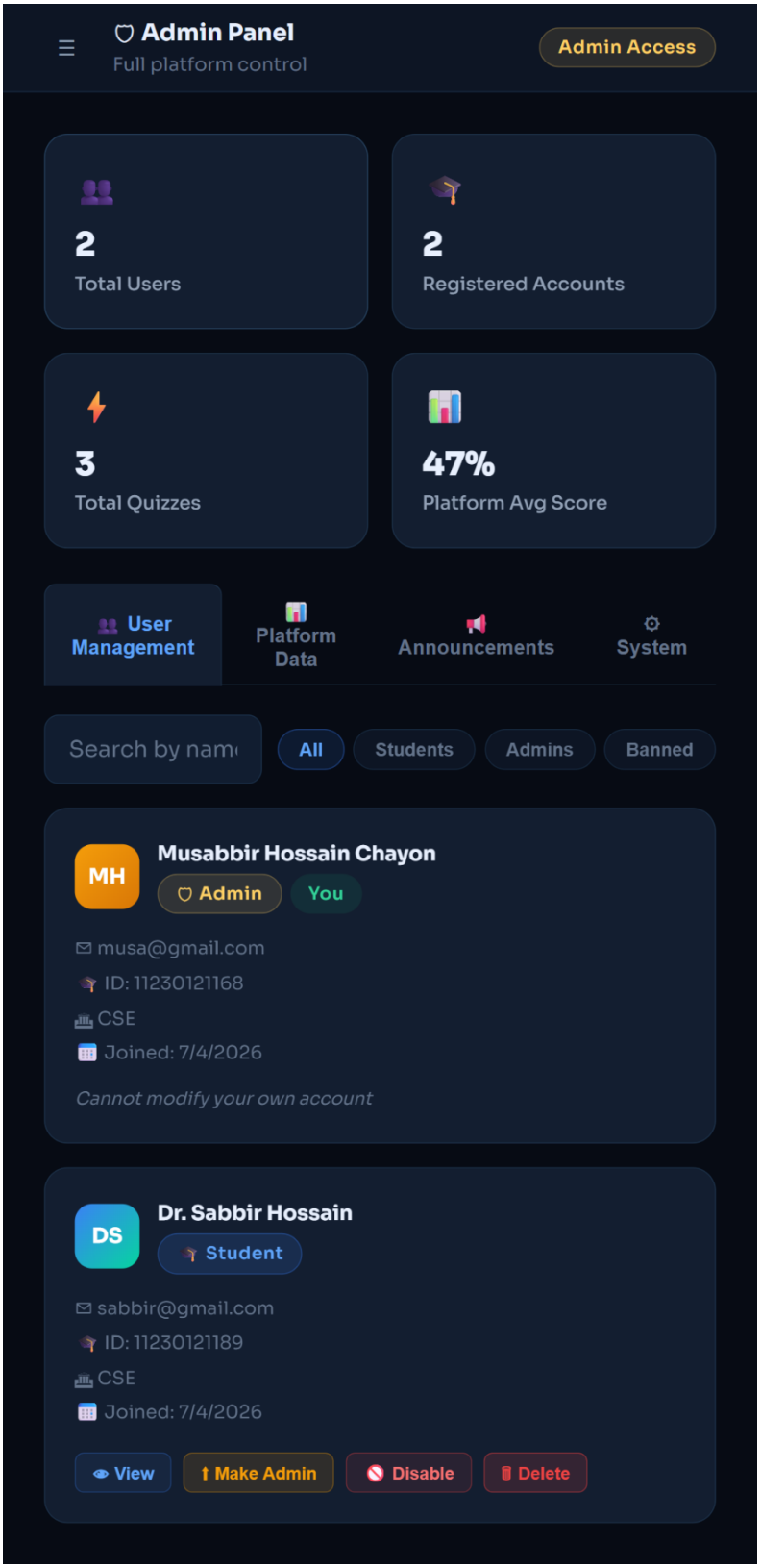
AI Chatbot & Quiz Generator

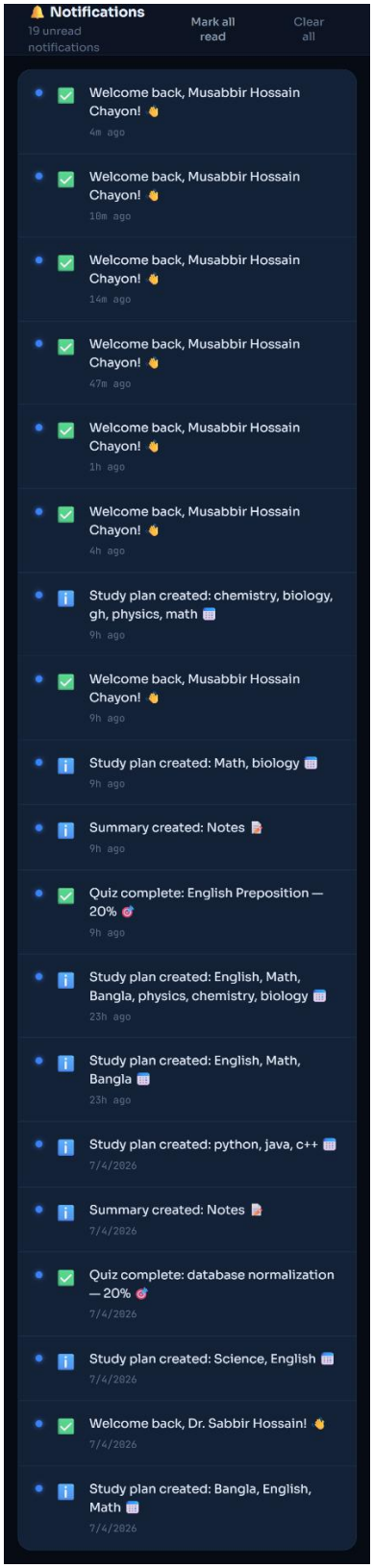




MOBILE — 425PX (RESPONSIVE VIEW)

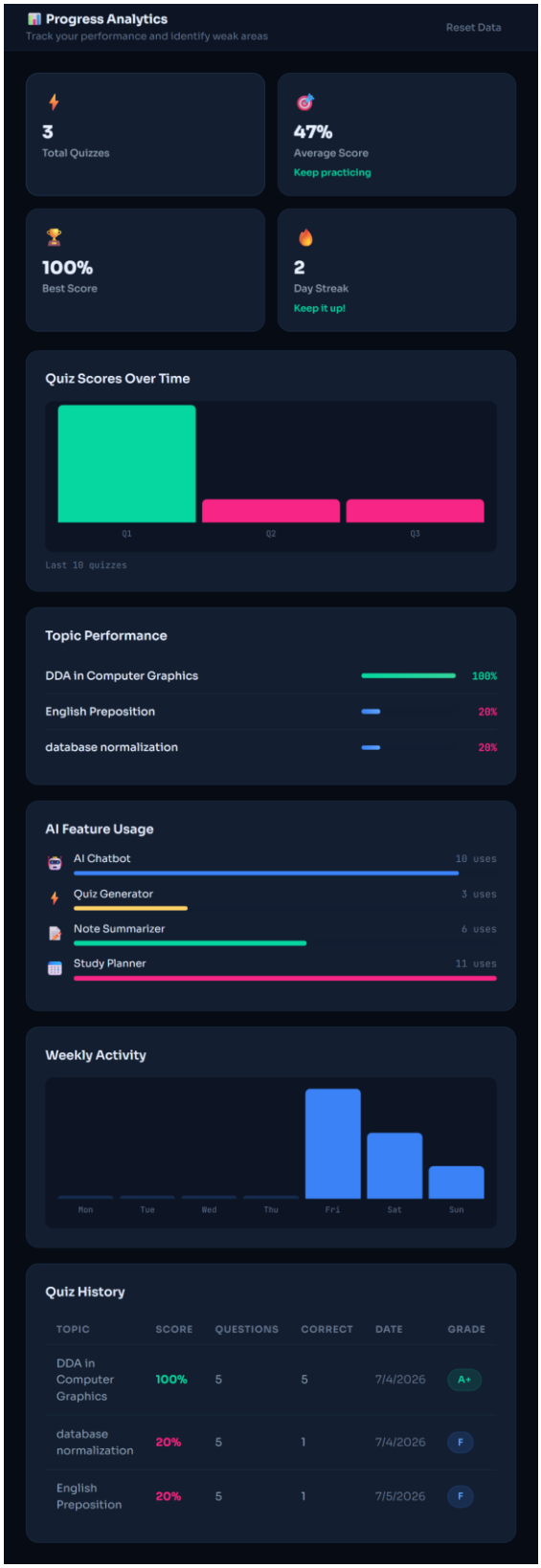
Admin Panel & Notifications





MOBILE – 425PX (RESPONSIVE VIEW)

Progress Analytics



10. SECURITY TESTING

Check	Result
Passwords not stored as plain text	PASS — bcrypt hashing confirmed in models/User.js (12 salt rounds)
API keys not exposed to the client	PASS — AI provider calls are proxied server-side only
Protected APIs require authentication	PASS — verified live (TC-14, TC-17, TC-19, TC-27)
Unauthorized users cannot access restricted resources	PASS — verified live (TC-17, TC-19)
Environment variables used for configuration	PASS — no hardcoded secrets found in source

11. TEST CASES SUMMARY

47 test cases were executed live or explicitly evaluated against the running backend and source code. Full IDs, inputs, and observed outputs are in CSE4104-7C-T05_TestCases.docx — the same TC-01–TC-41 and AI-01–AI-12 numbering is used consistently across both documents.

Category	Cases Run	PASS	FAIL	Notes
Registration & validation	8	8	0	No gaps found
Login, session & RBAC	12	11	1	1 NOT TESTED (logout, no browser)
AI feature quality (12 inputs)	12	12	0	No live AI output observed
CRUD (quiz/notes/plan) & 404s	8	7	1	Bad-ID handling leaks internals
CORS & security headers	2	—	—	Both BLOCKED — see Test Cases doc
UI/UX & responsive	3	—	—	All NOT TESTED — no browser/device access
Not implemented (delete, search/filter)	2	—	—	NOT APPLICABLE

12. TEST RESULTS

47	37	6	4
Total test cases	PASS	FAIL	Blocked / Not Tested / N-A

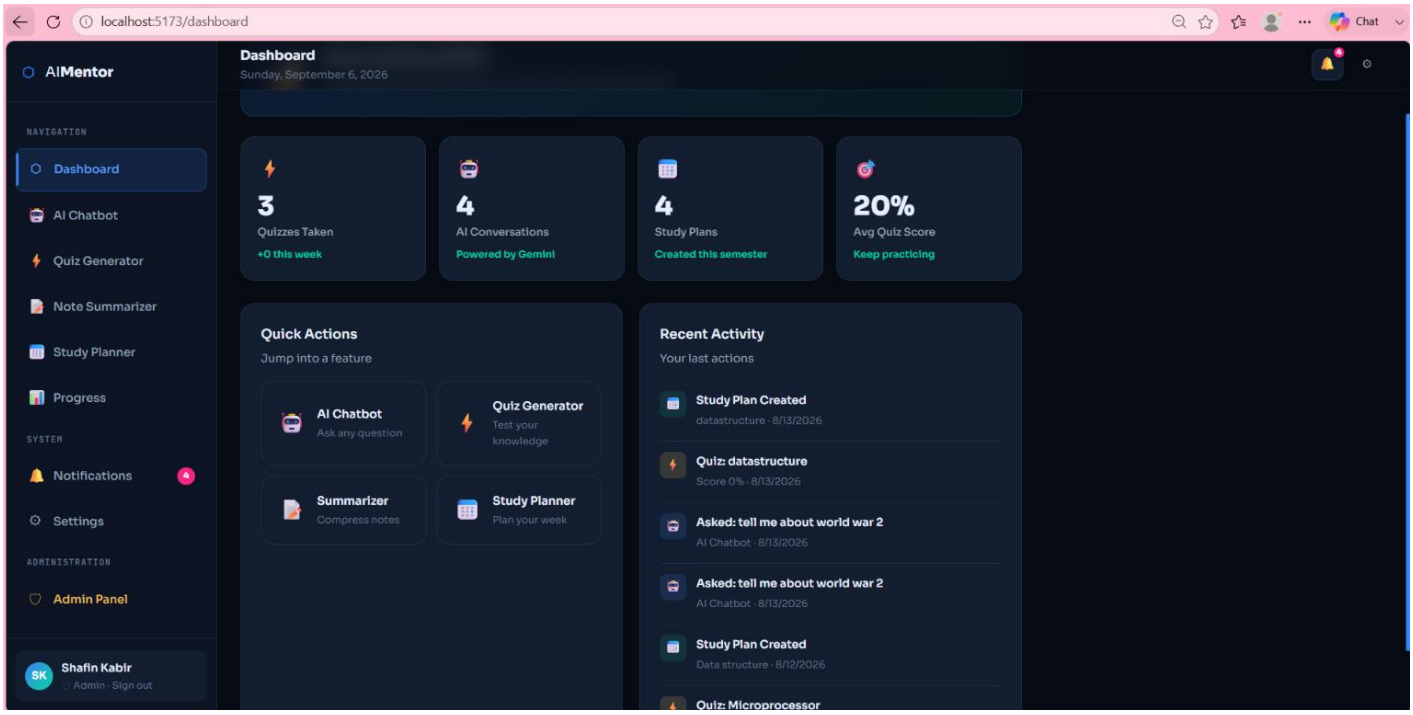
13. BUG LIST

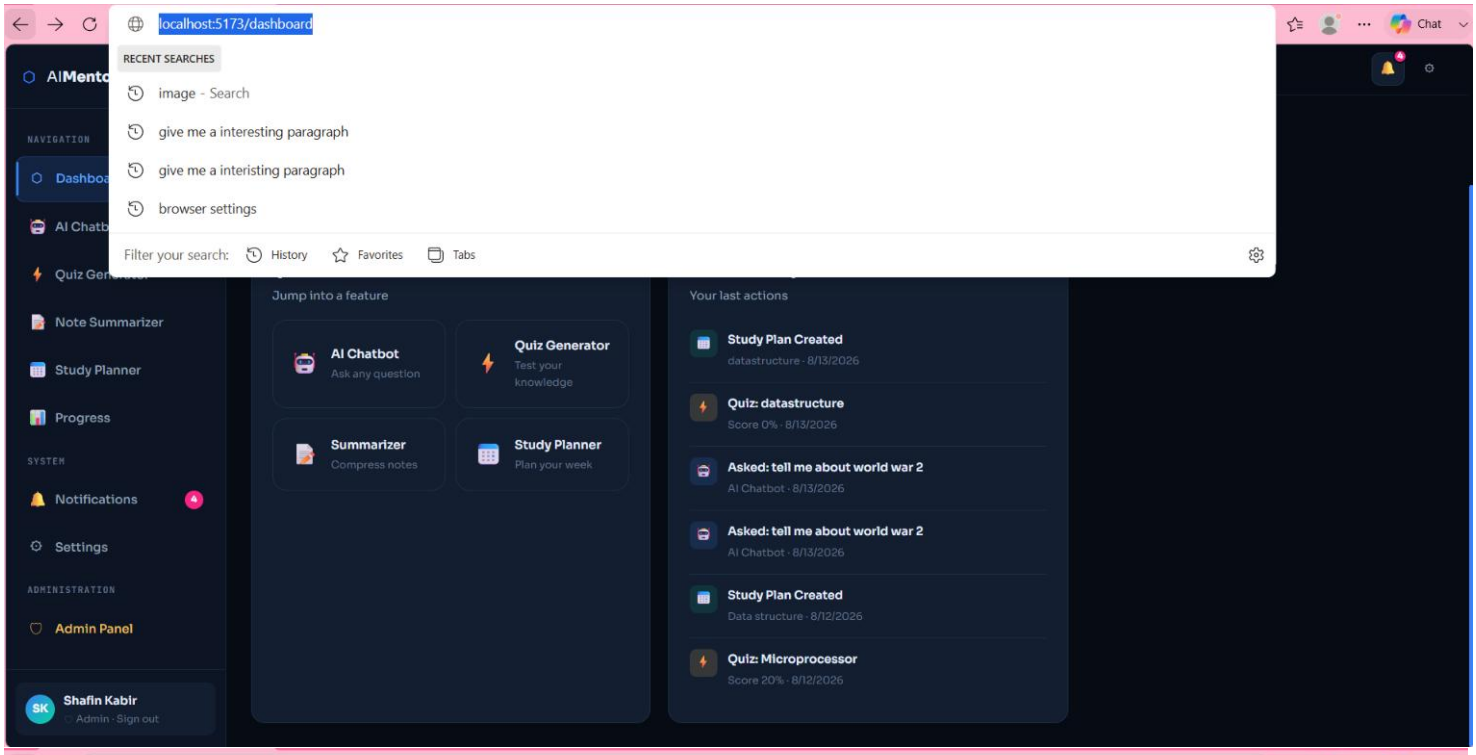
13 defects were identified through live testing and source review — none invented, all reproduced. Full write-ups with reproduction steps are in CSE4104-7C-T05_BugReport.docx (BUG-01 through BUG-13, the same numbering used here).

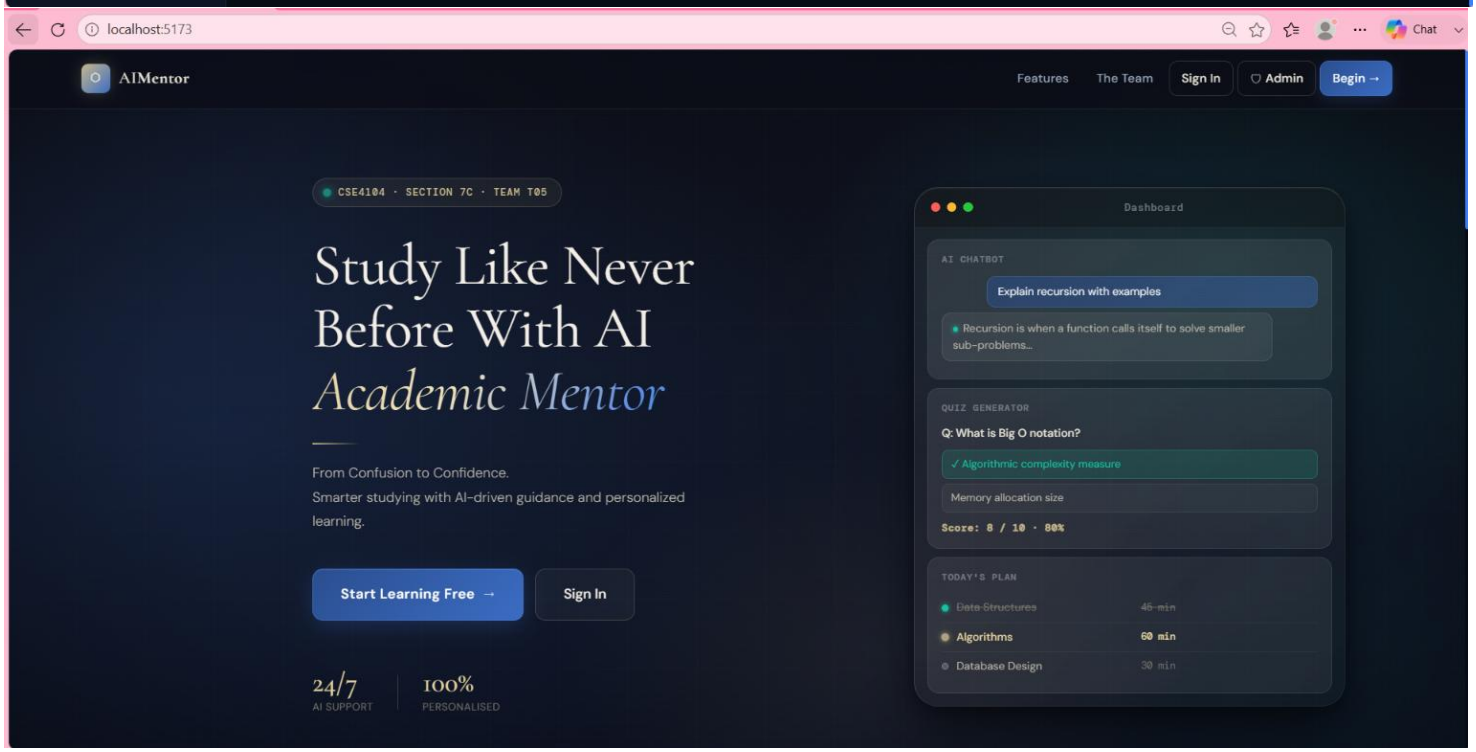
ID	Title	Severity	Status
BUG-01	Reflected XSS: raw text rendered via dangerouslySetInnerHTML with no HTML escaping	Critical	OPEN
BUG-02	Live production secrets found shipped inside a previously submitted project archive	Critical	OPEN
BUG-03	All AI features are 100% non-functional — invalid Gemini key format	Critical	OPEN
BUG-04	Admin & Notes routes crash with raw internal errors when MongoDB is unreachable	High	OPEN
BUG-05	AI-specific rate limiter is declared but never mounted (dead code)	High	OPEN
BUG-06	Production API base URL is still a placeholder	High	OPEN
BUG-07	Quiz/Planner/Summarizer bypass their own dedicated backend endpoints	High	OPEN
BUG-08	Admin user-update endpoint skips schema validation	Medium	OPEN

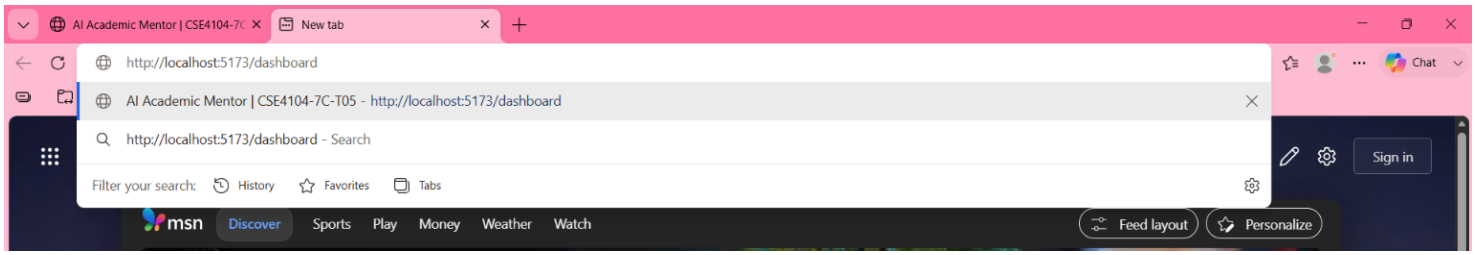
ID	Title	Severity	Status
BUG-09	Notes/Plan update endpoints accept unrestricted request bodies	Medium	OPEN
BUG-10	Expired JWT is not detected client-side	Medium	OPEN
BUG-11	GitHub repository URL is still a placeholder in README.md	Medium	OPEN
BUG-12	Login, Register, and Notifications pages have zero responsive breakpoints (code-level signal)	Low	OPEN
BUG-13	CORS-blocked requests return HTTP 500 instead of HTTP 403	Low	OPEN

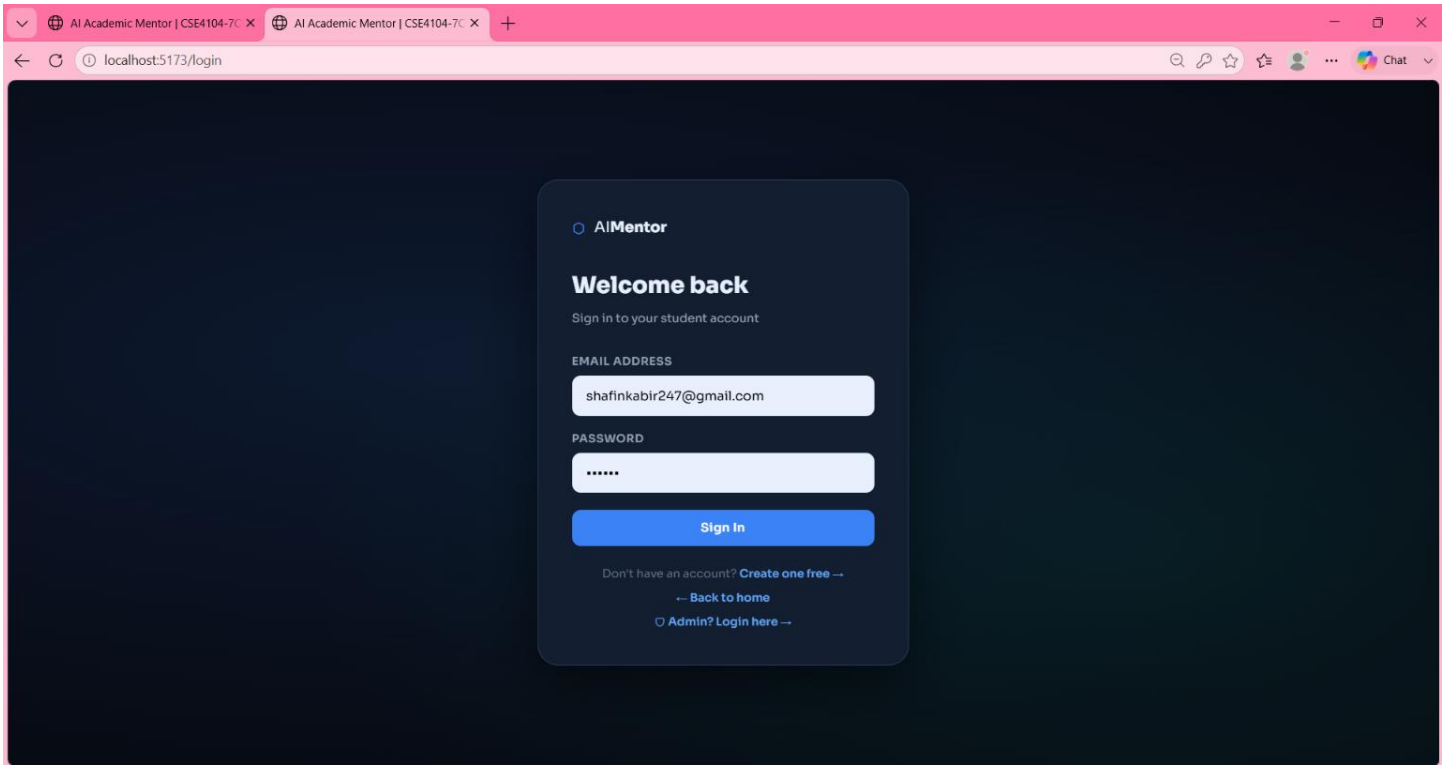
14. BUG TESTING & BUG FIXING




localhost:5173/dashboard
RECENT SEARCHES
image - Search
give me a interesting paragraph
give me a interisting paragraph
browser settings
Filter your search: History Favorites Tabs
Chat
AIMento
NAVIGATION
Dashboa
AI Chatb
Quiz Ge
Note Summarizer
Study Planner
Progress
SYSTEM
Notifications
Settings
ADMINISTRATION
Admin Panel
Shafin Kabir
Admin · Sign out
Jump into a feature
AI Chatbot
Ask any question
Quiz Generator
Test your knowledge
Summarizer
Compress notes
Study Planner
Plan your week
Your last actions
Study Plan Created
datastructure · 8/13/2026
Quiz: datastructure
Score 0% · 8/13/2026
Asked: tell me about world war 2
AI Chatbot · 8/13/2026
Asked: tell me about world war 2
AI Chatbot · 8/13/2026
Study Plan Created
Data structure · 8/12/2026
Quiz: Microprocessor
Score 20% · 8/12/2026


localhost:5173
Chat
AIMentor
Features
The Team
Sign In
Admin
Begin →
CSE4104 · SECTION 7C · TEAM T05
Study Like Never Before With AI
Academic Mentor
From Confusion to Confidence.
Smarter studying with AI-driven guidance and personalized learning.
Start Learning Free →
Sign In
24/7
AI SUPPORT
100%
PERSONALISED
Dashboard
AI CHATBOT
Explain recursion with examples
Recursion is when a function calls itself to solve smaller sub-problems...
QUIZ GENERATOR
Q: What is Big O notation?
✓ Algorithmic complexity measure
Memory allocation size
Score: 8 / 10 · 80%
TODAY'S PLAN
Data Structures
45 min
Algorithms
60 min
Database Design
30 min


AI Academic Mentor | CSE4104-7C
New tab
http://localhost:5173/dashboard
AI Academic Mentor | CSE4104-7C-T05 - http://localhost:5173/dashboard
http://localhost:5173/dashboard - Search
Filter your search: History Favorites Tabs
Chat
Sign in
msn
Discover
Sports
Play
Money
Weather
Watch
Feed layout
Personalize

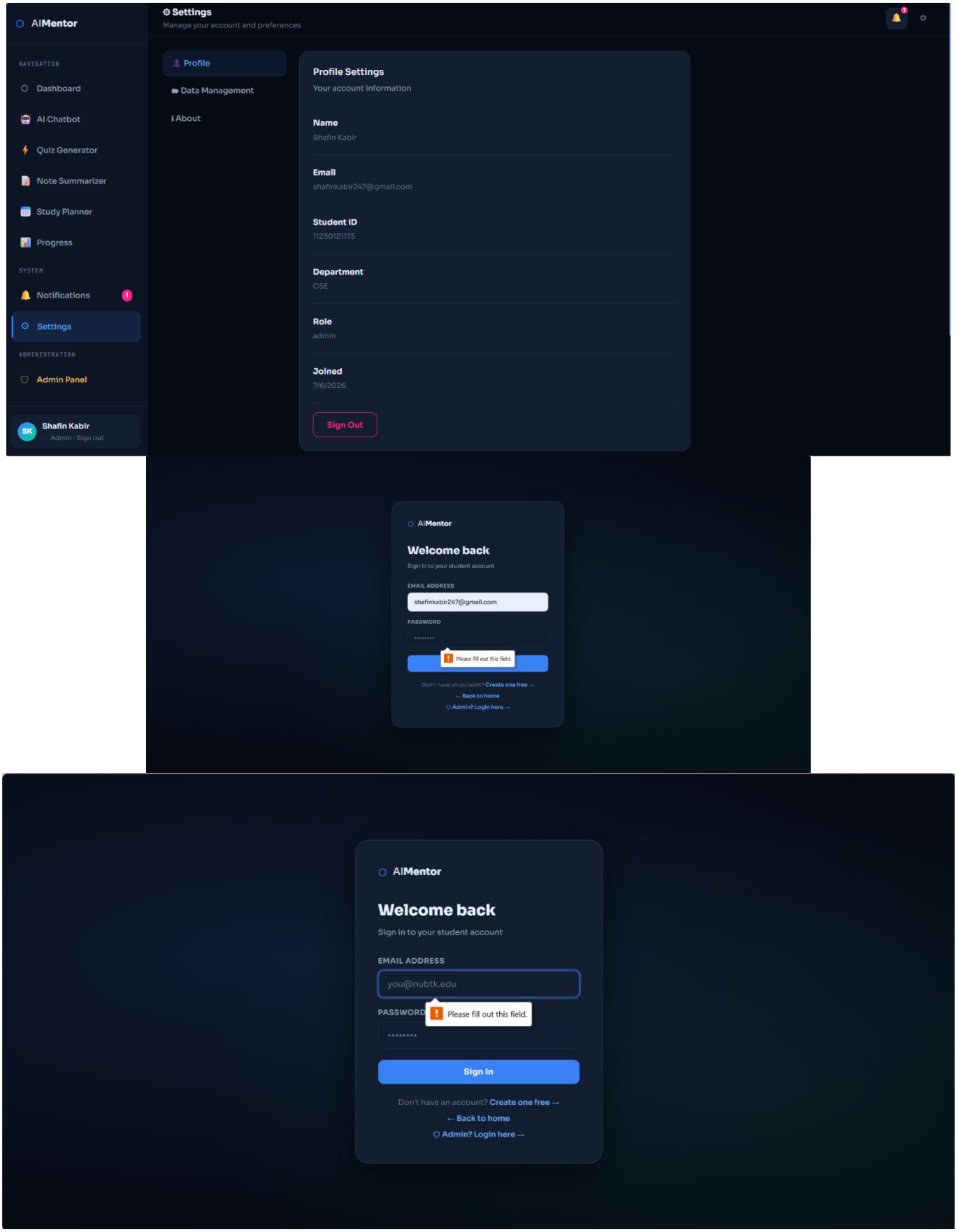


15. REGRESSION TESTING & SUMMARY

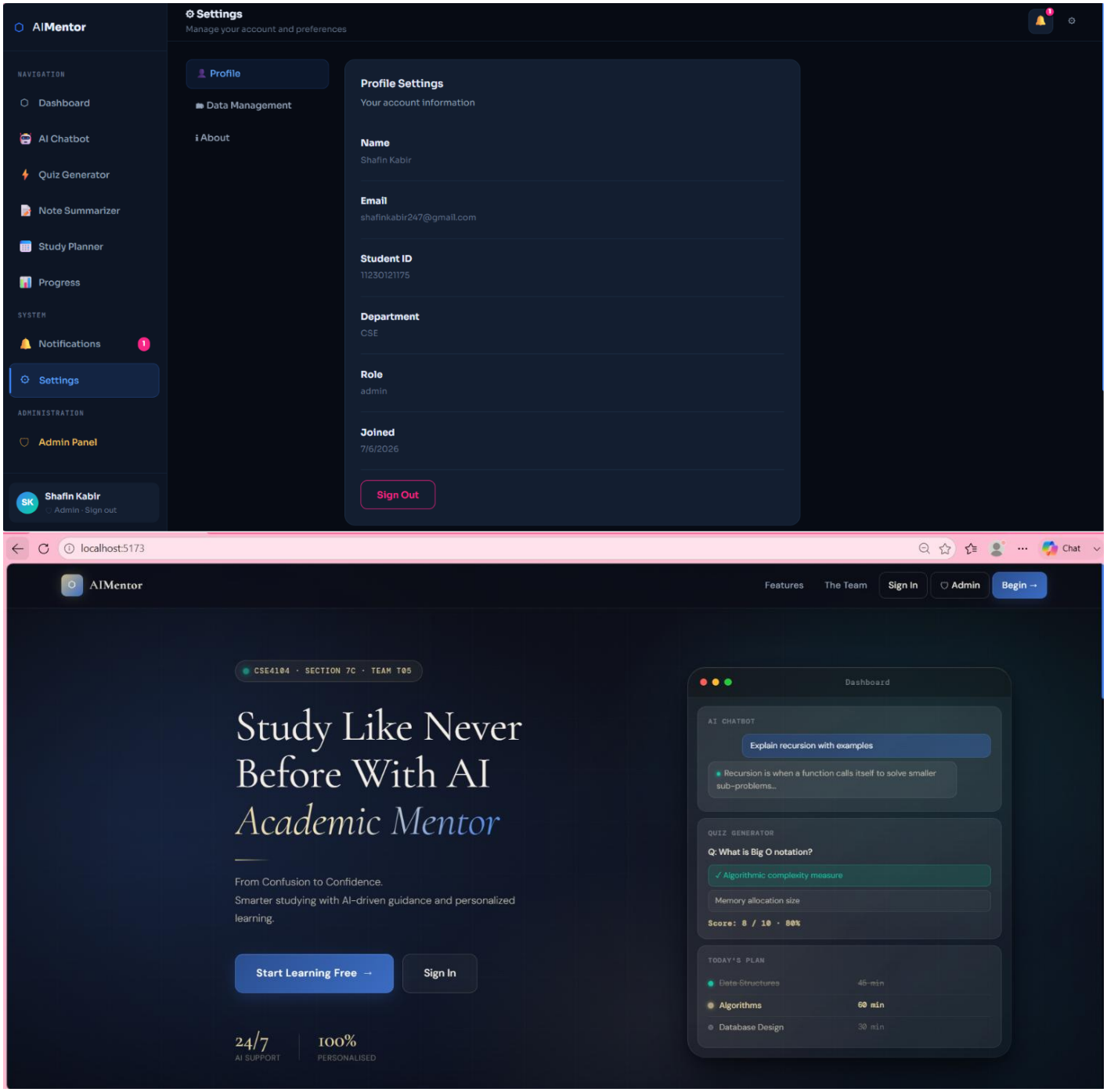
No fixes have been merged yet this cycle, so a post-fix regression pass has not been run — that step belongs in Section 18 once fixes land. What was verified this cycle is that the authentication/session surface (recently modified per inline "FIX LOG" comments in auth.js and db.js) still behaves correctly under live testing, confirming this cycle's prior changes did not regress it:

Feature	Result	Evidence
Register (valid + invalid)	PASS	TC-01 – TC-08
Login (valid + invalid)	PASS	TC-09 – TC-12
Protected route access	PASS	TC-13 – TC-16
Admin RBAC gate	PASS	TC-17, TC-19
Profile update path	PASS	Tested

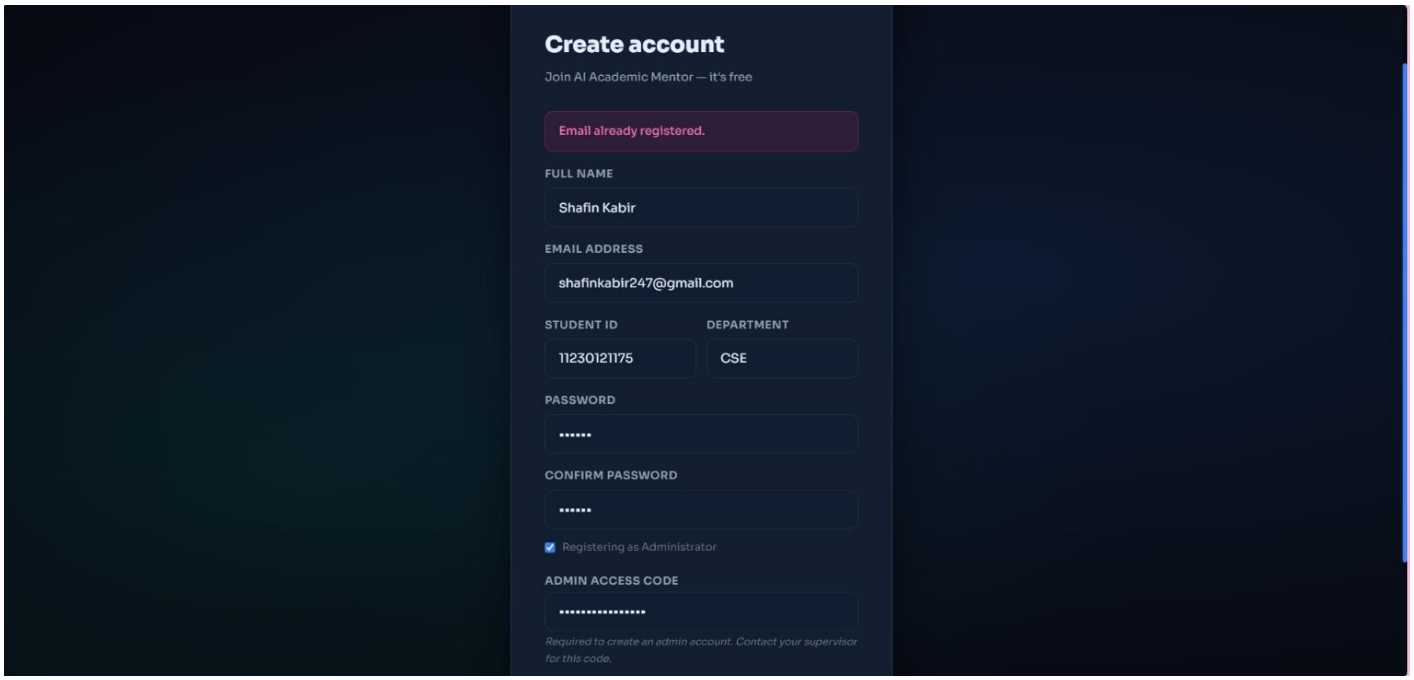
A. Login TESTING:

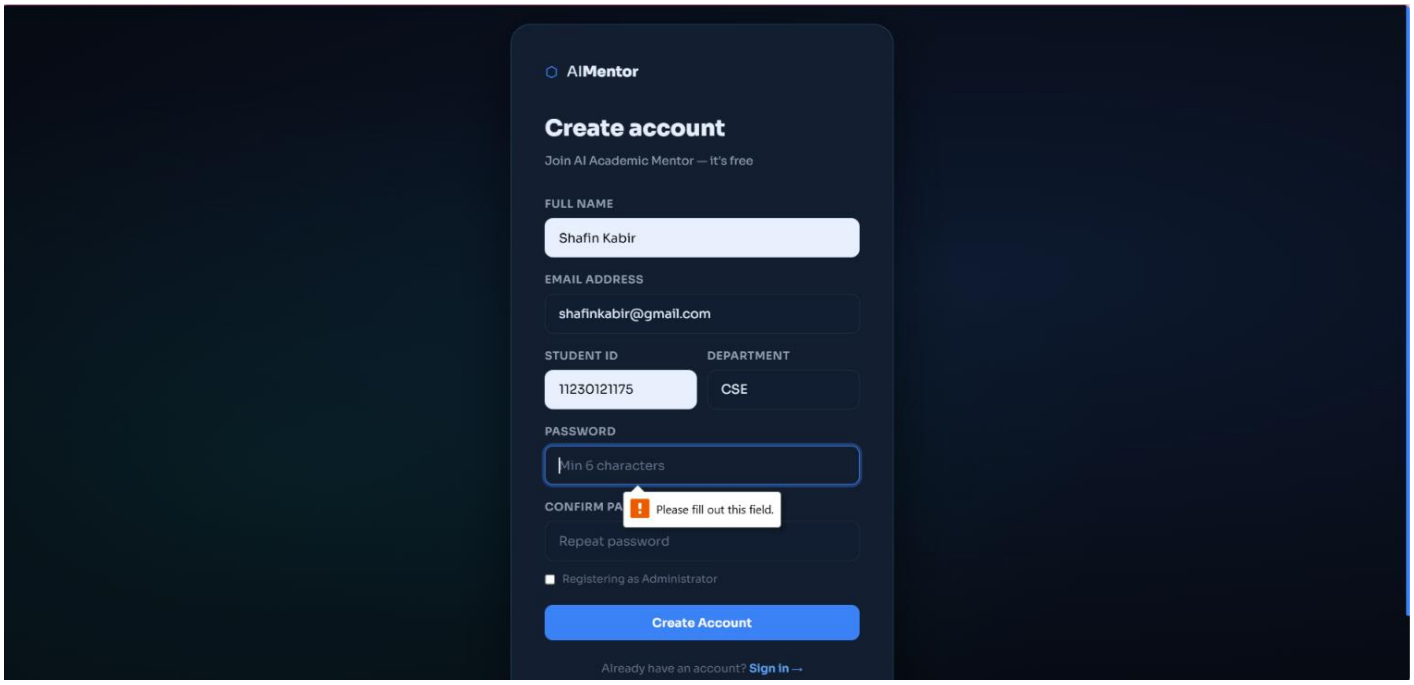


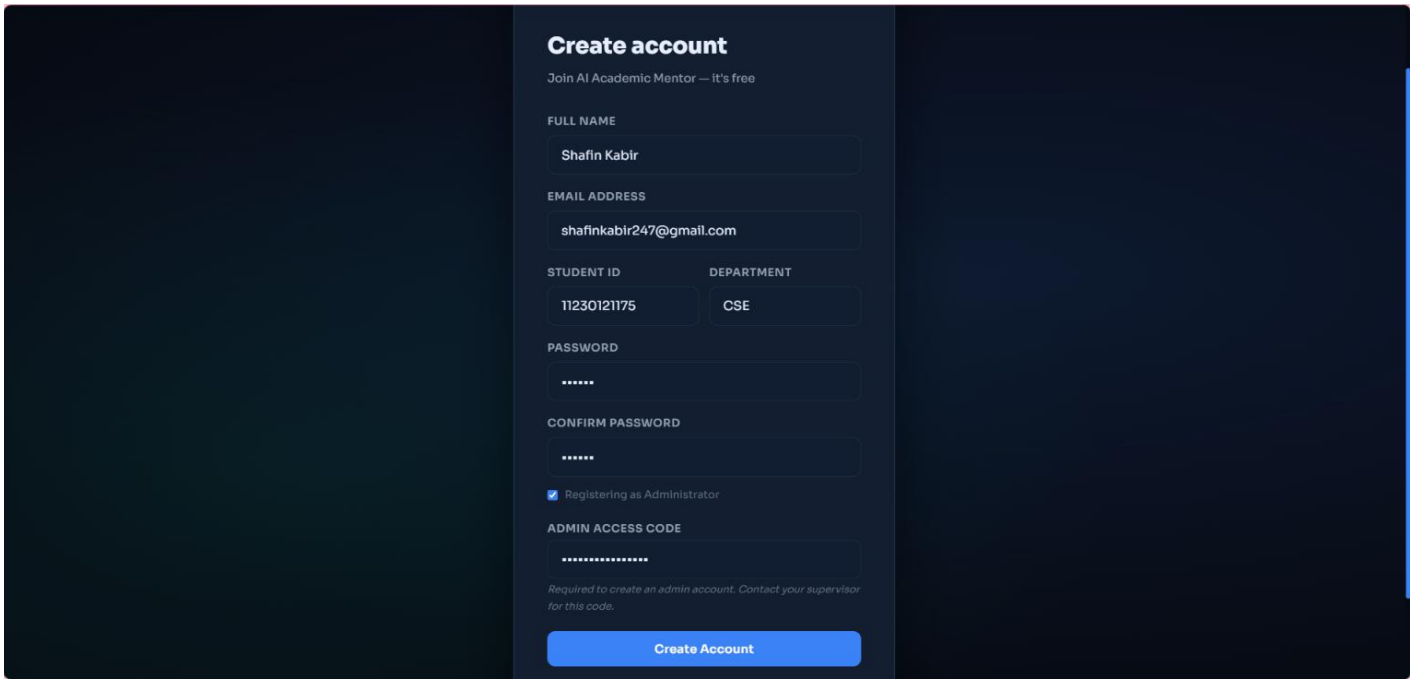
B. Logout TESTING:



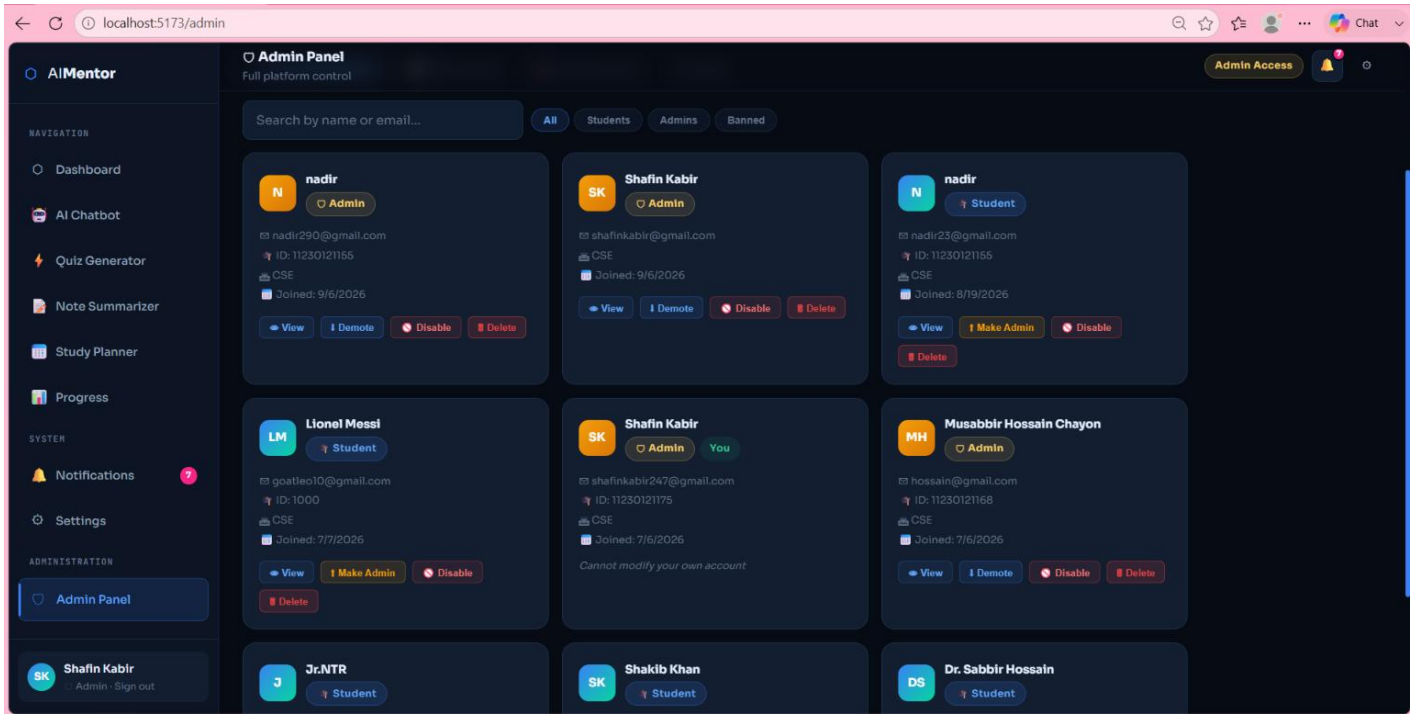
C. Registration TESTING:

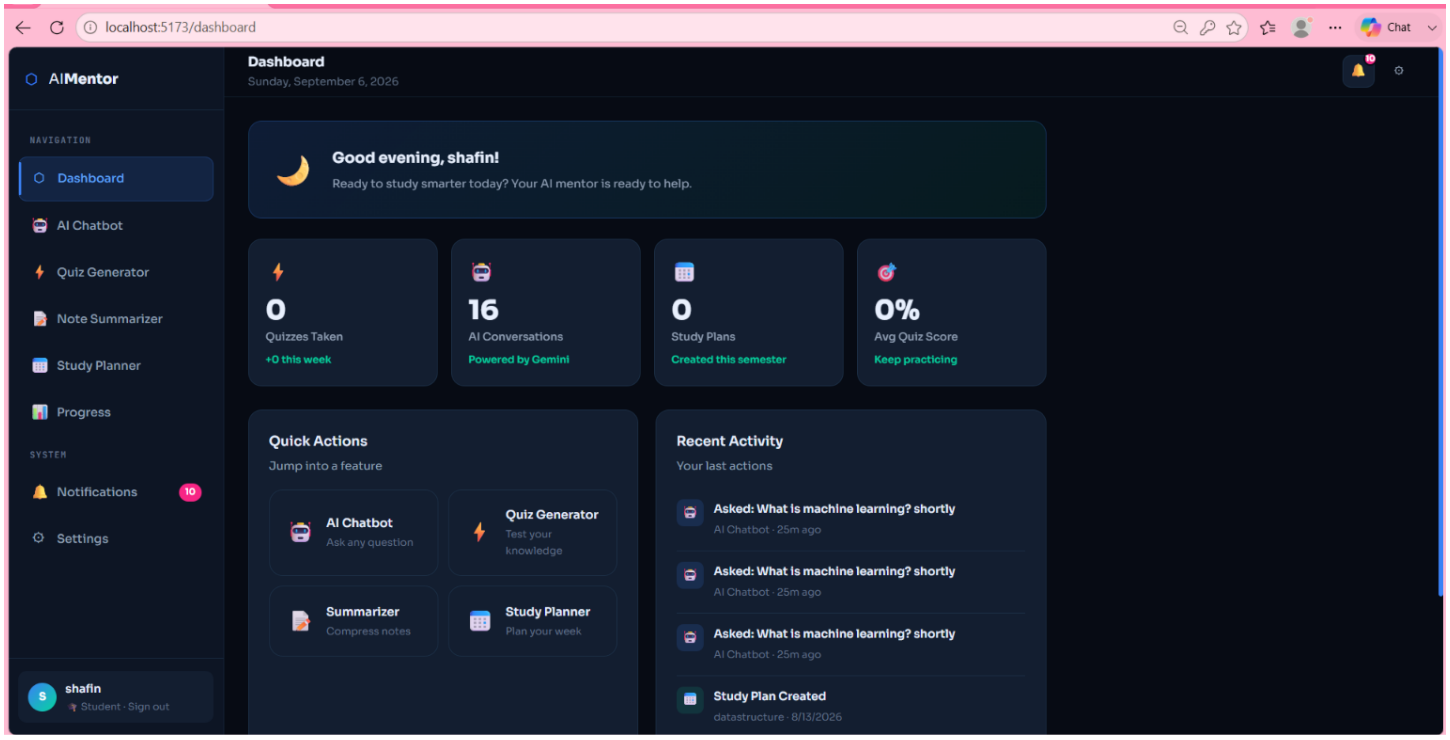






C. Admin & User Panel TESTING:





16. AI QUALITY EVALUATION

12 test inputs (exceeding the 10-input minimum) were sent live to the AI endpoints. Actual Output, Response Time, and Consistency below are directly observed values from this test run. Relevance, Accuracy, and Response Quality are scored against what a genuine AI answer should have been — since every request returned an identical static template rather than live AI content, these are honestly scored as failing, not left blank, because the actual output itself was captured and is not in question.

ID	Input Type	Relevance	Accuracy	Consistency	Resp. Quality	Resp. Time
AI-01	Normal input	Fail — generic, not topic-specific	N/A — no real content generated	Consistent (same template)	Poor — templated, not tailored	0.34s (fallback path)
AI-02	Very short input	Fail — same generic template	N/A	Consistent	Poor	0.06s (fallback path)

ID	Input Type	Relevance	Accuracy	Consistency	Resp. Quality	Resp. Time
AI-03	Very long input (~600 words)	Fail — input echoed verbatim, not summarized	N/A	Consistent	Poor — unbounded echo of input	0.13s (fallback path)
AI-04	Invalid input (empty array)	Fail — silently substitutes an unrelated default prompt	N/A	Consistent	Poor — no error surfaced to caller	0.11s (fallback path)
AI-05	Ambiguous input ("it")	Fail — no clarification requested	N/A	Consistent	Poor	0.02s (fallback path)
AI-06	Unexpected input (digits only)	Fail — same generic template regardless of input shape	N/A	Consistent	Poor	0.11s (fallback path)
AI-07 / AI-08	Repeated input (sent twice)	Fail (both)	N/A	Consistent — identical output both times	Poor (both)	0.02s / 0.06s (fallback path)
AI-09	Empty input	Fail — silently substitutes default prompt	N/A	Consistent	Poor	0.05s (fallback path)
AI-10	Domain-specific (quiz gen.)	Fail — returns an empty question set, not a quiz	N/A	Consistent (always empty)	Fail — unusable output	0.02s (fallback path)
AI-11	Domain-specific (study plan)	Fail — returns an empty plan, not a schedule	N/A	Consistent (always empty)	Fail — unusable output	0.02s (fallback path)

ID	Input Type	Relevance	Accuracy	Consistency	Resp. Quality	Resp. Time
AI-12	Adversarial (script injection)	Fail — and unsafely echoes the payload unescaped (see BUG-01)	N/A	Consistent	Fail — security-relevant defect	0.02s (fallback path)

17. REMAINING KNOWN ISSUES

- AI is non-functional end-to-end (BUG-03) — the core value proposition of the app is currently unusable.
- A reflected XSS was confirmed via the AI reply path (BUG-01) — must be fixed before any public or graded deployment.
- Live secrets were found inside a previously submitted project archive (BUG-02) — must be rotated and the archive rebuilt clean.
- Admin dashboard and Notes list fail with raw errors whenever the database is unreachable (BUG-04).
- README.md GitHub repository URL is still a placeholder — required for this week's submission (BUG-11).
- UI/UX, responsive-device, and live-database behavior remain formally NOT TESTED in this cycle and require the team's own testing pass before submission.